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Electric Vehicle System Engineering

UNIT 1 :Introduction to Electric Vehicles

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Unit 1: Introduction to Electric Vehicles

1.1 Introduction to Electric Vehicles

Road transport is being re-engineered around the electric machine. A vehicle that once relied on a single heat engine and a mechanical gearbox is increasingly a coordinated system of an electrochemical store, power-electronic converters, electric machines, embedded controllers and thermal circuits. This section sets out what an electric vehicle is, how the technology evolved, why it matters, how electrified vehicles are classified, and which components make up the system. The later sections then quantify each of these ideas.

1.1.1 Definition and Concept of Electric Vehicles

An *electric vehicle* (EV) is a road vehicle in which at least part of the propulsive force at the wheels is produced by one or more electric machines drawing energy from an on-board electrical source. The definition is deliberately broad because the source differs among vehicle types: a rechargeable battery, a fuel-cell stack, or a generator driven by an engine. When all the propulsion energy comes from a battery charged from the grid, the vehicle is a *pure* or *battery* electric vehicle.

Three engineering ideas distinguish an EV from a conventional vehicle.

1. **The energy path is mainly electrical.** Energy reaches the wheels through cables and converters rather than through a fuel line, combustion chamber and multi-speed gearbox. This allows the designer to place components with much greater freedom.

2. **The energy path is bidirectional.** An electric machine can act as a motor or as a generator, so kinetic energy can return to the battery during braking. A heat engine has no equivalent ability.

3. **The torque source is fast and controllable.** A power-electronic drive can change motor torque in milliseconds and can deliver full torque from standstill. Hence many EVs need only a single fixed reduction gear.

1.1.2 Historical Development of Electric Vehicles

The EV is older than the petrol car. Its history has three broad phases, an early rise, a long eclipse and a modern return, and each phase was driven by the energy-storage technology available at the time [2, 8].

Early development (1830s–1900). Small electric carriages were demonstrated in the 1830s using non-rechargeable cells. The practical turning point was the lead-acid rechargeable cell of

Gaston Planté in 1859, later improved by Camille Fauré's pasted plates. By the 1890s electric cabs operated in cities such as London and New York. In 1899 the electric car *La Jamais Contente* of Camille Jenatzy became the first road vehicle to exceed 100 km/h [2].

Decline (1910s–1960s). EVs were quiet, clean and easy to start, but their range was limited by the low specific energy of lead-acid batteries. Three developments favoured petrol vehicles: cheap oil, mass production of the Ford Model T from 1908, and the electric self-starter introduced in 1912, which removed the hand crank. Electric traction then survived mainly in niche roles such as delivery floats, industrial trucks and rail.

Renewed interest (1970s–1990s). The oil shocks of the 1970s and later urban air-quality rules revived research. California's zero-emission vehicle mandate of 1990 prompted vehicles such as the General Motors EV1 (1996), and the Toyota Prius (1997) brought power-split hybrids to mass production [8]. Power MOSFETs, IGBTs and microcontrollers made efficient AC drives possible over the same period.

Lithium-ion era (2008 onwards). Lithium-ion cells, first commercialised for portable electronics in 1991, changed the range equation. The Tesla Roadster (2008) and the Nissan Leaf (2010) showed that lithium-ion BEVs were viable, and costs have since fallen with scale. According to the International Energy Agency, global electric-car sales exceeded 20 million in 2025, about one in four new cars sold [27]. In India the Ministry of Heavy Industries' PM E-DRIVE scheme, with an outlay of Rs 10,900 crore, succeeded FAME-II in 2024 to support electric two- and three-wheelers, buses, trucks and charging infrastructure [28].

1.1.3 Need and Importance of Electric Mobility

The case for electric mobility rests on several linked technical and social drivers.

- **Energy efficiency.** A BEV typically converts about 70–80% of the electrical energy taken from the battery into energy at the wheels, whereas a petrol engine's tank-to-wheel efficiency in urban use is commonly about 20–25%. Section 1.4 examines these figures.
- **Local air quality.** A BEV has no tailpipe emissions, which matters most in dense cities. Non-exhaust particulates from brakes and tyres remain, although regenerative braking reduces brake wear.
- **Climate policy.** Life-cycle greenhouse-gas emissions depend on the grid mix and on battery manufacture, so the benefit grows as electricity generation decarbonises [29].
- **Energy security.** Electricity can be generated from domestic and renewable sources, which reduces dependence on imported crude oil. This is significant for India, which imports most of its crude.

- **Grid and system integration.** Parked EVs are a flexible load and, through vehicle-to-grid operation, may become a distributed storage resource.
- **Operating cost and new functionality.** Electricity per kilometre is usually cheaper than fuel and there are fewer wearing parts. Precise wheel-torque control also supports stability control and automated driving.

1.1.4 Classification of Electric Vehicles

Electrified vehicles are classified mainly by the *degree of electrification*, that is, by how much of the propulsion energy and power is electric, and by the energy source. Figure 1.1 shows the four principal families. Moving from left to right in the second row does not indicate increasing merit: each branch is defined by what supplies the traction energy. The BEV draws solely on grid-charged batteries; the HEV obtains all its energy from fuel; the PHEV adds a plug and a larger battery; and the FCEV replaces the battery as the principal energy store with hydrogen. The lower rows show how each family is subdivided by architecture or operating strategy.

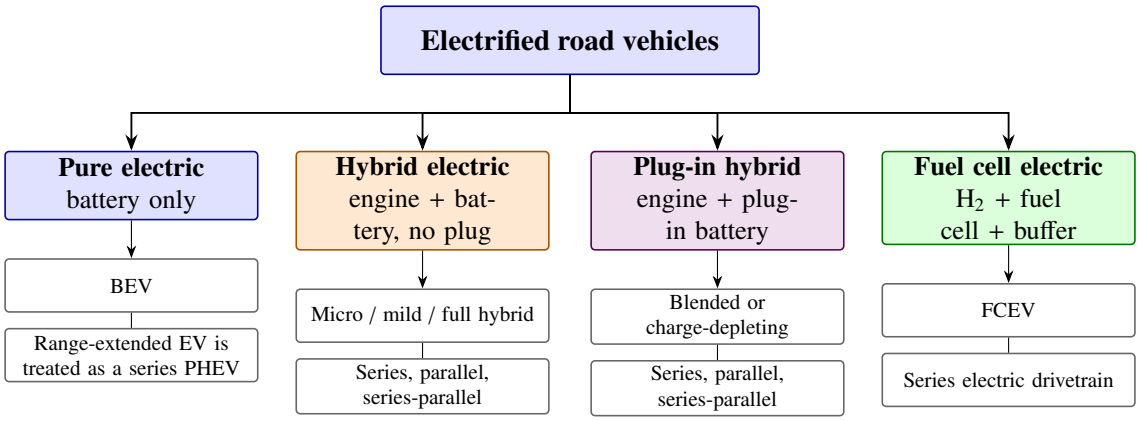


Figure 1.1: Classification of electrified road vehicles by energy source and degree of electrification.

Table 1.1 compares these categories. The key observation is that the battery's role changes with the category: it is the *energy* store in a BEV, a *power* buffer in an HEV or FCEV, and both in a PHEV. This determines the cell selection discussed in Section 1.5. Battery sizes are representative of passenger cars and are not limits.

Table 1.1: Classification of electric vehicles (representative passenger-car values).

Type	Energy source	Propulsion	Plug	Battery (kWh)	Battery role
BEV	Battery (grid)	Motor only	Yes	30–100	Energy store
HEV	Fuel	Engine + motor	No	1–2	Power buffer
PHEV	Fuel + grid	Engine + motor	Yes	8–25	Energy + power
FCEV	Hydrogen	Motor only	No[a]	1–2	Power buffer

[a]Hydrogen is refuelled; a few designs also offer battery plug-in charging.

1.1.4.1 Battery Electric Vehicles (BEV)

A BEV stores all of its energy electrochemically and is propelled only by electric motors. It has no engine, fuel tank or exhaust. The battery is recharged from the grid through an on-board charger for AC supply or directly from an off-board DC fast charger, and partly by regenerative braking. Its strengths are the highest well-to-wheel efficiency of the four families when the electricity is low-carbon, mechanical simplicity and quiet operation. Its limitations are the mass and cost of the battery and the charging time, because range scales almost linearly with installed energy. Sections 1.2 and 1.7 examine its architecture and sizing.

1.1.4.2 Hybrid Electric Vehicles (HEV)

An HEV combines an internal combustion engine (ICE) and at least one electric machine, and has no external charging. All net energy comes from fuel; the small battery only shifts energy in time. The hybrid gains efficiency in four ways: it stops the engine at idle, recovers braking energy, downsizes the engine, and runs it nearer its efficient operating region.

Hybrids are also graded by the electric machine's contribution. In a *micro hybrid* an uprated starter-generator provides stop-start. In a *mild hybrid*, often 48 V, the machine assists the engine but cannot normally propel the car alone. A *full hybrid* can drive short distances on electric power alone.

1.1.4.3 Plug-in Hybrid Electric Vehicles (PHEV)

A PHEV is a hybrid whose larger battery can be charged from the grid. It normally starts in *charge-depleting* mode, using grid energy, and changes to *charge-sustaining* mode, behaving like an HEV, when the battery reaches its lower SoC limit. A PHEV can cover most daily trips electrically and still have the range of a fuel vehicle, at the price of carrying two complete energy systems. Its real-world benefit depends strongly on how regularly it is charged. A range-extended EV is effectively a series PHEV whose engine only drives a generator.

1.1.4.4 Fuel Cell Electric Vehicles (FCEV)

An FCEV generates electricity on board in a fuel cell, normally a proton-exchange-membrane stack, by combining stored hydrogen with oxygen from air; the only exhaust product is water vapour. Traction is entirely electric, so its drivetrain resembles a BEV with the battery replaced by a stack and hydrogen tank plus a small buffer battery. Its advantages are fast refuelling and a high energy-to-mass ratio, which suits long-haul duty. Its limitations are sparse refuelling infrastructure, stack cost and durability, and the energy lost in producing, compressing and reconverting hydrogen.

1.1.5 Advantages and Limitations of EVs

The advantages and limitations below apply principally to BEVs and should be read as engineering trade-offs, not as absolutes.

Advantages.

- High tank-to-wheel efficiency and energy recovery during braking.
- Zero tailpipe emissions, and low noise and vibration.
- Full torque from standstill, with a wide constant-power speed range and a simple single-speed reduction.
- Fewer moving parts and no engine oil, clutch or exhaust after-treatment, which reduces routine maintenance.
- Low centre of gravity, because the battery sits in the floor, and improved packaging freedom.
- Precise and fast torque control, which benefits traction and stability control.

Limitations.

- The specific energy of batteries is far below that of liquid fuels, which increases vehicle mass for long range.
- High initial cost, dominated by the battery pack.
- Charging takes longer than refuelling, and charging infrastructure is unevenly distributed.
- Range falls in very hot or cold weather because of cabin conditioning and cell behaviour.
- Batteries degrade over time, and raw-material supply and end-of-life recycling need managing.
- High-voltage safety, thermal-runaway protection and electromagnetic compatibility need careful design.

1.1.6 Comparison of Conventional ICE Vehicles and Electric Vehicles

Table 1.2 compares the two vehicle concepts from an engineering perspective. Two contrasts deserve attention. First, the motor's torque-speed curve already matches the traction demand, so a multi-speed gearbox is usually unnecessary (Fig. 1.10). Second, the comparison of energy stores is reversed by efficiency: although petrol stores roughly fifty times as much energy per kilogram as a cell, a BEV uses each kilowatt-hour about three times as effectively, so the net penalty in pack mass is smaller than the raw figures suggest.

1.1.7 Major Components of an Electric Vehicle

Every EV is built from a small set of functional subsystems, listed in Table 1.3. Together they form three interacting domains: the *electrical* domain carrying power, the *mechanical* domain delivering torque to the road, and the *control* domain coordinating them. Figure 1.2 in the next section shows how they are connected.

Table 1.2: Comparison of ICE vehicles and BEVs.

Parameter	ICE vehicle	Battery electric vehicle
Prime mover	Petrol or diesel engine	Electric motor (PMSM, induction, others)
Energy store	Fuel, about 12 000 Wh/kg (petrol, LHV)	Cells, about 100–260 Wh/kg
Tank-to-wheel efficiency	About 20–25%	About 70–80%
Torque characteristic	Rises with speed; none at standstill	Maximum from zero speed
Transmission	Clutch or torque converter, multi-speed gearbox	Fixed reduction gear in most designs
Braking energy	Dissipated as heat	Partly regenerated
Refuelling	A few minutes	Tens of minutes (DC fast) to hours (AC)
Tailpipe emissions	CO_2, NO_x, CO, HC, particulates	None
Noise and vibration	Significant	Low; tyre and wind noise dominate
Maintenance	Oil, filters, belts, exhaust system	Mainly tyres, brake fluid, coolant
Cold-weather effect	Engine waste heat warms the cabin	Cabin heat draws on the battery

Table 1.3: EV powertrain components and their functions.

Component	Function	Typical technology
Traction battery	Stores energy and supplies or absorbs traction power	Lithium-ion pack, 300–800 V
Battery management system	Monitors cells, estimates state, protects the pack	Distributed master–slave electronics
Traction inverter	Converts DC to controlled variable-frequency AC and back	Si IGBT or SiC MOSFET VSI
Traction motor	Converts electrical power to torque and back	PMSM, induction, or wound-rotor machine
Reduction gear and differential	Multiplies torque; divides it between wheels	Fixed ratio about 7–12
On-board charger	Converts grid AC to controlled DC	Isolated AC–DC, 3–22 kW
DC–DC converter	Supplies the low-voltage network from the HV bus	Isolated buck, 1–3 kW
Vehicle control unit	Interprets driver demand; coordinates subsystems	Automotive microcontroller on CAN
Thermal management	Keeps battery, motor and inverter within limits	Liquid coolant loops, heat pump

1.2 Configuration of Electric Vehicles

The *configuration* of a vehicle describes how its energy sources, converters, machines and wheels are connected. The choice fixes which energy paths exist, and therefore which operating modes are possible, how large each component must be, and how the control problem is structured. Throughout this section electrical, mechanical and chemical paths use the colour convention given under the diagrams.

1.2.1 Basic Architecture of an Electric Vehicle

Any EV can be regarded as three subsystems [1]. The *electric propulsion subsystem* comprises the vehicle controller, power converter, motor, mechanical transmission and driven wheels. The *energy source subsystem* comprises the source, its management system and the charging unit. The *auxiliary subsystem* comprises power steering, climate control and the low-voltage supply. Inputs from the accelerator and brake pedals are interpreted by the controller as torque demands, and the converter regulates the power flowing between source and motor.

1.2.2 Battery Electric Vehicle (BEV) Configuration

Figure 1.2 shows the architecture of a typical single-motor BEV. The battery pack supplies DC power to the traction inverter, which converts it to controlled three-phase AC. The motor converts this into torque at speed ω_m, and the reduction gear and differential scale it to the wheels. The double-headed arrows along the main chain mark the path as bidirectional, as energy reverses during regenerative braking.

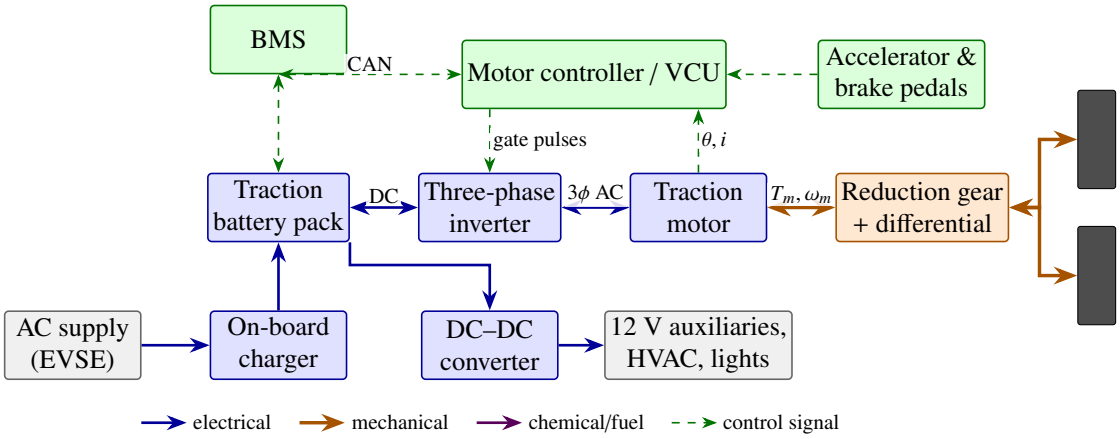


Figure 1.2: Basic architecture of a single-motor battery electric vehicle.

Three supporting branches complete the diagram. The on-board charger rectifies the AC supply delivered through the EVSE (electric vehicle supply equipment) and charges the pack under BMS supervision. The isolated DC–DC converter feeds the 12 V network and its loads, replacing the alternator of a conventional car. The dashed control paths show the motor controller receiving pedal commands and rotor-position and current feedback, while exchanging limits with the BMS over the CAN bus. Thus the controller never commands more than the battery can safely deliver or absorb.

1.2.3 Series Hybrid Electric Vehicle Configuration

In a series hybrid the engine has no mechanical link to the wheels, as Fig. 1.3 shows. Fuel drives the engine, the engine drives a generator, and the rectified output feeds a common DC link. The wheels are driven only by the traction motor, which draws on the DC link. The battery connects to the link and supplies or absorbs the difference between generated and demanded power.

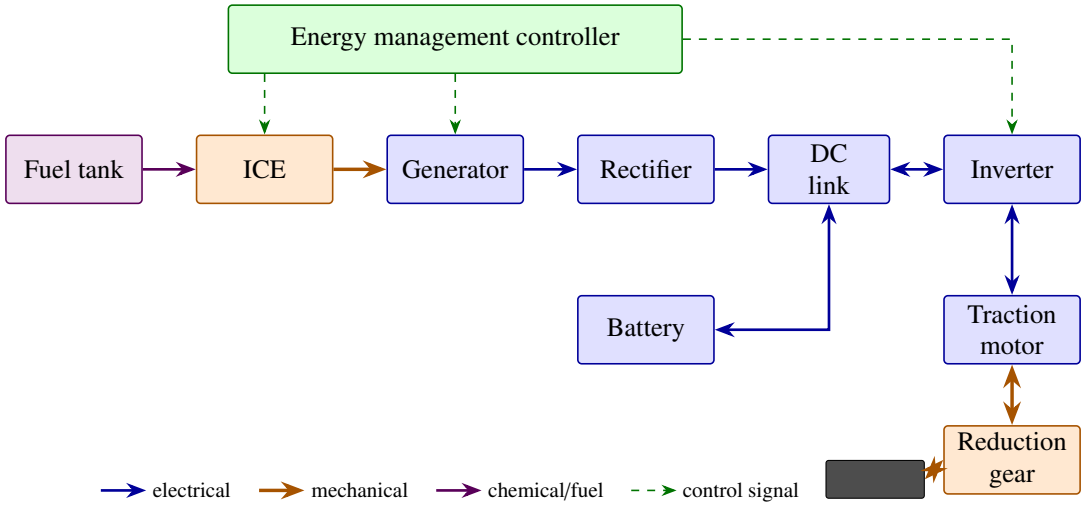


Figure 1.3: Series hybrid electric vehicle configuration.

Power balance at the DC link is the core of this configuration:

$$P_{\text{gen}} + P_{\text{bat}} = P_{\text{inv}}, \tag{1.1}$$

where P_{gen} (W) is the rectified generator power, P_{bat} (W) is the battery power (positive on discharge), and P_{inv} (W) is the inverter input power. Equation (1.1) means the engine's speed and torque are independent of the wheels, so the energy management controller shown in the figure can hold the engine in its best-efficiency region. The cost is the double energy conversion, mechanical to electrical to mechanical, and the need for three full-size machines. The arrangement suits stop-start duty such as city buses, and it forms the basis of range-extended EVs.

1.2.4 Parallel Hybrid Electric Vehicle Configuration

In a parallel hybrid, the engine and the motor both drive the wheels mechanically. In Fig. 1.4 their torques combine in a torque coupler, which may be a belt, gear pair or simply a common shaft. The clutch allows the engine to be disconnected for electric-only drive or regeneration. Because only one electric machine is required and engine power reaches the wheels without double conversion, the arrangement is compact and efficient at steady highway speeds.

For a torque coupler with ratios k_1 and k_2 from engine and motor, the output torque and speeds obey

$$T_{\text{out}} = k_1 T_e + k_2 T_m, \qquad \omega_{\text{out}} = \frac{\omega_e}{k_1} = \frac{\omega_m}{k_2}, \tag{1.2}$$

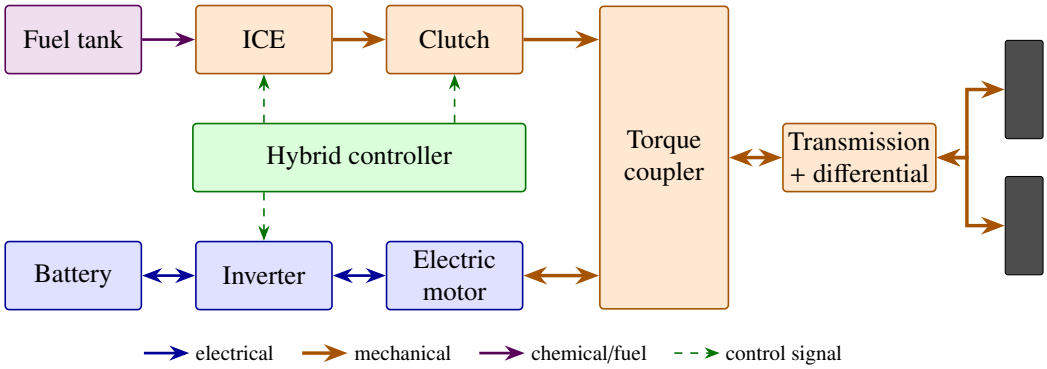


Figure 1.4: Parallel hybrid electric vehicle configuration with a torque coupler.

where T_e and T_m (N m) are the engine and motor torques and ω (rad/s) denotes speed. Equation (1.2) shows the essential limitation: engine speed is tied to wheel speed through the gear ratio, so only the engine *torque* can be chosen freely. The motor adds or subtracts torque to move the engine load towards an efficient point, but cannot free its speed. The motor can also act as a generator and charge the battery while the engine drives.

1.2.5 Series-Parallel Hybrid Configuration

The series-parallel (power-split) hybrid combines both paths. In the form shown in Fig. 1.5, a planetary gear set divides engine power into a *mechanical path* to the final drive and an *electrical path* through generator MG1, the inverters and traction motor MG2.

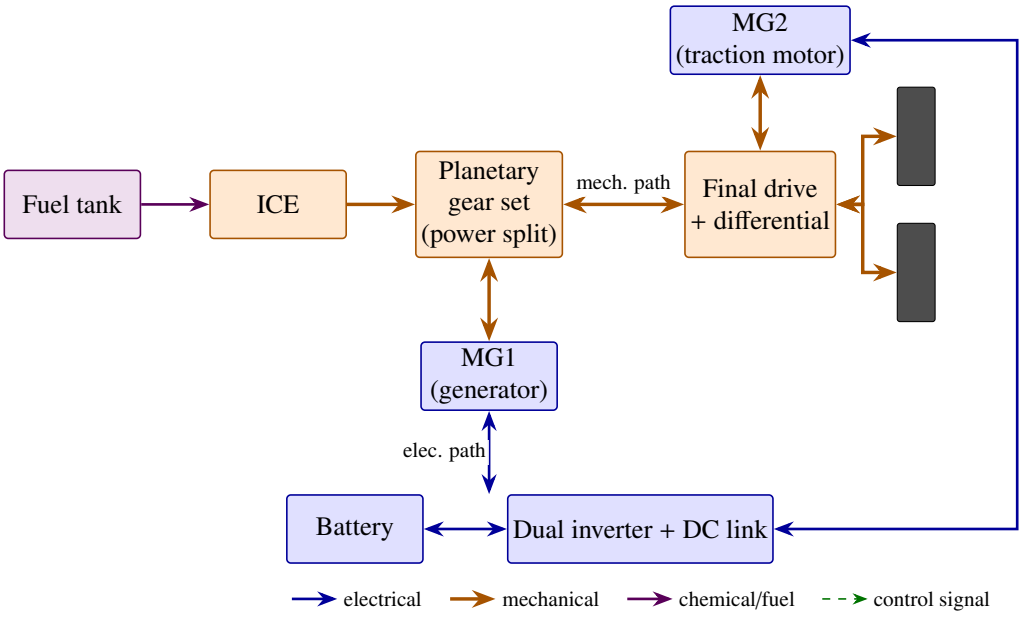


Figure 1.5: Series-parallel (power-split) hybrid configuration using a planetary gear set.

A planetary gear set with Z_s sun teeth and Z_r ring teeth imposes the Willis speed relation

$$Z_s\,\omega_s + Z_r\,\omega_r = (Z_s + Z_r)\,\omega_c, \tag{1.3}$$

with the engine on the carrier ω_c, MG1 on the sun ω_s and the output on the ring ω_r (rad/s). Equation (1.3) has two degrees of freedom. For any wheel speed, adjusting MG1 speed sets the engine speed, so the gear set works as an electrically controlled continuously variable transmission. This combines the engine-speed freedom of the series hybrid with the direct mechanical path of the parallel hybrid, at the price of greater mechanical and control complexity. Table 1.4 compares the configurations.

Table 1.4: Comparison of BEV and HEV configurations.

Feature	BEV	Series HEV	Parallel HEV	Series-parallel
Engine–wheel link	None (no engine)	None	Direct	Direct + electrical
Electric machines	1 or more	2	1	2
Engine speed freedom	–	Full	None	Full (e-CVT)
Engine torque freedom	–	Full	Full	Full
Conversion loss	Lowest	High at cruise	Low at cruise	Moderate
Best duty	All, range-limited	Urban stop-start	Highway	Mixed
Complexity	Low	Moderate	Moderate	High

The table shows why no hybrid architecture is universally best: each trades engine freedom against conversion loss and complexity, and the choice follows the intended duty cycle.

1.2.6 Fuel Cell Electric Vehicle Configuration

Figure 1.6 shows an FCEV. Hydrogen from a high-pressure tank and oxygen from the compressed air supply react in the stack to generate DC power. The stack voltage falls as load increases, so a unidirectional DC–DC converter regulates it to the DC link. A buffer battery on a bidirectional converter supplies acceleration peaks, absorbs regeneration, and allows the stack to change load slowly, which benefits stack durability. Downstream of the DC link the drivetrain is identical to that of a BEV. Effectively the FCEV is a series hybrid in which the fuel cell replaces the engine-generator set.

1.2.7 Front-Wheel, Rear-Wheel and All-Wheel Drive Configurations

The location of the drive unit determines how traction force reaches the road. Figure 1.7 shows plan views of the common layouts, where M denotes a motor and G its reduction gear.

- **Front-wheel drive, FWD** (a). A compact transverse unit combines steering and drive on the front axle. It is cost-effective for small cars and leaves cabin and boot space free, but traction falls as load transfers rearward under acceleration.
- **Rear-wheel drive, RWD** (b). Load transfer under acceleration increases normal force on the driven wheels, improving traction. Steering remains free of drive torque, and the arrangement is preferred for performance cars.

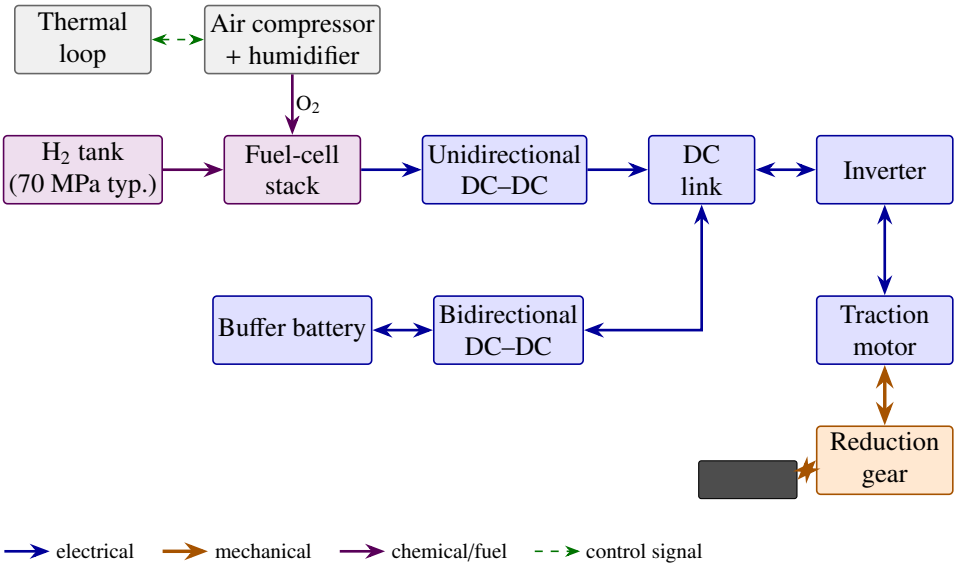


Figure 1.6: Fuel cell electric vehicle architecture with a hybridising buffer battery.

- **All-wheel drive, AWD** (c). A motor on each axle replaces the mechanical propeller shaft of a conventional 4WD. Torque can be shifted electronically between axles in milliseconds. The secondary motor can be disconnected or left unexcited at cruise to reduce losses.

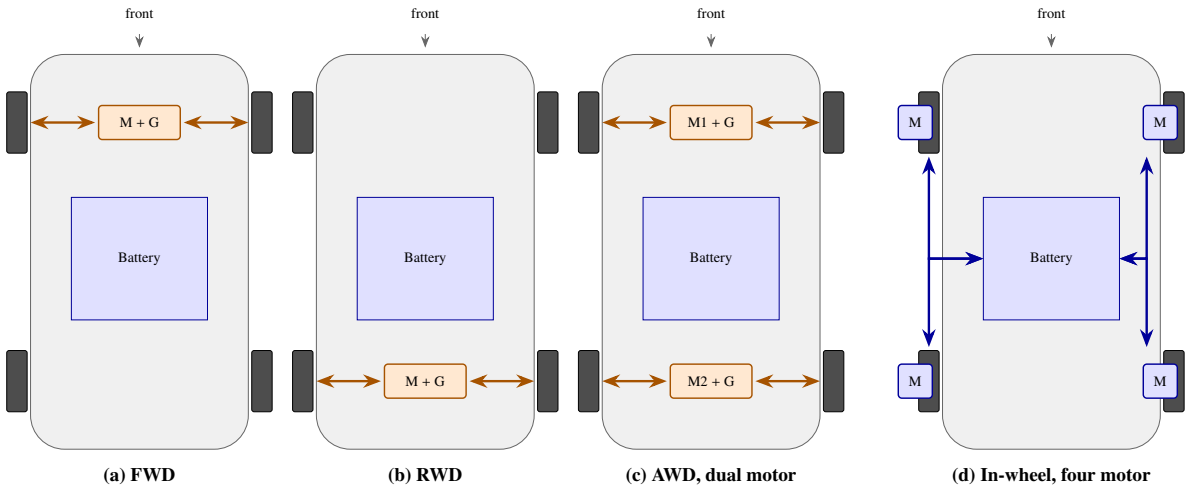


Figure 1.7: EV drive layouts: (a) front-wheel, (b) rear-wheel, (c) dual-motor all-wheel and (d) four-motor in-wheel drive.

The maximum tractive force on an axle is limited by tyre–road adhesion,

$$F_{t,\max} = \mu F_z, \tag{1.4}$$

where μ (dimensionless) is the adhesion coefficient and F_z (N) is the normal load on the driven axle. Equation (1.4) shows why driven-axle choice matters: it directly limits usable motor torque, however powerful the motor.

1.2.8 Single-Motor and Multi-Motor EV Architectures

A *single-motor* architecture uses one machine with a differential (Fig. 1.7a,b). It is the lowest-cost option and the efficiency map of a single machine is simple to manage. A *multi-motor* architecture uses two or more machines. Two axle motors (Fig. 1.7c) give AWD, redundancy and the chance to split torque so that both machines run efficiently. A motor at each driven wheel, or in-wheel motors (Fig. 1.7d), replaces the mechanical differential with an electronic one and allows torque vectoring, in which unequal left and right torque helps steer the car.

Multi-motor systems raise component count and cost and need robust fault management, because an unbalanced torque is safety-critical. In-wheel machines also increase unsprung mass, which degrades ride quality, and they must survive the harsh wheel environment.

1.2.9 Powertrain Components and Their Functions

Table 1.3 lists the functions of each component; here their interaction is considered. In motoring, the inverter imposes a voltage on the motor that produces the demanded torque; the gear reduces speed and multiplies torque; and the differential lets the wheels turn at different speeds in corners. Motor and inverter ratings are coupled by voltage and current: the back-EMF of the motor must stay below what the inverter can produce from the DC bus, and the phase current sets torque. The BMS and thermal system bound everything by supplying the allowable power of the battery and the temperature limits of the drive.

1.2.10 Mechanical and Electrical Power Transmission Paths

The two power paths are described by different pairs of variables whose products give power:

$$P_{\text{elec,DC}} = V_{dc} I_{dc}, \qquad P_{\text{elec,AC}} = \sqrt{3}\, V_{LL} I_L \cos\varphi, \qquad P_{\text{mech}} = T\omega, \tag{1.5}$$

where V_{dc} (V) and I_{dc} (A) are the DC-bus voltage and current, V_{LL} (V) and I_L (A) are the rms line voltage and current, $\cos\varphi$ is the power factor, T (N m) is torque and ω (rad/s) is rotational speed.

Electrical transmission allows flexible routing through cables and fast control, but demands high voltage to keep currents and I^2R loss small. Mechanical transmission is highly efficient and compact over short distances but constrains the layout. At each interface power is conserved apart from losses, so a reduction ratio G multiplies torque and divides speed:

$$T_w = G\,\eta_{tr}\,T_m, \qquad \omega_w = \frac{\omega_m}{G}, \tag{1.6}$$

where T_w (N m) and ω_w (rad/s) are the wheel torque and speed, and η_{tr} is the transmission efficiency. This relation underlies all the sizing in Section 1.7.

1.2.11 Regenerative Braking Configuration

During regenerative braking the motor runs as a generator, driven by the wheels. The inverter's diodes and switches actively rectify the machine's AC output and return DC power to the battery. Figure 1.8 shows the practical system. The brake-blending controller reads the pedal and splits the driver's demand into an electrical torque command T^*_{regen} and a friction command F^*_{fric}.

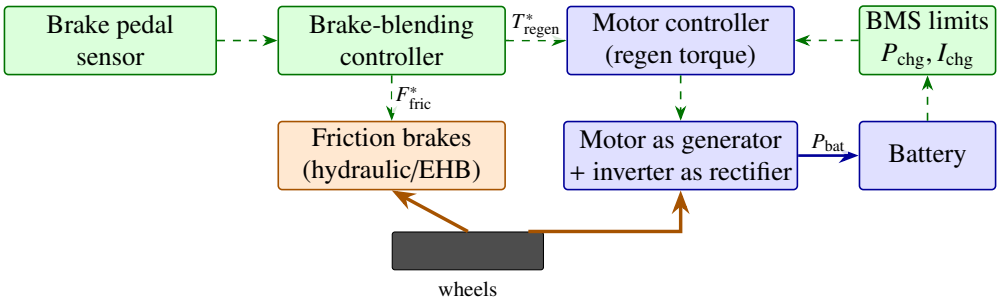


Figure 1.8: Regenerative braking system with blending of electrical and friction braking.

The friction brakes remain essential, for four reasons shown by the signals in the figure. The BMS limits charging power when the battery is cold or nearly full; the motor and inverter have power and torque limits; regeneration tapers off near standstill as the back-EMF collapses; and an emergency stop can demand more force than the drive can supply, especially on the non-driven axle. Section 1.3.14 quantifies the recoverable energy.

1.3 Performance of Electric Vehicles

The performance of a vehicle is what the driver experiences as acceleration, top speed, hill-climbing ability and range. All of these follow from one equation, Newton's second law applied along the road. This section derives that equation term by term, then uses it to find the torque, power and energy that the powertrain must deliver.

1.3.1 Vehicle Performance Characteristics

Vehicle performance is normally specified by the following measures.

- **Acceleration time**, usually from 0 to 100 km/h.
- **Maximum speed**, on a level road and sustained.
- **Gradeability**, the steepest slope the vehicle can climb at a given speed.
- **Driving range**, the distance on one charge under a stated cycle.
- **Energy consumption**, in Wh/km, which is the EV counterpart of fuel consumption.

The first three are power and torque requirements; the last two are energy requirements. This separation is important in design: motor and inverter follow from the former, and the battery energy from the latter.

1.3.2 Tractive Effort and Tractive Force

Figure 1.9 shows a vehicle climbing a slope of angle θ. The *tractive effort* F_t (N) is the forward force at the tyre–road contact produced by the powertrain. Opposing it are the rolling resistance forces F_r at the tyres ($F_{r,f}$ and $F_{r,r}$ at the front and rear axles), the aerodynamic drag F_a, the slope component of weight, and the inertial force F_i needed to accelerate the vehicle.

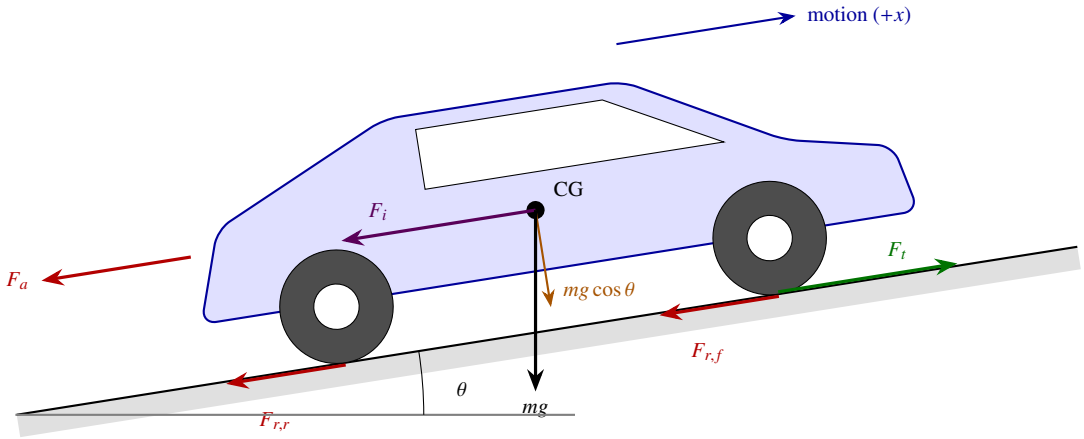


Figure 1.9: Forces acting on a vehicle moving up a slope of angle θ.

Assumptions. The model treats the vehicle as a point mass moving along the road, with no wheel slip and no lateral motion. Road grade is uniform over the short distance considered. Resistive coefficients are constant, and the air is still unless a wind speed is included.

Applying Newton's second law along the direction of motion gives

$$\delta m \frac{\mathrm{d}v}{\mathrm{d}t} = F_t - \left(F_r + F_a + F_g\right), \tag{1.7}$$

where m (kg) is the vehicle mass, v (m/s) its speed, δ (dimensionless) the rotational inertia factor defined below, and F_g (N) the grade force. Equation (1.7) is the central result: whatever tractive force exceeds the resistances accelerates the vehicle.

1.3.3 Rolling Resistance

Rolling resistance arises mainly from *hysteresis* in the tyre. The tyre deforms as it enters the contact patch and recovers as it leaves, but recovers less energy than was put in. Consequently the normal pressure is greater in the leading half of the patch, so the resultant ground reaction lies ahead of the axle and creates a moment opposing rotation. This moment is equivalent to a retarding force proportional to the normal load:

$$F_r = C_{rr}\, m\, g \cos\theta, \tag{1.8}$$

where C_{rr} (dimensionless) is the rolling resistance coefficient, $g = 9.81$ m/s^2 is gravitational acceleration, and θ (rad) is the road angle. For a passenger car on asphalt C_{rr} is about 0.008–

0.015, and low-rolling-resistance EV tyres lie at the lower end. It rises with speed, tyre under-inflation and rough surfaces. On slopes up to about 10% $\cos\theta \approx 1$, so F_r is almost constant and depends mainly on mass. It dominates at low speed.

1.3.4 Aerodynamic Drag Resistance

A moving vehicle transfers momentum to the air. Consider air of density ρ (kg/m^3) approaching at relative speed v. The dynamic pressure is $\frac{1}{2}\rho v^2$, so a body of frontal area A (m^2) experiences a force proportional to $\frac{1}{2}\rho v^2 A$. A dimensionless drag coefficient C_d, which includes pressure drag due to flow separation and skin friction, makes this exact:

$$F_a = \tfrac{1}{2}\,\rho\, C_d\, A\,(v + v_w)^2, \tag{1.9}$$

where v_w (m/s) is the headwind speed (zero in still air). At sea level $\rho \approx 1.225$ kg/m^3. Typical modern EV values are $C_d \approx 0.22$–0.32 and $A \approx 2.0$–2.6 m^2.

The quadratic dependence is decisive. Aerodynamic power grows with the *cube* of speed, $P_a = F_a v$, so doubling speed needs eight times the aerodynamic power. For this reason highway range is much shorter than urban range in EVs, the reverse of the usual ICE pattern, and why EV designers pursue low $C_d A$.

1.3.5 Gradient Resistance

On a slope the weight mg resolves into a component normal to the road and a component along it. The component along the road opposes uphill motion:

$$F_g = m\, g \sin\theta. \tag{1.10}$$

Road grade is quoted as a percentage, the rise per 100 units of horizontal run, $G_r = 100\tan\theta$. For small angles, $\sin\theta \approx \tan\theta$, so $F_g \approx mgG_r/100$. Thus a 1600 kg vehicle on a 10% slope has $F_g \approx 1560$ N. On a downhill section F_g is negative and helps the motion, and that energy can be recovered. The force is conservative: the potential energy gained on the climb can in principle be recovered on the descent.

1.3.6 Acceleration Resistance

The inertial term must account for the rotating parts as well as for translation. Let the rotor have inertia J_m (kg m^2), and the wheels a total inertia J_w (kg m^2). With wheel radius r (m), no slip gives $\omega_w = v/r$ and $\omega_m = Gv/r$. Equating total kinetic energy to that of an equivalent mass m_{eq},

$$\begin{aligned} \tfrac{1}{2} m_{\text{eq}} v^2 &= \tfrac{1}{2} m v^2 + \tfrac{1}{2} J_w \frac{v^2}{r^2} + \tfrac{1}{2} J_m \frac{G^2 v^2}{r^2}, \\ \delta \equiv \frac{m_{\text{eq}}}{m} &= 1 + \frac{J_w + G^2 J_m}{m\, r^2}. \end{aligned} \tag{1.11}$$

The acceleration resistance is then

$$F_i = \delta\, m\, a, \qquad a = \frac{\mathrm{d}v}{\mathrm{d}t}, \tag{1.12}$$

where a (m/s^2) is the acceleration. Equation (1.11) shows the effect of the gear ratio, since rotor inertia is multiplied by G^2. A value of $\delta \approx 1.03$–1.10 is typical of single-speed EVs; these notes use $\delta = 1.05$.

1.3.7 Total Road-Load Force

Rearranging Eq. (1.7) and substituting Eqs. (1.8)–(1.12) gives the total tractive force required:

$$F_t = \underbrace{C_{rr} m g \cos\theta}_{\text{rolling}} + \underbrace{\tfrac{1}{2}\rho C_d A v^2}_{\text{aerodynamic}} + \underbrace{m g \sin\theta}_{\text{grade}} + \underbrace{\delta m \frac{\mathrm{d}v}{\mathrm{d}t}}_{\text{inertial}}. \tag{1.13}$$

The first three terms form the *road load* [1, 3], which must be overcome at constant speed. The last term is the demand for acceleration. Dimensional check: each term has units of kg m/s^2, that is, newtons. Force F_t can be negative when the vehicle decelerates; this is the braking force, available in part for regeneration. Throughout this unit a reference vehicle is used with the *assumed* parameters of Table 1.5. It is a design example, not the specification of a real car.

Table 1.5: Assumed parameters of the reference BEV used in the examples.

Parameter	Symbol	Assumed value
Design mass (with occupants)	m	1600 kg
Rolling resistance coefficient	C_{rr}	0.010
Drag coefficient	C_d	0.30
Frontal area	A	2.3 m^2
Air density	ρ	1.225 kg/m^3
Wheel radius	r	0.32 m
Rotational inertia factor	δ	1.05
Transmission ratio	G	9.5
Transmission efficiency	η_{tr}	0.95

Example 1: Total tractive force for a BEV

Problem. The reference BEV of Table 1.5 climbs a 3% grade at 60 km/h and accelerates at 1.0 m/s^2. Find the total tractive force.

Given. Table 1.5; $v = 60$ km/h $= 16.67$ m/s; $a = 1.0$ m/s^2; $\tan\theta = 0.03$.

Required. F_t.

Governing equation. Eq. (1.13).

Solution. The road angle is $\theta = \tan^{-1}(0.03) = 1.718°$, with $\cos\theta = 0.99955$ and $\sin\theta = 0.02999$.

$$
\begin{aligned}
F_r &= 0.010 \times 1600 \times 9.81 \times 0.99955 = 156.9 \text{ N},\\
F_a &= 0.5 \times 1.225 \times 0.30 \times 2.3 \times 16.67^2 = 117.4 \text{ N},\\
F_g &= 1600 \times 9.81 \times 0.02999 = 470.7 \text{ N},\\
F_i &= 1.05 \times 1600 \times 1.0 = 1680.0 \text{ N},\\
F_t &= 156.9 + 117.4 + 470.7 + 1680.0 = 2425 \text{ N}.
\end{aligned}
$$

Result. $F_t \approx 2.42$ kN.

Interpretation. Acceleration accounts for 69% of the force and grade for 19%. At moderate speed, mass is by far the dominant factor, which is why city consumption is sensitive to vehicle mass.

Example 2: Rolling resistance and aerodynamic drag

Problem. For the same vehicle on a level road, compare rolling and aerodynamic resistance at a steady 100 km/h, and find the speed at which they are equal.

Given. $v = 100$ km/h $= 27.78$ m/s; $\theta = 0$; $a = 0$.

Required. F_r, F_a, the road-load power at the wheels, and the crossover speed.

Governing equations. Eqs. (1.8) and (1.9); $P = Fv$.

Solution.

$$
\begin{aligned}
F_r &= 0.010 \times 1600 \times 9.81 = 157.0 \text{ N},\\
F_a &= 0.5 \times 1.225 \times 0.30 \times 2.3 \times 27.78^2 = 326.1 \text{ N},\\
P_w &= (157.0 + 326.1) \times 27.78 = 13.42 \text{ kW}.
\end{aligned}
$$

Equating the two forces gives the crossover speed:

$$
v_c = \sqrt{\frac{2C_{rr}mg}{\rho C_d A}} = \sqrt{\frac{2 \times 157.0}{1.225 \times 0.30 \times 2.3}} = 19.27 \text{ m/s} = 69.4 \text{ km/h}.
$$

Result. $F_r = 157$ N, $F_a = 326$ N, $P_w = 13.4$ kW, $v_c \approx 69$ km/h.

Interpretation. Above roughly 70 km/h, aerodynamics dominate. At 100 km/h it accounts for two-thirds of the road load, so C_dA matters more than mass for highway range.

1.3.8 Torque and Power Requirements

The wheel torque follows directly from the tractive force acting at radius r, and Eq. (1.6) takes it back to the motor shaft:

$$T_w = F_t\, r, \qquad T_m = \frac{F_t\, r}{G\,\eta_{tr}} \quad (F_t \geq 0), \tag{1.14}$$

$$\omega_m = \frac{G\,v}{r}, \qquad n_m = \frac{60}{2\pi}\omega_m, \tag{1.15}$$

where n_m (rpm) is the motor speed. The mechanical power at the wheels and at the motor shaft are

$$P_w = F_t\, v, \qquad P_m = T_m\omega_m = \frac{F_t v}{\eta_{tr}}. \tag{1.16}$$

Each kilowatt at the wheels thus needs $1/\eta_{tr}$ kilowatts from the motor. When $F_t < 0$, power flows from the wheels to the motor, so the efficiency multiplies instead: $T_m = F_t\, r\, \eta_{tr}/G$.

A traction motor delivers constant torque up to its *base speed*, then constant power as field weakening reduces flux. Figure 1.10 shows the corresponding tractive effort plotted with road-load curves. This shape closely matches traction needs, with high force for starting and climbing and declining force at speed. The vertical gap between the tractive curve and road load is the surplus available for acceleration, and the crossing with the road-load line sets the speed limit on that grade.

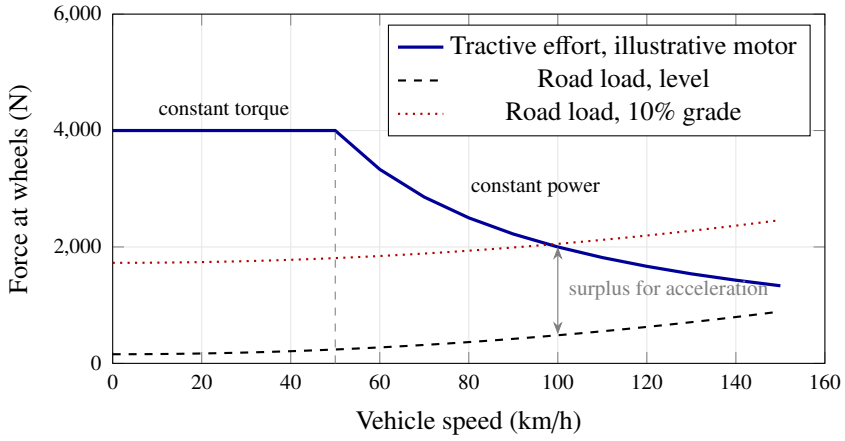


Figure 1.10: Illustrative tractive effort of a single-speed EV and road-load curves for the reference vehicle.

Example 3: Motor torque and power requirement

Problem. Find the motor torque, speed and power for the operating point of Example 1.

Given. $F_t = 2425$ N, $v = 16.67$ m/s, $r = 0.32$ m, $G = 9.5$, $\eta_{tr} = 0.95$.

Required. T_w, T_m, n_m, P_w, P_m.

Governing equations. Eqs. (1.14)–(1.16).

Solution.

$$
\begin{aligned}
T_w &= 2425 \times 0.32 = 776.0 \text{ N m}, \\
T_m &= \frac{776.0}{9.5 \times 0.95} = 86.0 \text{ N m}, \\
\omega_m &= \frac{9.5 \times 16.67}{0.32} = 494.8 \text{ rad/s} \quad \Rightarrow \quad n_m = 4725 \text{ rpm}, \\
P_w &= 2425 \times 16.67 = 40.4 \text{ kW}, \\
P_m &= 86.0 \times 494.8 = 42.5 \text{ kW}.
\end{aligned}
$$

Result. The motor delivers 86 N m at 4725 rpm, or 42.5 kW.

Interpretation. The reduction gear turns a modest motor torque into 776 N m at the wheels. High-speed, low-torque motors are smaller for the same power, which explains the trend to high-speed machines with large G.

1.3.9 Vehicle Acceleration and Maximum Speed

Solving Eq. (1.7) for acceleration gives the available acceleration at speed v:

$$
a(v) = \frac{F_t(v) - F_r - F_a(v) - F_g}{\delta m}. \tag{1.17}
$$

Since $a = \mathrm{d}v/\mathrm{d}t$, the time to reach speed v_f from rest is found by separating variables:

$$
t_{\text{acc}} = \int_0^{v_f} \frac{\delta m \, \mathrm{d}v}{F_t(v) - F_r - F_a(v) - F_g}. \tag{1.18}
$$

The integral is usually evaluated numerically, because $F_t(v)$ is piecewise, with constant torque below base speed v_b and constant power above it. If resistive forces are neglected, the integral has a closed form. With tractive power limit P at the wheels, $F_t = P/v_b$ for $v < v_b$ and $F_t = P/v$ above. Integrating both regions gives

$$
t_{\text{acc}} \approx \frac{\delta m \, (v_f^2 + v_b^2)}{2P} \quad \Longrightarrow \quad P \approx \frac{\delta m \, (v_f^2 + v_b^2)}{2 \, t_{\text{acc}}}. \tag{1.19}
$$

This classic estimate [1] shows that lowering base speed (widening the constant-power range) cuts the required peak power, because the motor reaches full power earlier in the run.

The *maximum speed* is reached when tractive force equals road load, or when the motor reaches its speed limit, whichever occurs first:

$$
v_{\text{max,motor}} = \frac{\omega_{m,\max} \, r}{G}, \qquad F_t(v_{\max}) = F_r + F_a(v_{\max}). \tag{1.20}
$$

1.3.10 Gradeability

Gradeability is the steepest grade that can be held at constant speed. Setting $a = 0$ in Eq. (1.13) and neglecting $\cos\theta \approx 1$ gives

$$\sin\theta_{\max} = \frac{F_t(v) - F_r - F_a(v)}{mg}, \qquad G_{r,\max} = 100\tan\theta_{\max}\ (\%). \tag{1.21}$$

At low speed F_a is negligible and F_t is at its peak, so gradeability depends on peak torque and gear ratio. In practice tyre adhesion, Eq. (1.4), and thermal limits of the motor may set a lower limit.

1.3.11 Driving Range

Range is the usable energy divided by the energy consumed per unit distance. If the battery has nominal energy E_{bat} (Wh), usable fraction DoD, and the vehicle consumes e_c (Wh/km) measured at the battery, then

$$R = \frac{E_{\text{bat}}\,\text{DoD}}{e_c}. \tag{1.22}$$

For steady driving at speed v (m/s), with drivetrain efficiency η_d and auxiliary load P_{aux} (W), the consumption in Wh/km is

$$e_c = \frac{1}{3.6}\left(\frac{F_{\text{road}}}{\eta_d} + \frac{P_{\text{aux}}}{\eta_b\, v}\right), \tag{1.23}$$

because 1 N of force over 1 km corresponds to 1000 J = 1/3.6 Wh, and where F_{road} (N) is the road load and η_b the battery efficiency. The second term shows that auxiliaries cost more per kilometre at low speed. Range therefore peaks at a moderate speed: too slow and auxiliaries dominate, too fast and drag dominates.

1.3.12 Energy Consumption

The energy demanded at the wheels over a trip is the time integral of wheel power:

$$E_w = \int_0^T F_t(t)\, v(t)\, \mathrm{d}t. \tag{1.24}$$

Substituting Eq. (1.13) separates the energy into its physical sinks. Over a trip that starts and ends at rest and at the same altitude, the inertial and grade terms integrate to zero, since $\int \delta m v\, \mathrm{d}v = 0$ and $\int mg\sin\theta\, \mathrm{d}s = 0$. So, *with ideal energy recovery*, only rolling and aerodynamic losses would remain:

$$E_{w,\text{ideal}} = \int_0^T (F_r + F_a)\, v\, \mathrm{d}t. \tag{1.25}$$

Equation (1.25) expresses the fundamental limit of regenerative braking. In practice recovery is imperfect, so each stop costs energy. Section 1.6 evaluates these integrals for real speed profiles.

Example 4: Battery capacity and driving range

Problem. The reference BEV cruises at a steady 80 km/h on a level road with a 500 W auxiliary load. It has a 60 kWh, 400 V pack with 90% usable energy. Estimate consumption, range, battery capacity in ampere-hours, and the discharge C-rate. Take a battery-to-wheel efficiency of $\eta_d = 0.8045$ and $\eta_b = 0.97$.

Given. $v = 22.22$ m/s; $E_{\text{bat}} = 60$ kWh; $V = 400$ V; DoD $= 0.90$.

Required. e_c, R, Q, and the C-rate.

Governing equations. Eqs. (1.22), (1.23), and (1.38) of Section 1.5.

Solution.

$$
\begin{aligned}
F_{\text{road}} &= 157.0 + 0.5 \times 1.225 \times 0.30 \times 2.3 \times 22.22^2 = 157.0 + 208.7 = 365.7 \text{ N}, \\
P_w &= 365.7 \times 22.22 = 8.13 \text{ kW}, \\
P_b &= \frac{8126}{0.8045} + \frac{500}{0.97} = 10\,101 + 515 = 10.62 \text{ kW}, \\
e_c &= \frac{P_b}{v} = \frac{10\,616 \text{ W}}{80 \text{ km/h}} = 132.7 \text{ Wh/km}, \\
R &= \frac{60\,000 \times 0.90}{132.7} = 407 \text{ km}, \\
Q &= \frac{60\,000}{400} = 150 \text{ A h}, \quad I = \frac{10\,616}{400} = 26.5 \text{ A}, \quad \text{C-rate} = \frac{26.5}{150} = 0.18\text{C}.
\end{aligned}
$$

Result. $e_c \approx 133$ Wh/km; $R \approx 407$ km; $Q = 150$ A h; about C/5.6.

Interpretation. Steady cruise loads a traction pack very lightly. It is peaks, not cruising, that stress the battery, as Section 1.7 shows. The figure is an idealised steady result; a real cycle consumes more because of stops, as found later.

1.3.13 Drivetrain Efficiency

Drivetrain efficiency is the product of the efficiencies of the components in series. From battery terminals to wheels,

$$\eta_d = \eta_b \, \eta_{\text{inv}} \, \eta_{\text{mot}} \, \eta_{tr}. \tag{1.26}$$

With assumed average values $\eta_b = 0.97$, $\eta_{\text{inv}} = 0.97$, $\eta_{\text{mot}} = 0.90$ and $\eta_{tr} = 0.95$, the result is $\eta_d = 0.8045$. These are *operating-point dependent*. A motor can reach above 95% near its best point but much less at very low load, and inverter losses vary with current and switching frequency. A fixed value is adequate for early estimates, but detailed design uses efficiency maps of $\eta(T, \omega)$.

1.3.14 Regenerative Braking Performance

The energy available for recovery in a stop from speed v_0 is the kinetic energy less what road load dissipates during the stop. The kinetic energy is

$$E_k = \tfrac{1}{2}\,\delta\, m\, v_0^2. \tag{1.27}$$

The fraction reaching the battery depends on how much of the braking the electric drive actually provides, represented by k_r, and on the reverse-path efficiency:

$$E_{\text{regen}} = k_r\, \eta_{tr}\, \eta_{\text{mot}}\, \eta_{\text{inv}}\, \eta_b\, E_{\text{brake}}, \tag{1.28}$$

where E_{brake} (Wh) is the energy absorbed by the braking system at the wheels and k_r (dimensionless, $0 \le k_r \le 1$) is the share captured electrically. A common performance index is the *regeneration effectiveness*, the ratio of battery energy recovered to braking energy available. For example, from 60 km/h the reference vehicle has $E_k = 64.8$ Wh. With $k_r = 0.65$ and $\eta = 0.8045$ for the reverse path, a stop returns about 52% of its braking energy to the battery. Regeneration is limited by battery charge acceptance, by low-speed fade, and by the need for stable braking, as described in Fig. 1.8.

1.4 Energy Flow Analysis

Energy flow analysis traces where energy goes as it travels between the battery and the road. It identifies the largest losses, shows which component improvements offer the largest gain, and connects the vehicle to the wider energy system. It rests on one principle, the first law of thermodynamics: at every component, input power equals output power plus loss.

$$P_{\text{in}} = P_{\text{out}} + P_{\text{loss}}, \qquad \eta = \frac{P_{\text{out}}}{P_{\text{in}}}. \tag{1.29}$$

1.4.1 Energy Flow in Battery Electric Vehicles

Energy in a BEV flows along a single chain, but in both directions. In *motoring*, energy flows from battery to wheels and each component adds a loss. In *regenerative braking*, the chain reverses and the same components are traversed in the opposite order, so losses are incurred twice for any energy that is stored and reused. In addition, the auxiliary branch draws power continuously, regardless of whether the vehicle is moving.

1.4.2 Battery-to-Wheel Energy Conversion

Figure 1.11 is the energy flow diagram during motoring. The numbers are percentages of the energy released by the cells, computed with the assumed average efficiencies of Eq. (1.26). Starting from 100 units of chemical energy, resistive heating in the cells leaves 97.0 units at the terminals. The inverter then passes 94.1 units, the motor delivers 84.7, and the gear and

differential deliver 80.4 to the wheels. The largest single loss is in the motor; Eq. (1.26) implies a chain efficiency of 80.4%.

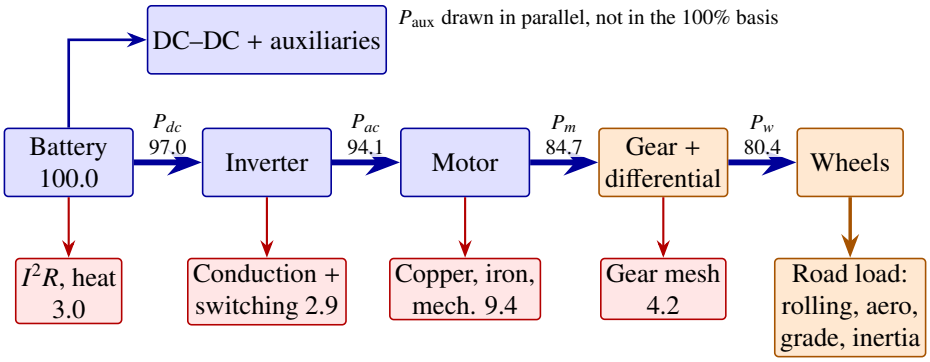


Figure 1.11: Energy flow during motoring, in percent of cell energy (assumed average efficiencies).

1.4.3 Power Flow Through DC-DC Converters, Inverters and Electric Motors

Applying Eq. (1.29) to the chain gives the power at each stage for a demanded wheel power P_w:

$$P_m = \frac{P_w}{\eta_{tr}}, \qquad P_{ac} = \frac{P_m}{\eta_{\mathrm{mot}}}, \qquad P_{dc} = \frac{P_{ac}}{\eta_{\mathrm{inv}}}, \tag{1.30}$$

$$P_{\mathrm{bat}} = P_{dc} + \frac{P_{\mathrm{aux}}}{\eta_{\mathrm{DCDC}}}. \tag{1.31}$$

Here P_m, P_{ac} and P_{dc} (W) are the motor shaft power, the AC power into the motor and the DC power into the inverter; P_{bat} (W) is the battery terminal power and P_{aux} (W) the low-voltage auxiliary load. Some vehicles place a boost DC–DC converter between the battery and the inverter so that the motor sees a higher bus voltage than the pack. In that case, a further factor η_{boost} enters Eq. (1.30), which is a cost paid for reduced motor current and better high-speed capability. The HVAC system normally connects directly to the HV bus because of its power level.

On the DC side current follows from voltage, $I_{dc} = P_{\mathrm{bat}}/V_{dc}$; on the AC side, it follows from the third relation in Eq. (1.5). Hence a 100 kW demand at 400 V means 250 A, whereas at 800 V it means 125 A, which reduces cable mass and I^2R loss by a factor of four for the same conductor. This is the main motivation for 800 V platforms.

1.4.4 Mechanical Transmission and Wheel Power

The mechanical chain contains bearing and gear-mesh losses, normally about 3–5% in a single-speed unit [2], and a churning loss in the lubricant, which rises with speed. At the wheel the power $P_w = F_t v$ does not itself represent loss. It supplies the road load, which then dissipates it as rolling hysteresis and turbulent air, or stores it as kinetic or potential energy. Some of the stored portion is what regenerative braking can reclaim.

1.4.5 Energy Losses in Different EV Components

Each component loses energy through distinct physical mechanisms, summarised below.

- **Battery.** Ohmic and polarisation losses, modelled as $P_{\text{loss}} \approx I^2 R_{\text{int}}$, where R_{int} (Ω) is the pack internal resistance. Losses rise with the square of current and at low temperature.
- **Inverter.** Conduction loss, proportional to current through the device on-state drop, and switching loss, proportional to switching frequency. Section 1.8.17 gives expressions.
- **Motor.** Copper loss $3I^2R_s$ in the stator, iron loss from hysteresis and eddy currents (rising with frequency), and mechanical friction and windage.
- **Transmission.** Gear-mesh friction, bearing friction and oil churning.
- **Auxiliaries.** Heating and air-conditioning can draw several kilowatts, which is often comparable to the traction demand in slow traffic.

1.4.6 Regenerative Braking Energy Flow

Figure 1.12 shows the reverse flow. The vehicle's kinetic and potential energy reach the wheels, and any share taken by friction brakes is rejected as heat. Of the part entering the electrical path, normalised to 100, the gear passes 95.0, the machine generates 85.5, the inverter delivers 82.9 and the cell stores 80.4. The percentages use the same assumed average efficiencies as motoring.

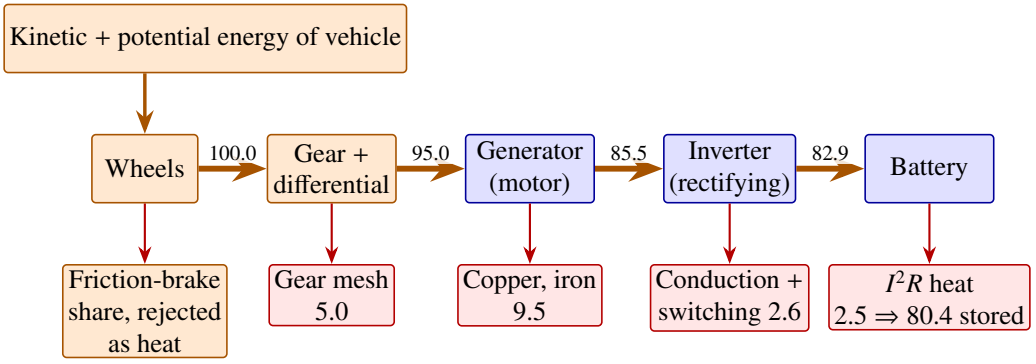


Figure 1.12: Energy flow during regenerative braking, in percent of energy entering the driveline at the wheels.

Energy that is stored must later return through the motoring chain. So a round trip from wheels to battery and back to wheels has efficiency

$$\eta_{\text{rt}} = \eta_{\text{regen}}\,\eta_{\text{motoring}} = 0.804 \times 0.804 = 0.647. \tag{1.32}$$

Equation (1.32) is a useful design insight: even with perfect blending only about 65% of braking energy at the wheels is returned to the wheels later, so anticipating a stop and coasting is better than braking and re-accelerating.

1.4.7 Well-to-Wheel and Tank-to-Wheel Energy Concepts

Vehicle comparisons need clear boundaries. *Tank-to-wheel* (TTW) efficiency covers on-board conversion; for a BEV it is taken from the charging port or battery to the wheel. *Well-to-tank* (WTT) efficiency covers production and delivery of the energy carrier. *Well-to-wheel* (WTW) efficiency is their product, covers the entire chain, and is the appropriate basis for comparing different energy carriers. Argonne National Laboratory's GREET model is the standard life-cycle tool for this analysis [29].

$$\eta_{\mathrm{WTW}} = \eta_{\mathrm{WTT}}\,\eta_{\mathrm{TTW}}. \tag{1.33}$$

Figure 1.13 applies Eq. (1.33) with assumed, illustrative efficiencies. For the BEV, the assumed 40% generation efficiency is what a thermal-dominated grid might provide; a renewable-rich grid would be better. On this basis the BEV delivers 25.9 units to the wheels against 19.0 for the ICE vehicle, so the BEV's advantage narrows markedly from its TTW lead. Electricity generation therefore matters strongly: it is why a BEV's emissions benefit depends on the regional grid.

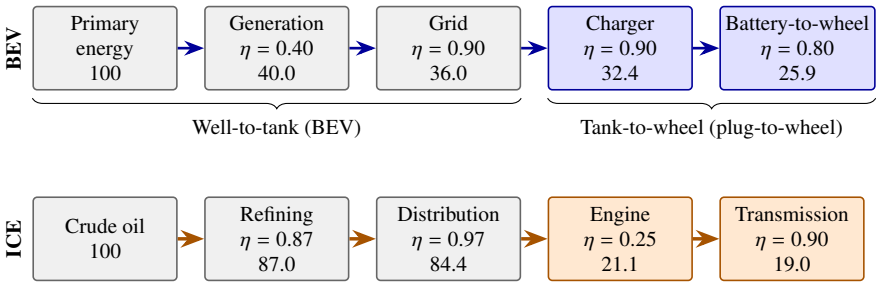


Figure 1.13: Illustrative well-to-wheel chains for a BEV and an ICE vehicle (assumed efficiencies; values in percent).

1.4.8 Energy Efficiency Analysis

Efficiency can be stated per instant or over a trip. Trip efficiency uses energy integrals rather than a ratio of powers:

$$\bar{\eta} = \frac{\int P_{\mathrm{out}}\,\mathrm{d}t}{\int P_{\mathrm{in}}\,\mathrm{d}t}. \tag{1.34}$$

This matters since components are often used far from their rated point. A motor spending much of an urban cycle at 10% torque can have a cycle-average efficiency several points below its peak. The value of this analysis lies in sensitivity. Since Eq. (1.26) is a product, a one-point improvement in any efficiency in the chain improves the total proportionally, but improvements upstream in the battery also reduce the heat that the thermal system must remove. Reductions in demand, such as lower mass, lower C_dA and lower C_{rr}, multiply through the entire chain, so each watt saved at the wheel saves about 1.24 W at the battery ($1/0.8045$).

1.4.9 Energy Flow Diagrams Under Motoring and Regenerative Braking Conditions

Figures 1.11 and 1.12 should be read together. They show the same components in reverse order, with arrow directions marking the sign of power. Sign conventions follow: battery power positive on discharge, motor torque positive when it accelerates the vehicle. Table 1.6 relates sign of torque and speed to operating mode. A traction drive normally operates in all four quadrants, and the inverter is what enables it.

Table 1.6: Four-quadrant operation of an EV drive.

Quadrant	Speed	Torque	Power flow	Mode
I	+	+	Battery → wheels	Forward motoring
II	+	−	Wheels → battery	Forward regeneration
III	−	−	Battery → wheels	Reverse motoring
IV	−	+	Wheels → battery	Reverse regeneration

1.5 Introduction to Traction Batteries

1.5.1 Introduction to Traction Batteries

A traction battery is a rechargeable electrochemical energy store that supplies the propulsion power of an EV. It is the largest, heaviest and costliest component in a BEV, and it sets the range, much of the performance, and the service life. A *cell* is the basic electrochemical unit. Cells are grouped into *modules*, and modules into a *pack* together with the structure, cooling, sensing, safety devices and control electronics. Virtually all modern traction batteries are lithium-ion, which is why this section focuses on that chemistry. Figure 1.14 shows the hierarchy from cell to pack.

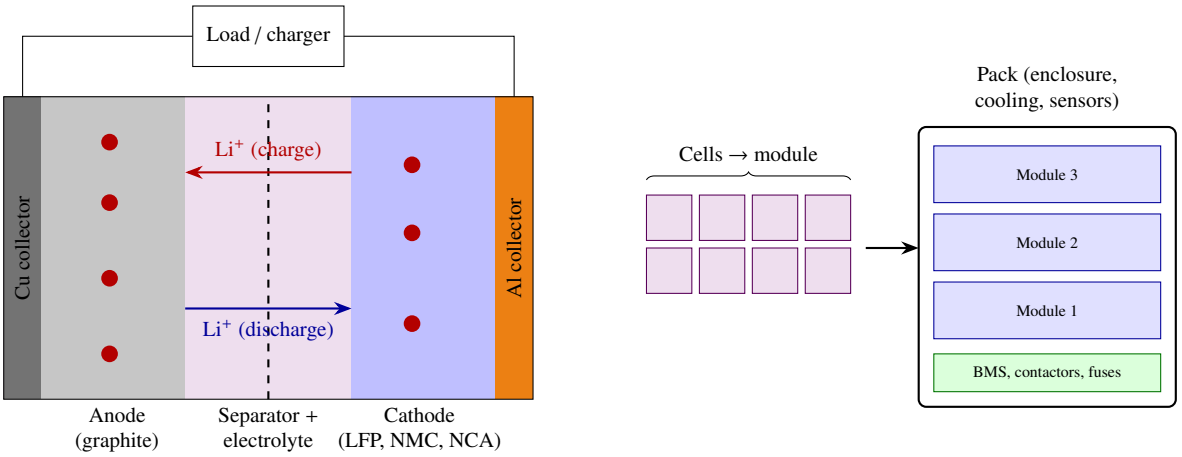


Figure 1.14: Lithium-ion cell operation and the hierarchy from cell to module to pack.

1.5.2 Requirements of EV Traction Batteries

Compared with consumer cells, a traction pack must meet a demanding and conflicting set of requirements.

- High specific energy and energy density, for range within limited mass and volume.
- High specific power, for acceleration, regeneration and fast charging.
- Long cycle and calendar life, typically eight years or more under warranty.
- Safety under abuse, including crash, overcharge, short circuit and thermal runaway; standards such as UN R100 and India's AIS-156 apply [22, 23].
- Operation from below freezing to high summer temperatures.
- Low cost per kilowatt-hour, recyclability and a secure supply of materials.

1.5.3 Battery Characteristics and Specifications

Table 1.7 defines the principal performance parameters. Their definitions follow common usage in the literature [2, 7], and test methods are standardised, for example in IEC 62660 and ISO 12405 [24, 25].

Table 1.7: Battery performance parameters.

Parameter	Unit	Meaning	Pack-level typical
Nominal voltage	V	Mean voltage during discharge	300–800
Capacity	A h	Charge deliverable under stated conditions	50–200
Energy	kWh	Capacity times voltage	30–100
Specific energy	Wh/kg	Energy per unit mass	100–200
Energy density	Wh/L	Energy per unit volume	150–350
Specific power	W/kg	Power per unit mass	300–1000
C-rate	h^{-1}	Current normalised by capacity	0.3–3
Internal resistance	mΩ	Determines voltage drop and heat	50–150
Round-trip efficiency	%	Energy out divided by energy in	90–97

1.5.4 Energy Density and Power Density

Energy density and power density are volumetric measures:

$$\rho_E = \frac{E}{\mathcal{V}}, \qquad \rho_P = \frac{P}{\mathcal{V}}, \tag{1.35}$$

where ρ_E (Wh/L) and ρ_P (W/L) are the densities, E (Wh) the energy, P (W) the power and $\mathcal{V}$ (L) the volume. Energy density determines how much range fits under the floor; power density determines how small the battery can be for a given peak power.

1.5.5 Specific Energy and Specific Power

The gravimetric equivalents are

$$e_s = \frac{E}{M}, \qquad p_s = \frac{P}{M}, \tag{1.36}$$

where e_s (Wh/kg) and p_s (W/kg) are specific energy and specific power, and M (kg) is the cell or pack mass. It is essential to state the basis: pack-level values are about 60–75% of cell-level values because of the structure, cooling and electronics, and cell-to-pack designs aim to narrow this gap. A chart of e_s against p_s is a Ragone plot. Within any cell family, these two trade off: thin electrodes support high power, whereas thick electrodes hold more energy.

1.5.6 Nominal Voltage and Capacity

Cell *nominal voltage* depends on chemistry, about 3.6–3.7 V for NMC and NCA and 3.2 V for LFP. *Capacity* Q (A h) is the charge deliverable between the voltage limits at a specified current and temperature. Stored energy is thus the integral of voltage over charge and is approximately

$$E = \int_0^{Q} v(q)\,\mathrm{d}q \approx V_{\mathrm{nom}}\,Q. \tag{1.37}$$

1.5.7 C-Rate

C-rate expresses current relative to capacity, so that packs of different sizes can be compared:

$$C_{\mathrm{rate}} = \frac{I}{Q}, \qquad t_{\mathrm{nom}} \approx \frac{1}{C_{\mathrm{rate}}}, \tag{1.38}$$

where I (A) is the current, Q (A h) the rated capacity and t_{nom} (h) the nominal time to discharge. At 1C a 150 A h pack delivers 150 A for one hour, and at 2C, 300 A for roughly half an hour. Delivered capacity falls at high C-rate because of internal voltage drop, which is why capacity is quoted at a stated rate.

1.5.8 State of Charge (SoC)

State of charge is the remaining charge as a fraction of the present full capacity, the battery's "fuel gauge". *Coulomb counting* integrates the measured current:

$$\mathrm{SoC}(t) = \mathrm{SoC}(t_0) - \frac{1}{3600\,Q}\int_{t_0}^{t} \eta_c\, i(\tau)\,\mathrm{d}\tau, \tag{1.39}$$

where i (A) is positive on discharge, Q is in A h, τ (s) is the time variable, and η_c is the coulombic efficiency, which is close to one for lithium-ion. Coulomb counting drifts with sensor offset and needs a correct initial value, so practical BMSs combine it with a voltage-based model, often in a Kalman filter [7]. Estimation is harder for LFP because its voltage curve is very flat over most of the SoC range, as Fig. 1.15 shows.

1.5.9 State of Health (SoH)

State of health measures ageing relative to a new cell. It may be defined on capacity or on resistance:

$$\mathrm{SoH}_Q = \frac{Q_{\mathrm{now}}}{Q_{\mathrm{new}}} \times 100\%, \qquad \mathrm{SoH}_R = \frac{R_{\mathrm{EoL}} - R_{\mathrm{now}}}{R_{\mathrm{EoL}} - R_{\mathrm{new}}} \times 100\%, \tag{1.40}$$

where R_{EoL} (Ω) is the resistance defining end of life. A traction battery is conventionally at end of life when capacity falls to about 80% of the initial value. Capacity fade cuts range; resistance growth cuts power and raises heat.

1.5.10 Depth of Discharge (DoD)

Depth of discharge is the fraction removed from the fully charged state, DoD = 1 − SoC for a single discharge from full. Designers also use a *usable window*, for example 5–95% SoC, so that the BMS never exposes the extremes, which cause the most stress. Shallower average cycling greatly improves life, and so a pack sized with a margin may outlast the vehicle.

1.5.11 Cycle Life and Calendar Life

Cycle life is the number of charge–discharge cycles before end of life under specified conditions; it depends on DoD, C-rate and temperature. *Calendar life* is the ageing that occurs with time alone, mainly by growth of the solid-electrolyte interphase on the anode. Calendar ageing accelerates with temperature and with high state of charge, so storing a vehicle at high SoC in heat is harmful. A useful sizing estimate links cycle life and lifetime distance:

$$D_{\mathrm{life}} \approx N_{\mathrm{cyc}}\, R\,, \tag{1.41}$$

where N_{cyc} is the number of full-equivalent cycles and R (km) the usable range per cycle. For example, 1500 cycles at 300 km corresponds to 450 000 km.

1.5.12 Charging and Discharging Characteristics

Figure 1.15 shows illustrative curves. On the left, cell voltage falls with discharged capacity. NMC falls smoothly from about 4.2 V, whereas LFP stays nearly flat near 3.3 V before dropping steeply at each end. Higher current displaces the curve down, because of the voltage drop across internal resistance:

$$v_t = \mathrm{OCV}(\mathrm{SoC}) - i\,R_{\mathrm{int}}, \tag{1.42}$$

where v_t (V) is the terminal voltage and OCV is the open-circuit voltage. The drop also ends discharge earlier at high rates, reducing usable capacity.

The right of the figure shows the standard charging protocol. In the *constant-current* (CC) phase, the voltage rises to the upper limit; in the *constant-voltage* (CV) phase, the charger holds that limit and current tapers as the cell approaches full. The CV taper is why fast charging is usually quoted as 10–80%, and why the last 20% takes disproportionately long. Charge rate is further restricted at low temperature, when lithium plating on the anode would damage the cell.

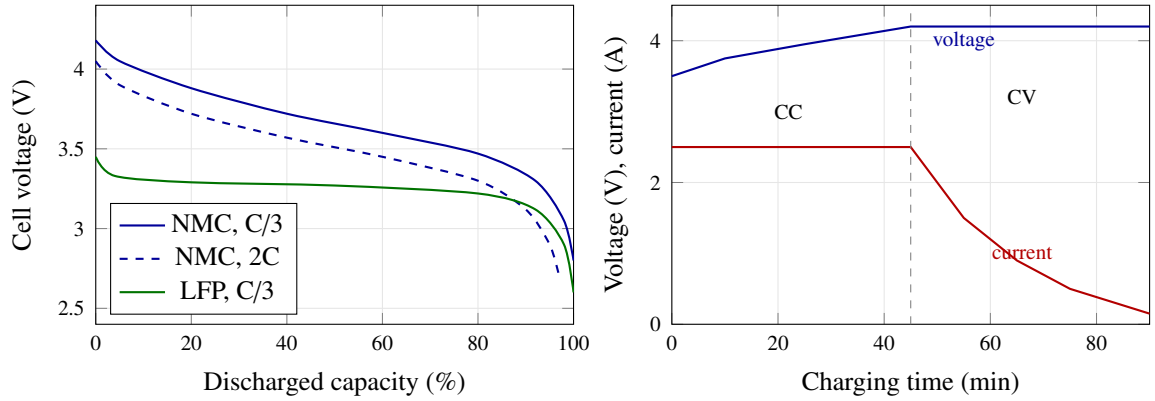


Figure 1.15: Illustrative discharge curves (left) and constant-current, constant-voltage charging (right). Curves are qualitative and not measured data.

1.5.13 Battery Pack Configuration

Pack design selects cell format and arrangement. Three formats are used: *cylindrical* cells, such as the 2170 and 4680 sizes; *prismatic* cells in a rigid can; and *pouch* cells in a laminated film. Cylindrical cells are robust and standardised but have a low packing factor; prismatic cells pack well and are often used in cell-to-pack designs; pouch cells are light but need external support. Whatever the format, the pack must also provide contactors, fuses, a pre-charge circuit, current and voltage sensing, and structural crash protection, as listed in Fig. 1.14.

1.5.14 Series and Parallel Cell Connections

Series connection adds voltage and parallel connection adds capacity, as shown in Fig. 1.16. For N_s series elements and N_p parallel cells,

$$V_{\text{pack}} = N_s V_{\text{cell}}, \quad Q_{\text{pack}} = N_p Q_{\text{cell}}, \quad E_{\text{pack}} = N_s N_p V_{\text{cell}} Q_{\text{cell}}, \quad R_{\text{pack}} = \frac{N_s}{N_p} R_{\text{cell}}. \tag{1.43}$$

Energy depends only on the total number of cells, so the same cells can be arranged for different voltages. The internal resistance relation shows that higher voltage (more series cells) raises pack resistance, but the current for a given power falls in proportion, so the I^2R loss in the pack stays the same for a given cell count. In the figure, parallel cells are tied first and then connected in series (N_pPN_sS), which shares current among neighbours and needs one voltage sense per level. Series strings tied in parallel give string isolation but demand more monitoring.

Because the pack is only as strong as its weakest series element, cell-to-cell variation in capacity and resistance must be managed. This is one of the functions of the BMS.

Illustration: a pack from cells. To build about 400 V and 60 kWh from 3.65 V, 50 A h cells of 2.0 mΩ resistance, take $N_s = 400/3.65 = 109.6$, rounded up to 110, so that $V_{\text{pack}} = 401.5$ V. The required capacity is $60\,000/401.5 = 149.4$ A h, so $N_p = 3$, giving $Q_{\text{pack}} = 150$ A h and $E_{\text{pack}} = 60.2$ kWh with 330 cells. Resistance is $R_{\text{pack}} = (110/3) \times 2.0 = 73.3$ mΩ. At 300 A

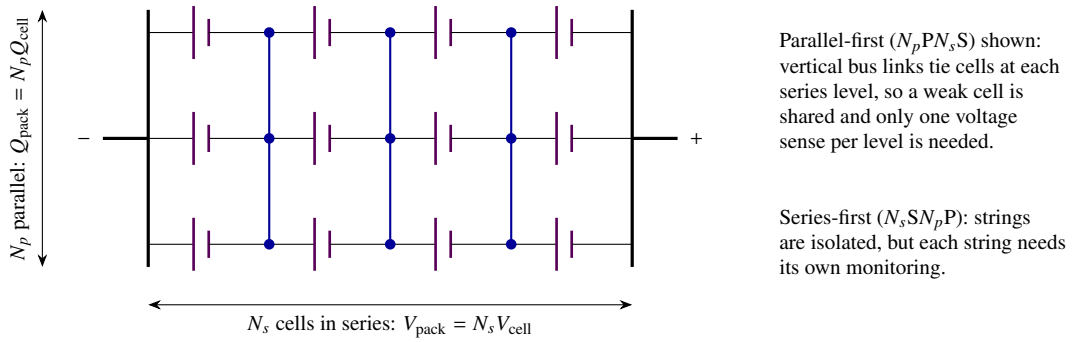


Figure 1.16: Series-parallel connection of cells in a battery pack (3P4S shown).

the voltage sag is 22 V and the heat generated is 6.6 kW, so thermal management is necessary at peak power even though this is a small fraction of the output.

1.5.15 Lithium-Ion Battery Chemistry

A lithium-ion cell (Fig. 1.14, left) stores energy by moving lithium ions between two host materials, a mechanism called *intercalation*. The anode is commonly graphite, sometimes with silicon added, on a copper foil. The cathode is a lithium metal oxide or phosphate on an aluminium foil. A porous separator keeps the electrodes apart while the organic liquid electrolyte conducts ions. During discharge, lithium ions leave the anode and intercalate into the cathode, while electrons travel through the external circuit and deliver power. Charging reverses the process.

For a layered-oxide cathode, the reactions on discharge are

$$\text{Anode:} \quad \mathrm{Li}_x\mathrm{C}_6 \;\rightarrow\; \mathrm{C}_6 + x\,\mathrm{Li}^+ + x\,e^-, \tag{1.44}$$

$$\text{Cathode:} \quad \mathrm{Li}_{1-x}\mathrm{MO}_2 + x\,\mathrm{Li}^+ + x\,e^- \;\rightarrow\; \mathrm{LiMO}_2, \tag{1.45}$$

where M denotes transition metals such as Ni, Mn, Co or Al. The cell voltage is the difference between the electrochemical potentials of the two electrodes, so the cathode chemistry defines most of the voltage and capacity [14].

1.5.16 LFP, NMC, NCA and LTO Battery Technologies

The dominant technologies differ mainly in their cathode, except for LTO, which replaces the graphite anode. Table 1.8 summarises them. Nominal voltages and specific energies are representative figures for commercial cells from a published review [15]; the cycle-life and other entries are qualitative, and actual values vary by manufacturer and test condition.

- **LFP** avoids nickel and cobalt, has a very stable olivine structure and long life, and is now widely used thanks to cell-to-pack designs that offset its lower energy. It also has a flat voltage curve, which complicates SoC estimation, and its low-temperature charging is poor.

Table 1.8: Comparison of traction lithium-ion chemistries (commercial cell level) [15].

Property	LFP	NMC	NCA	LTO
Cathode / anode	$LiFePO_4$ / graphite	$LiNiMnCoO_2$ / graphite	$LiNiCoAlO_2$ / graphite	NMC or LMO / $Li_4Ti_5O_{12}$
Nominal voltage (V)	3.2	3.6–3.7	3.65	2.2–2.4
Specific energy (Wh/kg)	100–140	150–220	200–260	50–80
Cycle life	High	High	Medium	Very high
Thermal stability	High	Moderate	Low	Very high
Fast-charge ability	Good	Good	Moderate	Excellent
Cost	Low	Moderate	Moderate-high	High
Typical use	Mass-market cars, buses, 2W/3W	Passenger cars	Long-range cars	Buses, high-cycle fleets

- **NMC** balances energy, power and life. High-nickel grades such as NMC811 offer more energy at the expense of thermal stability.
- **NCA** has very high specific energy, but it demands stricter thermal management.
- **LTO** uses a titanate anode that has no SEI growth or lithium plating, which gives outstanding life, safety and fast charge, but its low cell voltage gives poor specific energy.

The main observation from the table is that no chemistry dominates: higher specific energy generally accompanies lower thermal stability. Selection is a trade-off for each application, as discussed below.

1.5.17 Battery Management System (BMS)

The BMS is the embedded system that keeps every cell inside its safe operating area and tells the vehicle what the battery can do. Figure 1.17 shows a distributed architecture. Slave boards attached to each module measure cell voltages and temperatures and perform cell balancing. They communicate over an isolated link with a master controller. The master also reads pack current, runs estimation and fault logic, and controls the contactors and pre-charge. It reports allowable charge and discharge power to the VCU and charger over CAN, and requests cooling or heating from the thermal loop.

The main functions are the following.

1. **Measurement:** cell voltage, pack current, temperature, insulation resistance.
2. **Estimation:** SoC, SoH and state of power.
3. **Protection:** over-voltage, under-voltage, over-current, over-temperature and isolation faults, by derating or by opening contactors.

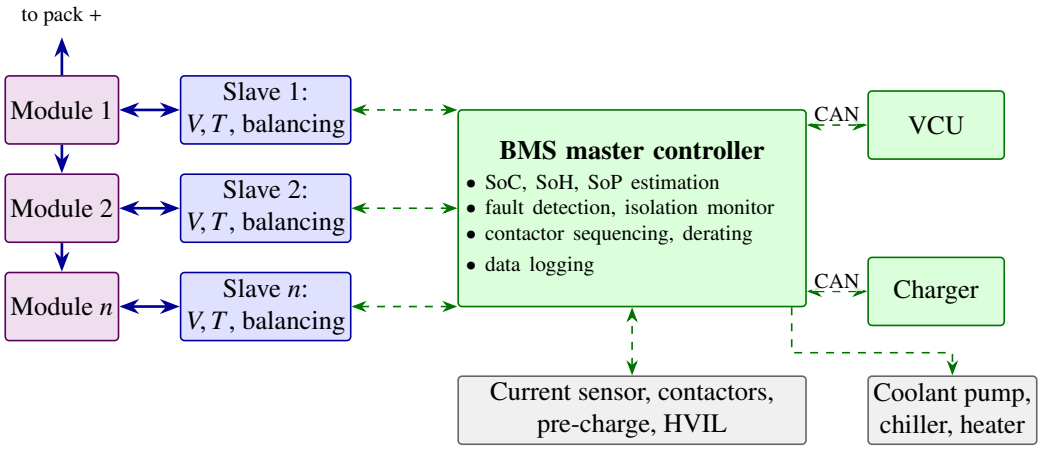


Figure 1.17: Distributed battery management system architecture.

4. **Balancing:** passive balancing bleeds excess charge from high cells through resistors; active balancing transfers charge between cells.

5. **Communication and logging:** data exchange with the vehicle and charger, and diagnostic storage. In India, AIS-156 specifies safety requirements for EV battery systems [22].

1.5.18 Battery Thermal Management

Lithium-ion cells perform best within a narrow temperature window, commonly about 15–35 °C. Too cold, and resistance rises and fast charging risks lithium plating. Too hot, and ageing accelerates and, past the onset temperature, thermal runaway may start. The heat generated in a pack is approximately

$$\dot{Q}_{\text{gen}} \approx I^2 R_{\text{int}}, \tag{1.46}$$

and the lumped thermal balance is

$$Mc_p \frac{\mathrm{d}T}{\mathrm{d}t} = \dot{Q}_{\text{gen}} - hA_s\,(T - T_c)\,, \tag{1.47}$$

where $\dot{Q}_{\text{gen}}$ (W) is the heat generation, M (kg) the pack mass, c_p (J/kg K) the specific heat, T (K) the pack temperature, h (W/m^2 K) the heat transfer coefficient, A_s (m^2) the cooled surface area and T_c (K) the coolant temperature. Equation (1.47) shows two design levers, thermal mass and the hA_s product. Air cooling has a low h and is confined to smaller or low-power packs; liquid cooling through cold plates or channels is the standard for cars; immersion cooling offers still better performance and is emerging for fast-charging applications. Heaters or heat pumps warm the pack in cold climates.

1.5.19 Battery Selection Considerations for EV Applications

Battery selection follows the duty. The principal considerations are as follows.

- **Energy requirement:** range target and vehicle consumption (Eq. (1.22)).

- **Power requirement:** the peak discharge and charge power expressed as C-rate.
- **Voltage:** set by the motor and inverter design and by fast-charging needs.
- **Life:** daily distance, climate, and warranty target.
- **Safety and regulation:** chemistry stability and homologation requirements.
- **Packaging and mass:** available floor area and height.
- **Cost and supply:** cell price and material availability.

A simple rule is that the *power-to-energy ratio* (in h^{-1}, numerically a C-rate) identifies the class of cell needed. A long-range car with 100 kW and 60 kWh needs only 1.7C, so a high-energy cell suffices. A small hybrid with 30 kW and 1.5 kWh needs 20C, so it needs a high-power cell. In Indian conditions, high ambient temperature and cost sensitivity favour LFP for electric three-wheelers, buses and mass-market cars, whereas NMC remains common where range per kilogram matters most.

1.6 Drive Cycle and Its Importance in System Analysis

1.6.1 Definition of a Drive Cycle

A *drive cycle* is a standardised or measured record of vehicle speed against time, $v(t)$, that represents a type of driving. It defines the accelerations, cruises, decelerations and stops that a vehicle must follow. A *regulatory* cycle is performed on a chassis dynamometer under controlled conditions so that vehicles can be compared on an equal basis. Sometimes the term also includes a road gradient profile, $\theta(t)$, but the standard chassis cycles assume a level road.

1.6.2 Importance of Drive Cycles in EV Design

Section 1.3 gave instantaneous equations; a drive cycle provides the time history that turns them into design figures. It is used in the following ways.

- **Regulation and labelling:** energy consumption and range are homologated on a stated cycle.
- **Component sizing:** peak and root-mean-square torque and power fix motor and inverter ratings, and cycle energy fixes the battery.
- **Efficiency evaluation:** operating-point distribution over a motor map predicts average efficiency.
- **Thermal and durability design:** current profiles drive thermal models and battery ageing tests.

- **Control development:** energy management and regeneration strategies are compared over identical conditions, for example in NREL's ADVISOR or MathWorks Powertrain Blockset [17, 30].

No regulatory cycle exactly represents real driving; the choice of cycle changes the declared range, so the same car has different ranges under MIDC, WLTC and EPA tests. Quoted range should always be read together with its cycle.

1.6.3 Speed-Time Representation of Drive Cycles

A drive cycle is a sampled time series, usually at 1 Hz, plotted as speed against time. Figure 1.18 shows the *illustrative* cycle used in this unit, constructed for teaching and clearly not an official test cycle. Its upper trace has low-speed urban trips, then a suburban section and a short high-speed portion. The lower trace is wheel power for the reference vehicle, computed with Eq. (1.13). Comparing the two is instructive. Power spikes occur in acceleration phases, and are highest when accelerating at high speed; cruising needs only a modest plateau that grows with speed; braking gives negative power that is available for regeneration.

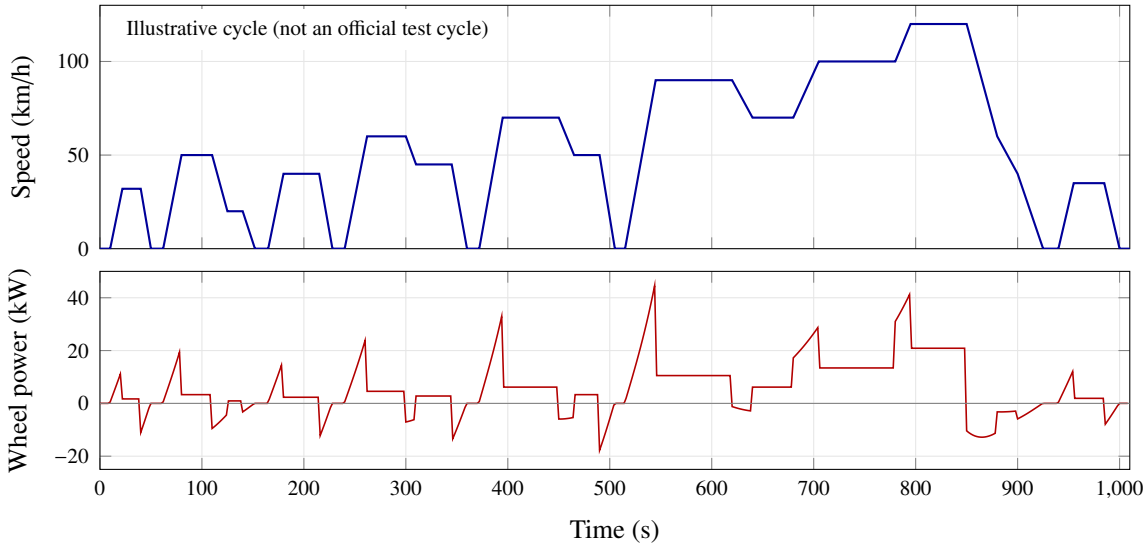


Figure 1.18: Illustrative speed-time drive-cycle profile (top) and the resulting wheel power of the reference BEV (bottom). This is teaching data, not an official test cycle.

Cycles may be classed as *modal*, built from straight-line accelerations and constant speeds such as NEDC, or *transient*, derived from real driving with continuously changing speed such as WLTC and UDDS. Modal cycles are easy to analyse by hand but understate real transient demand.

1.6.4 Standard Driving Cycles

The following subsections describe the cycles named in the syllabus. Table 1.9 collects their key statistics, compiled from the regulatory or published sources [16, 18–21]. Rounded values are quoted; refer to the standards for official time series.

1.6.4.1 Indian Driving Cycle (IDC)

The IDC is a short modal cycle for two- and three-wheelers in India, based on observed low-speed traffic. It comprises six repeats of a 108 s cycle, giving 648 s and about 3.9 km at an average of about 21.9 km/h, with a peak speed of about 42 km/h [20]. It is widely quoted for electric scooters, so its low speed gives ranges well above what a rider would obtain on faster roads.

1.6.4.2 Modified Indian Driving Cycle (MIDC)

The MIDC is the Indian cycle for passenger cars, with a structure like NEDC: four repetitions of a 195 s urban section followed by a 400 s extra-urban section. The total duration is 1180 s and the distance about 10.7 km. The extra-urban maximum speed is 90 km/h [21], a limit that differs from NEDC. Because it is gentle, MIDC ranges are generally optimistic compared with WLTC ranges for the same car.

1.6.4.3 New European Driving Cycle (NEDC)

NEDC is a modal cycle, used in Europe until replaced by WLTP from 2017, and comprises four repeats of the urban ECE-15 section and one extra-urban EUDC section. It lasts 1180 s and covers about 11 km, with a 120 km/h peak and a mean speed of about 34 km/h. Figure 1.19 gives a simplified illustration of this urban-plus-extra-urban structure. Its smooth, constant-acceleration segments are not representative of real driving, which was the principal reason for replacing it.

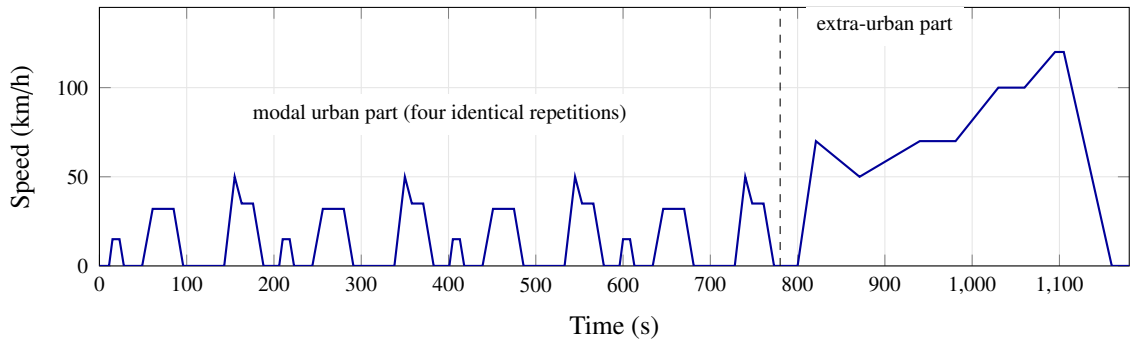


Figure 1.19: Simplified illustration of the modal urban plus extra-urban structure of NEDC-type cycles (illustrative; not official data; MIDC limits the extra-urban peak to 90 km/h).

1.6.4.4 Worldwide Harmonized Light Vehicles Test Cycle (WLTC)

WLTC was developed under UNECE from real-world driving data worldwide, and is part of UN GTR No. 15 [16, 18]. Three vehicle classes are defined by power-to-mass ratio; nearly all cars are Class 3 (above 34 W/kg). The Class 3 cycle lasts 1800 s, covers about 23.3 km, reaches 131.3 km/h, and averages 46.5 km/h. It has four phases of increasing speed: Low (589 s), Medium (433 s), High (455 s) and Extra-High (323 s).

1.6.4.5 EPA FTP-75 Drive Cycle

The Federal Test Procedure FTP-75 is the US city cycle for light-duty vehicles. It comprises a cold-start transient phase, a stabilised phase, a 10-minute soak and a hot-start repeat of the first phase, with total driven time 1874 s, about 17.8 km, a peak of 91.2 km/h, and an average of 34.1 km/h [19].

1.6.4.6 Urban Dynamometer Driving Schedule (UDDS)

UDDS, also known as FTP-72 or LA-4, was derived from a Los Angeles commuter route and forms the first two phases of FTP-75. It lasts 1369 s, covers about 12 km, averages 31.5 km/h and peaks at 91.2 km/h [19]. For BEVs, it is a part of the multi-cycle range procedure in SAE J1634 [26].

1.6.4.7 Highway Fuel Economy Test (HWFET)

HWFET represents US rural and highway driving below 60 mph. It lasts 765 s, covers 16.45 km, averages 77.7 km/h and peaks at 96.4 km/h, with no stops apart from the ends [19]. It exposes the aerodynamic term of Eq. (1.13).

1.6.5 Comparison of Different Drive Cycles

Table 1.9: Standard drive-cycle comparison (rounded values).

Cycle	Duration (s)	Distance (km)	Mean (km/h)	Max (km/h)	Type	Application
IDC	648	3.9	21.9	42	Modal	Indian 2W/3W
MIDC	1180	10.7	32.6	90	Modal	Indian cars
NEDC	1180	11.0	33.6	120	Modal	Former EU
WLTC (Cl. 3)	1800	23.3	46.5	131.3	Transient	Global light duty
FTP-75	1874	17.8	34.1	91.2	Transient	US city
UDDS	1369	12.0	31.5	91.2	Transient	US urban
HWFET	765	16.45	77.7	96.4	Transient	US highway

Three observations stand out from Table 1.9. Mean speed varies by a factor of over three, between IDC and HWFET, so aerodynamic energy changes radically. The MIDC mean speed is well below that of WLTC, so MIDC range overstates highway capability. And the modal cycles share a similar duration but have much lower transient acceleration content than WLTC.

1.6.6 Calculation of Acceleration from a Drive Cycle

Let the cycle be sampled as $v_k = v(t_k)$, $k = 0, 1, \ldots, N$, with step $\Delta t = t_{k+1} - t_k$ (s). Acceleration is obtained from the backward or forward difference,

$$a_k = \frac{v_{k+1} - v_k}{\Delta t}. \tag{1.48}$$

Sampled data contains noise, so light filtering is commonly applied before differentiating, since numerical differentiation amplifies noise.

1.6.7 Calculation of Distance Travelled

Distance is the integral of speed. The trapezoidal rule gives

$$s = \int_0^T v(t)\,\mathrm{d}t \approx \sum_{k=0}^{N-1} \frac{v_k + v_{k+1}}{2}\,\Delta t. \tag{1.49}$$

1.6.8 Calculation of Instantaneous Traction Force

At each step, apply Eq. (1.13) with the mid-interval speed $\bar{v}_k = (v_k + v_{k+1})/2$:

$$F_{t,k} = C_{rr}\, mg \cos\theta_k\, \sigma_k + \tfrac{1}{2}\rho C_d A\, \bar{v}_k^2 + mg \sin\theta_k + \delta m\, a_k, \tag{1.50}$$

where $\sigma_k = 1$ when the vehicle is moving and 0 when stationary, so that rolling resistance is not counted at standstill.

1.6.9 Calculation of Instantaneous Power Demand

The instantaneous wheel power and the battery power are

$$P_{w,k} = F_{t,k}\,\bar{v}_k, \qquad P_{b,k} = \begin{cases} \dfrac{P_{w,k}}{\eta_d} + \dfrac{P_{\text{aux}}}{\eta_b}, & P_{w,k} \geq 0, \\[2ex] k_r\,\eta_r\,P_{w,k} + \dfrac{P_{\text{aux}}}{\eta_b}, & P_{w,k} < 0, \end{cases} \tag{1.51}$$

where η_d is the motoring efficiency, η_r the regeneration-path efficiency, and k_r the fraction of braking energy captured. Equation (1.51) treats efficiency as constant; a detailed model would look up $\eta_{\text{mot}}(T_k, \omega_k)$.

1.6.10 Calculation of Energy Consumption

Summing battery power over the cycle gives

$$E_b = \frac{1}{3600} \sum_{k=0}^{N-1} P_{b,k}\,\Delta t \quad \text{(Wh)}, \qquad e_c = \frac{E_b}{s/1000} \quad \text{(Wh/km)}. \tag{1.52}$$

Algorithm 1 gives the data processing sequence and Algorithm 2 the calculation of the force at each sample.

Example 5: Energy consumption using a drive cycle

Problem. The reference BEV completes a single urban trip: 0–60 km/h in 15 s at constant acceleration, 30 s at 60 km/h, 60–0 km/h in 15 s, then 10 s of idling. Find the distance, peak force and power, and battery energy per kilometre, for a level road. Assume $\eta_d = \eta_r = 0.8045$, $k_r = 0.65$, $P_{\text{aux}} = 500$ W, $\eta_b = 0.97$.

Algorithm 1 Drive-cycle data processing

Require: Time and speed vectors t_k, v_k (km/h), $k = 0, \ldots, N$
Ensure: Acceleration a_k, distance s, cycle statistics

Convert speed: $v_k \leftarrow v_k / 3.6$ (m/s)
Initialise $s \leftarrow 0$, $t_{\text{idle}} \leftarrow 0$, $a_{\max} \leftarrow 0$
for $k = 0$ **to** $N - 1$ **do**
    $\Delta t \leftarrow t_{k+1} - t_k$
    $a_k \leftarrow (v_{k+1} - v_k)/\Delta t$
    $\bar{v}_k \leftarrow (v_k + v_{k+1})/2$
    $s \leftarrow s + \bar{v}_k \, \Delta t$
    if $\bar{v}_k = 0$ **then**
        $t_{\text{idle}} \leftarrow t_{\text{idle}} + \Delta t$
    end if
    $a_{\max} \leftarrow \max(a_{\max}, a_k)$
end for
$v_{\text{mean}} \leftarrow s/(t_N - t_0)$
return a_k, s, v_{mean}, $\max_k v_k$, $a_{\max}$, t_{idle}

Algorithm 2 Calculation of vehicle tractive force

Require: $\bar{v}_k$, a_k, θ_k; parameters $m, g, C_{rr}, \rho, C_d, A, \delta$
Ensure: $F_{t,k}$, $P_{w,k}$ for all samples

for $k = 0$ **to** $N - 1$ **do**
    if $\bar{v}_k > 0$ **then**
        $F_r \leftarrow C_{rr} m g \cos\theta_k$
    else
        $F_r \leftarrow 0$
    end if
    $F_a \leftarrow \frac{1}{2} \rho C_d A \bar{v}_k^2$
    $F_g \leftarrow m g \sin\theta_k$
    $F_i \leftarrow \delta m \, a_k$
    $F_{t,k} \leftarrow F_r + F_a + F_g + F_i$
    $P_{w,k} \leftarrow F_{t,k} \, \bar{v}_k$
end for
return $F_{t,k}$, $P_{w,k}$

Given. Table 1.5; $v_c = 16.67$ m/s; $T = 70$ s.

Required. s, $F_{\max}$, $P_{\max}$, E_b, e_c.

Governing equations. Eqs. (1.48)–(1.52).

Solution. *Acceleration:* $a = \pm 16.67/15 = \pm 1.111$ m/s^2.

Distance: segments are $\frac{1}{2}(16.67)(15) = 125$ m, $16.67 \times 30 = 500$ m and 125 m, so $s = 750$ m.

Force: let $k_a = \frac{1}{2}\rho C_d A = 0.4226$ kg/m. At the end of acceleration,

$$F = 157.0 + 0.4226(16.67)^2 + 1.05(1600)(1.111) = 157.0 + 117.4 + 1866.7 = 2141 \text{ N},$$

which gives the peak wheel power $P_{\max} = 2141 \times 16.67 = 35.7$ kW.

Wheel energy: rolling energy is $F_r s$; aerodynamic energy in the ramps is $k_a v_c^3 t/4$, and in the cruise $k_a v_c^3 t$; kinetic energy is $\frac{1}{2}\delta m v_c^2$.

Segment	Rolling (Wh)	Aero (Wh)	Kinetic (Wh)	Net wheel (Wh)
Acceleration	5.45	2.04	+64.81	+72.30
Cruise	21.80	16.30	0	+38.10
Deceleration	5.45	2.04	−64.81	−57.33

Battery energy:

$$E_{\text{trac}} = \frac{72.30 + 38.10}{0.8045} = 137.2 \text{ Wh},$$
$$E_{\text{regen}} = 0.65 \times 0.8045 \times 57.33 = 30.0 \text{ Wh},$$
$$E_{\text{aux}} = \frac{500 \times 70}{0.97 \times 3600} = 10.0 \text{ Wh},$$
$$E_b = 137.2 - 30.0 + 10.0 = 117.3 \text{ Wh}, \qquad e_c = \frac{117.3}{0.75} = 156 \text{ Wh/km}.$$

Result. $s = 750$ m, $F_{\max} = 2.14$ kN, $P_{\max} = 35.7$ kW, and $e_c \approx 156$ Wh/km.

Interpretation. The steady 60 km/h value with auxiliaries is about 103 Wh/km, so the stop-start pattern adds about 50%. Without regeneration, E_b would be 147.2 Wh, or 196 Wh/km. Regeneration thus saves 20% on this stop-start trip.

1.6.11 Battery Energy Requirement Estimation

Cycle consumption transforms into a battery specification. For a target range R (km), with usable window DoD and a margin k_{age} for ageing and reserve, the nominal battery energy is

$$E_{\text{bat}} = \frac{R\, e_c}{\text{DoD}\,(1 - k_{\text{age}})}. \tag{1.53}$$

Applying Algorithms 1 and 2 at 1 s steps to the illustrative cycle of Fig. 1.18 gives the statistics of Table 1.10. The positive wheel energy of 1.95 kWh comprises 0.66 kWh of rolling and 0.93 kWh of aerodynamic loss over the cycle, and the remaining 0.36 kWh is the energy dissipated in braking, which is the net amount available for regeneration. Regeneration reduces consumption from 169.8 to 157.3 Wh/km, a 7.3% saving, smaller than in Example 5 because this cycle includes fewer stops per kilometre and faster cruising.

Table 1.10: Results for the illustrative drive cycle and reference BEV ($k_r = 0.65$, $\eta_d = \eta_r = 0.8045$, $P_{\text{aux}} = 500$ W).

Quantity	Value	Quantity	Value
Duration	1010 s	Positive wheel energy	1.951 kWh
Distance	15.14 km	Negative wheel energy	−0.360 kWh
Mean speed	54.0 km/h	Battery traction energy	2.425 kWh
Maximum speed	120 km/h	Regenerated to battery	0.188 kWh
Maximum acceleration	0.85 m/s^2	Auxiliary energy	0.145 kWh
Maximum deceleration	0.93 m/s^2	Net battery energy	2.381 kWh
Peak wheel power	44.6 kW	Consumption	157.3 Wh/km
Peak braking power (wheel)	17.8 kW	Consumption, no regen	169.8 Wh/km

1.6.12 Drive-Cycle-Based EV Component Sizing

The cycle results feed component ratings in two ways. Peak values set short-term limits, while root-mean-square values indicate heating. Copper loss is proportional to i^2, and torque to current in the constant-flux region, so the thermal load on a motor over a cycle follows the rms torque:

$$T_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T T_m^2(t)\,\mathrm{d}t}. \tag{1.54}$$

Over the illustrative cycle, the motor torque of the reference vehicle peaks at 64 N m, with $T_{\text{rms}} = 25.8$ N m at speeds up to about 9450 rpm. A standard cycle alone rarely sets the peak rating, however, as customers also expect acceleration and hill-climbing performance. Section 1.7 treats this explicitly.

1.7 Component Selection for a Given Drive Cycle for a BEV

1.7.1 Introduction to Drive-Cycle-Based Component Sizing

Component sizing translates a vehicle's mission into ratings for the motor, gear, inverter, battery and auxiliaries. Under-sizing fails the customer; over-sizing adds mass, cost and, since heavier vehicles need more energy, a further spiral of size. Figure 1.20 gives the systematic method used in this section. Targets and vehicle parameters enter at the top. The drive cycle gives the energy and thermal demand, and performance targets give the peak demand. The components are then chosen in order, and the design is checked against constraints. The feedback arrow

is important: battery mass depends on battery energy, which depends on vehicle mass, so the process is iterative.

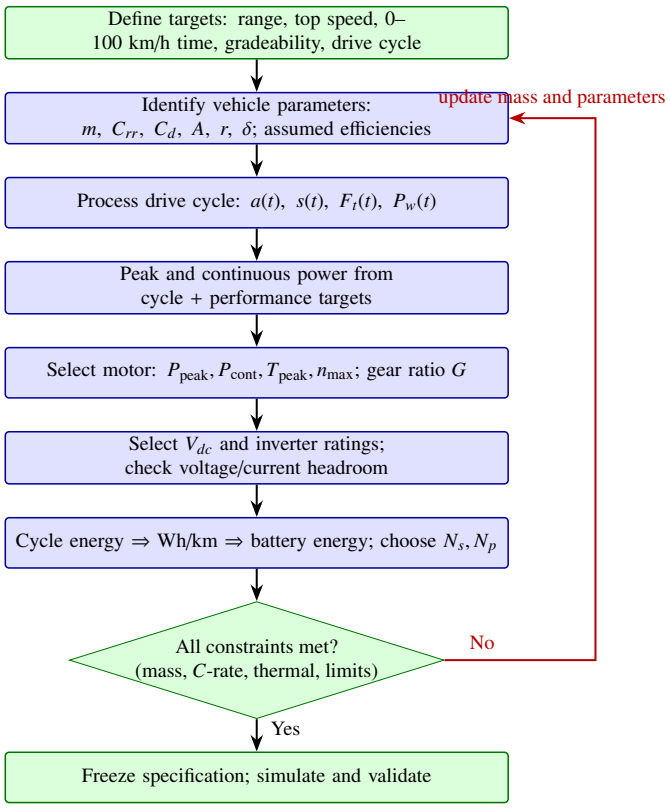


Figure 1.20: Drive-cycle-based component sizing flowchart for a BEV.

1.7.2 Vehicle Parameter Identification

The input data fall into three groups: the *mission* (range, top speed, acceleration, gradeability, chosen cycle, climate); the *vehicle* parameters of Eq. (1.13); and the *efficiency assumptions*, later refined with component maps. At concept stage, values come from benchmarks and from coast-down data on similar vehicles.

1.7.3 Selection of Vehicle Mass

Design mass adds payload to curb weight. Curb mass can be written as glider mass plus powertrain mass plus battery mass:

$$m = m_{\text{glider}} + m_{\text{drive}} + \frac{E_{\text{bat}}}{e_{s,\text{pack}}} + m_{\text{payload}}. \tag{1.55}$$

Because E_{bat} grows with consumption, and consumption with m, Eq. (1.55) couples mass and energy. The *design* mass for performance is usually the gross vehicle mass, while range is often rated at curb mass plus a defined load.

1.7.4 Selection of Rolling Resistance Coefficient

C_{rr} is chosen from tyre data, typically 0.008–0.012 for low-rolling-resistance passenger tyres on good roads. A conservative value should be used for poor road surfaces or for commercial tyres. Tyre grip, comfort and noise compete with low C_{rr}, so it is not freely reducible.

1.7.5 Selection of Aerodynamic Drag Coefficient

C_d depends on body shape and is established by CFD and wind-tunnel testing. Benchmarks are about 0.22–0.30 for sedans and 0.30–0.38 for SUVs. Closed front grilles, flat underbodies, aero wheels and active shutters are the standard measures in BEVs, because the BEV needs far less cooling air than an ICE car.

1.7.6 Selection of Frontal Area

Frontal area A can be estimated from height H and track width or body width W (m) by

$$A \approx 0.85\, H\, W. \tag{1.56}$$

Since the battery usually lies under the floor, it tends to raise roof height and A, an effect that designers seek to limit.

1.7.7 Selection of Wheel Radius

Rolling radius r comes from the tyre designation. A 215/55 R17 tyre, for instance, has a free radius $0.5(17 \times 0.0254) + 0.215 \times 0.55 = 0.334$ m, and a dynamic rolling radius of about 3–5% less under load. Larger wheels raise rolling diameter, so they reduce motor speed but increase the motor torque required at the same force; the gear ratio compensates.

1.7.8 Calculation of Road-Load Forces

Equation (1.13) is evaluated for two purposes: over the drive cycle, using Algorithm 2, and at the performance corners listed below.

1.7.9 Calculation of Peak Traction Power

Peak power is the largest of the demands from three sources: the cycle peak; the acceleration target, by Eq. (1.19) or by numerical solution of Eq. (1.18); and the gradeability target,

$$P_{\text{peak}} = \max\left\{\frac{P_{w,\text{cycle}}^{\max}}{\eta_{tr}},\; P_{\text{acc}},\; P_{\text{grade}}\right\}. \tag{1.57}$$

1.7.10 Calculation of Continuous Traction Power

Continuous power is the highest power that must be held indefinitely without overheating. It is commonly set by sustained top-speed cruise or a long climb:

$$P_{\text{cont}} = \max\left\{P_{\text{rms,cycle}},\ \frac{F_{\text{road}}(v_{\max})\,v_{\max}}{\eta_{tr}},\ \frac{F_{\text{road}}(v_g,\theta_g)\,v_g}{\eta_{tr}}\right\}, \tag{1.58}$$

where v_g and θ_g denote the sustained-climb speed and slope. In machine terms, a motor's peak-to-continuous ratio of about 2:1 for tens of seconds is typical of liquid-cooled traction machines.

1.7.11 Motor Torque and Speed Selection

The key choices are peak torque and maximum speed. Their relation to power is

$$T_{\text{peak}} = \frac{P_{\text{peak}}}{\omega_b}, \qquad \text{CPSR} = \frac{\omega_{\max}}{\omega_b}, \tag{1.59}$$

where ω_b (rad/s) is base speed and CPSR is the constant-power speed ratio. High-speed machines are smaller for the same power, but demand more gear reduction and need bearings and rotors suitable for the speed. A wide CPSR reduces peak power requirements, but it requires a design that can weaken the field, and it raises the inverter's rating relative to the motor's power.

1.7.12 Motor Power Rating Selection

The chosen motor must have a peak-power and continuous-power envelope covering all design points, including the operating cloud of the drive cycle. A margin of about 10–20% is added for losses of output as the motor heats, for voltage sag at low SoC, and for vehicle mass growth during development. Table 1.12 lists the parameters that must be fixed.

1.7.13 Gear Ratio Selection

With a single fixed ratio, G is bounded on both sides. The top-speed limit sets an upper bound and required wheel torque a lower bound:

$$\frac{F_{t,\max}\,r}{\eta_{tr}T_{\text{peak}}} \le G \le \frac{\omega_{m,\max}\,r}{v_{\max}}. \tag{1.60}$$

If the two bounds overlap, a value is chosen to balance performance and efficiency; if not, a higher-speed or higher-torque motor, or a two-speed gearbox, is required.

1.7.14 Inverter Voltage and Current Rating Selection

Device voltage is fixed by the maximum DC bus, including overshoot on turn-off. Current is fixed by the motor's peak phase current at minimum voltage. For a sinusoidal three-phase

machine,

$$I_{\text{rms}} = \frac{P_{\text{peak}}}{\sqrt{3}\, V_{LL}\, \eta_{\text{mot}} \cos\varphi}, \qquad V_{LL,\max} = \frac{V_{dc,\min}}{\sqrt{2}} \ \text{(SVPWM)}. \tag{1.61}$$

The right-hand expression follows from the SVPWM limit derived in Section 1.10. Device ratings are then selected with margins such as

$$V_{\text{device}} \geq 1.3\text{–}1.5\, V_{dc,\max}, \qquad I_{\text{device,pk}} \geq 1.25\ \sqrt{2}\, I_{\text{rms}}. \tag{1.62}$$

1.7.15 Battery Voltage Selection

Bus voltage is a system-level compromise. Higher voltage reduces current, cable mass and copper loss, and supports faster DC charging. It also demands more series cells, more insulation and higher-voltage devices. Passenger cars generally use nominal voltages of about 350–400 V, or about 800 V in fast-charging platforms; two-wheelers commonly use 48–96 V. The designer must check that the motor's back-EMF at top speed can be handled at the minimum pack voltage.

1.7.16 Battery Capacity Selection

Capacity follows from energy and voltage, and the cell configuration from Eq. (1.43):

$$Q_{\text{pack}} = \frac{E_{\text{bat}}}{V_{\text{nom}}}, \qquad N_s = \left\lceil \frac{V_{\text{nom}}}{V_{\text{cell}}} \right\rceil, \qquad N_p = \left\lceil \frac{Q_{\text{pack}}}{Q_{\text{cell}}} \right\rceil. \tag{1.63}$$

The result must satisfy the current limit: $I_{\max} \leq C_{\max} Q_{\text{pack}}$.

1.7.17 Battery Energy Requirement

Energy is obtained from Eq. (1.53) with the cycle's e_c. A pack sized for energy may need a check for power: the battery must be sized by *whichever* requirement is larger,

$$E_{\text{bat}} \geq \max\left\{ \frac{R\, e_c}{\text{DoD}(1 - k_{\text{age}})},\ \frac{P_{\text{bat,max}}}{(C_{\max} \text{ in h}^{-1})} \right\}. \tag{1.64}$$

For long-range BEVs, the energy term governs; for low-range or high-power vehicles, the power term can.

1.7.18 DC-DC Converter Selection

The DC–DC converter is sized for the total low-voltage load, which includes ECUs, lights, pumps, power steering and infotainment, typically 1–3 kW for cars. Its output current is $I_{LV} = P_{\text{aux}}/V_{LV}$. It must operate across the full input range between $V_{\min}$ and $V_{\max}$ of the pack, and provide galvanic isolation between HV and chassis-referenced LV networks.

1.7.19 Regenerative Braking Considerations

Regenerative braking influences sizing in three respects. The motor and inverter are normally symmetric in torque, but the battery's charge limit is usually smaller than its discharge limit, so *charge acceptance* often sets the maximum regen power. Thus at high SoC or low temperature regeneration may be cut sharply. The friction brake system must still handle a full stop without regeneration. And brake blending must respect tyre adhesion distribution and stability; the front axle takes more brake force, which is a reason that RWD vehicles cannot regenerate all braking.

1.7.20 Safety Margins and Design Constraints

Beyond performance, the design must satisfy the following constraints.

- Thermal limits of motor windings, magnets, semiconductor junctions and cells.
- Tyre adhesion limit, Eq. (1.4).
- Motor voltage limit at top speed and minimum SoC.
- Package and mass budget, including axle load limits.
- HV safety: insulation, interlocks, contactors, and crash disconnection.
- Margins for ageing: capacity fade, resistance rise, and bus-capacitor degradation.

Algorithm 3 assembles the method as a procedure.

1.7.21 Complete Numerical Case Study of BEV Component Selection

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Example 6: Selection of motor, inverter and battery for a specified BEV drive cycle

Problem. Select the motor, gear, inverter, battery and DC–DC converter for a compact BEV with the reference data of Table 1.5 and the following targets: range 300 km on the illustrative cycle of Fig. 1.18; top speed 150 km/h; 0–100 km/h in at most 10 s; sustained 6% grade at 80 km/h. All values are assumed design inputs.

Given. Cycle results in Table 1.10. $\eta_{\mathrm{mot}} = 0.92$ at the peak-power point, $\eta_{\mathrm{inv}} = 0.97$, power factor 0.85. Maximum motor speed 12 000 rpm; base speed 4000 rpm. Auxiliary load 2.5 kW peak on the 12 V network; 500 W average.

Required. Ratings for all main components.

Governing equations. Eqs. (1.18), (1.20), (1.53) and (1.57)–(1.64).

Step 1: Gear ratio. From the speed bound in Eq. (1.60),

$$G \le \frac{(12\,000 \times 2\pi/60)(0.32)}{41.67} = 9.65.$$

Algorithm 3 BEV component selection based on a drive cycle

Require: Cycle $v(t)$; targets R, $v_{\max}$, $t_{0\text{–}100}$, grade; parameters of Table 1.5; efficiencies
Ensure: $P_{\text{peak}}, P_{\text{cont}}, T_{\text{peak}}, G, V_{dc}, I_{\text{inv}}, E_{\text{bat}}, N_s, N_p$

```
Set initial mass m ← m_0
repeat
    Run Algorithms 1 and 2 to obtain F_{t,k}, P_{w,k}, s
    Compute P_acc from Eq. (1.18) and P_grade from Eq. (1.13)
    P_peak ← (1 + k_m) max{P^max_{w,cycle}/η_tr, P_acc, P_grade}
    P_cont ← (1 + k_m) max{P_rms, P(v_max), P(v_g, θ_g)}
    Choose ω_max, CPSR; T_peak ← P_peak/ω_b
    Pick G satisfying Eq. (1.60)
    Compute e_c from Eqs. (1.51)–(1.52)
    E_bat ← Eq. (1.64)
    Choose V_nom; N_s, N_p from Eq. (1.63); recompute E_bat
    Size inverter from Eqs. (1.61)–(1.62)
    m_new ← Eq. (1.55)
    if |m_new − m| < ε and all constraints met then
        converged ← true
    else
        m ← m_new
    end if
until converged
return final specification
```

Choose $G = 9.5$, so the speed-limited top speed is 152 km/h. This meets the target at a motor speed of 11 812 rpm.

Step 2: Peak power. The cycle peak is $44.6/0.95 = 46.9$ kW at the motor. For acceleration, the base vehicle speed is $v_b = (4000 \times 2\pi/60)(0.32)/9.5 = 14.1$ m/s. The estimate of Eq. (1.19) with 10 s gives

$$P \approx \frac{1.05 \times 1600\,(27.78^2 + 14.1^2)}{2 \times 10} = 81.5 \text{ kW at the wheels, or } 85.8 \text{ kW at the motor.}$$

This neglects road load. A numerical integration of Eq. (1.18), with road load included, gives 10.26 s at 90 kW and 9.16 s at 100 kW. Choose $P_{\text{peak}} = 100$ kW, providing a 0.8 s margin for voltage sag and mass growth.

Step 3: Motor torque.

$$T_{\text{peak}} = \frac{100\,000}{4000 \times 2\pi/60} = 238.7 \text{ N m} \;\Rightarrow\; \text{select } 240 \text{ N m}.$$

At peak torque, the wheel force is $240 \times 9.5 \times 0.95/0.32 = 6.77$ kN. The theoretical gradeability from Eq. (1.21) is about 46% at 30 km/h. The adhesion limit with $\mu = 0.8$ and 55% of weight on the driven axle is $0.8 \times 0.55 \times 15\,696 = 6.91$ kN, so torque is usable on dry roads but traction control is required on wet ones.

Step 4: Continuous power. The sustained cases at the motor shaft are

Case	Motor power (kW)
Cycle rms	11.9
Cruise, 120 km/h	22.0
Cruise, 150 km/h	39.1
6% grade at 80 km/h	30.5

With a 25% margin on the largest, $P_{\text{cont}} = 48.9$ kW, so select 50 kW. Peak-to-continuous is 2:1, which is typical for liquid cooling. Note that the cycle rms is far below this: the highway and climb cases size the machine thermally.

Step 5: Battery energy. From Table 1.10, $e_c = 157.3$ Wh/km. With DoD = 0.90 and $k_{\text{age}} = 0.10$,

$$E_{\text{bat}} = \frac{300 \times 157.3}{0.90 \times 0.90} = 58.3 \text{ kWh}.$$

Step 6: Pack configuration. Choose a 400 V class with 3.65 V, 50 A h NMC prismatic cells (4.2 V maximum, 3.0 V minimum). $N_s = 110$ and $N_p = 3$, as derived in the pack illustration of Section 1.5. Thus $V_{\text{nom}} = 401.5$ V, $Q = 150$ A h, and $E = 60.2$ kWh. The bus varies between 330 and 462 V. The range is $0.90 \times 60\,225/157.3 = 345$ km when new, or 310 km with 10% fade.

Step 7: Battery power check. At peak power and minimum voltage,

$$P_{dc} = \frac{100\,000}{0.92 \times 0.97} = 112.1 \text{ kW}, \quad I = \frac{112.1 + 1.5}{0.330} = 344 \text{ A} = 2.3\text{C}.$$

A 2.3C peak is well within the capability of EV cells, so energy governs battery size. The cycle's braking peak of 17.8 kW at the wheel returns $17.8 \times 0.8045 = 14.3$ kW, or about 0.36C.

Step 8: Inverter. From Eq. (1.61), $V_{LL} = 330/\sqrt{2} = 233$ V. Then

$$I_{\text{rms}} = \frac{100\,000}{\sqrt{3} \times 233.3 \times 0.92 \times 0.85} = 316 \text{ A}, \qquad I_{\text{pk}} = 447 \text{ A}.$$

Using margins of Eq. (1.62): voltage rating $\geq 1.4 \times 462 = 647$ V and current rating $\geq 1.25 \times 447 = 559$ A. Select a 750 V, 600 A six-pack power module. A 750 V device leaves headroom for turn-off overshoot.

Step 9: DC–DC converter. A 2.5 kW load at 13.5 V requires 185 A. Select a 2.5 kW isolated converter, 330–462 V input, 13.5 V output.

Result. Table 1.11 and Table 1.12 state the specification.

Interpretation. Each component was fixed by a different constraint: motor peak by acceleration, not by the cycle; continuous by top-speed cruise; gear by top speed; battery by range; inverter by peak current at low SoC. A cycle alone would have recommended a roughly 50 kW

motor, half of what acceleration demands. As the design mass of 1600 kg included a pack assumption of about 350 kg (170 Wh/kg pack), the mass loop of Fig. 1.20 converges with no change at this stage.

Table 1.11: BEV component selection specifications for the case study (design values).

Component	Selected value	Sizing driver
Motor peak power	100 kW	0–100 km/h target
Motor continuous power	50 kW	150 km/h cruise + margin
Motor peak torque	240 N m	Peak power at base speed
Gear ratio	9.5	Top speed at 12 000 rpm
Battery	110S3P, 60.2 kWh, 401.5 V	300 km on the cycle
Battery peak current	344 A (2.3C)	Peak power at 330 V
Inverter	750 V, 600 A peak	Peak current, maximum bus
DC–DC converter	2.5 kW, 13.5 V	Low-voltage load

Table 1.12: Motor and inverter selection parameters for the case study.

Motor parameter	Value	Inverter parameter	Value
Type	PMSM (assumed)	Topology	2-level VSI
Base speed	4000 rpm	DC bus range	330–462 V
Maximum speed	12 000 rpm	Device rating	750 V
CPSR	3:1	Peak phase current	447 A
Peak / cont. power	100 / 50 kW	Rated current	600 A peak
Peak torque	240 N m	Modulation	SVPWM
Power factor (assumed)	0.85	Switching frequency	10 kHz

Tables 1.11 and 1.12 are the outcome of a first design loop. A next step would simulate the design with motor and inverter efficiency maps to verify the assumed averages, and then iterate through thermal models. The inverter chosen here now becomes the subject of the following sections.

1.8 Three-Phase Inverter

1.8.1 Introduction to Power Inverters

An *inverter* is a power-electronic converter that converts DC power into AC power with controllable voltage, frequency and phase. It works by switching semiconductor devices on and off rapidly, so that the output is a sequence of voltage pulses whose average over a short interval follows the desired waveform. Devices operate as near-ideal switches, either fully on or fully

off, which keeps the loss small even at high power: a switch ideally has either zero voltage or zero current, and so dissipates no power [4].

1.8.2 Role of Inverters in EV Powertrains

The traction inverter sits between the DC battery and the AC motor (Fig. 1.2). It performs four functions.

1. It produces the variable-frequency, variable-voltage supply that sets motor speed and torque.
2. It carries out the fast current control that makes torque response nearly instantaneous.
3. It reverses power flow for regenerative braking, with no extra hardware, because each switch has an antiparallel diode.
4. It protects the machine and battery against over-current, over-voltage and over-temperature.

Its efficiency directly enters Eq. (1.26), and it must withstand high current, automotive vibration and coolant temperatures. Silicon IGBTs have long been the norm at 400 V, and SiC MOSFETs are spreading quickly because their lower switching loss suits higher voltage and higher frequency.

1.8.3 Classification of Inverters

Inverters are classified according to several attributes.

- **By number of output phases:** single-phase, three-phase, or multiphase.
- **By source type:** voltage source (VSI) or current source (CSI).
- **By number of output levels:** two-level, or multilevel such as neutral-point-clamped.
- **By control method:** six-step (square wave) or pulse-width modulated.
- **By commutation:** hard-switched or soft-switched (resonant).

1.8.4 Single-Phase and Three-Phase Inverters

A single-phase full-bridge has two legs, and its output alternates across one load. Instantaneous single-phase power pulsates at twice the output frequency. A balanced three-phase system instead delivers constant instantaneous power,

$$p(t) = v_a i_a + v_b i_b + v_c i_c = 3 V_{\mathrm{rms}} I_{\mathrm{rms}} \cos\varphi, \tag{1.65}$$

where v_x and i_x are the instantaneous phase voltages and currents. Equation (1.65) is the reason three-phase machines produce smooth torque and are universal in EV traction, with single-phase inverters confined to low-power uses such as portable loads or vehicle-to-load outlets.

1.8.5 Voltage Source Inverter (VSI)

A VSI is fed from a stiff DC voltage, usually a battery in parallel with a low-impedance capacitor. Its output voltage waveform is set by the switching pattern, while its output current depends on the load. All traction inverters are of this type, because a battery is naturally a voltage source. The DC-link capacitor supplies the high-frequency current pulses drawn by the switching bridge, so that the battery and its cables see a smooth current. A leg must never have both switches on: that would short-circuit the DC link, an event called *shoot-through*.

1.8.6 Current Source Inverter (CSI)

A CSI is fed from a stiff DC current, obtained by placing a large inductor in series with the source. Its switches steer this current through the phases, so that its output current waveform is defined and the voltage depends on the load. Its switches must block reverse voltage, and it has inherent short-circuit robustness. However, the large DC inductor is heavy and lossy, and the regulation of current needs an additional front-end stage. Table 1.13 summarises the comparison, and it shows why the VSI dominates in EVs.

Table 1.13: Comparison of VSI and CSI.

Feature	VSI	CSI
DC source	Stiff voltage	Stiff current
DC-link storage	Capacitor	Inductor
Impressed output	Voltage	Current
Critical fault	Leg shoot-through	Open DC circuit
Switch requirement	Reverse-conducting	Reverse-blocking
Regeneration	Reverses DC current	Reverses DC voltage
EV traction use	Standard	Rare

1.8.7 Two-Level Three-Phase VSI

A two-level VSI connects each motor terminal to one of two voltages at any instant, positive or negative DC rail. It is simple, compact and well understood, and its six switches make it the most widely used traction topology. Multilevel inverters reduce device stress and ripple but are justified mainly at higher voltages, so they are seldom used in 400 V cars, though they are of growing interest at 800 V.

1.8.8 Three-Phase Bridge Configuration

Figure 1.21 shows the complete circuit. The battery of voltage V_{dc} feeds the positive and negative DC rails. Two series capacitors C_{dc} represent the DC link and provide a fictitious midpoint O for analysis; in a real inverter a single capacitor is normally used and O is only a reference. Three legs are connected across the rails. The midpoints a, b and c of the legs drive the motor's three windings, shown as resistance, inductance and back-EMF in a star connection with neutral n isolated.

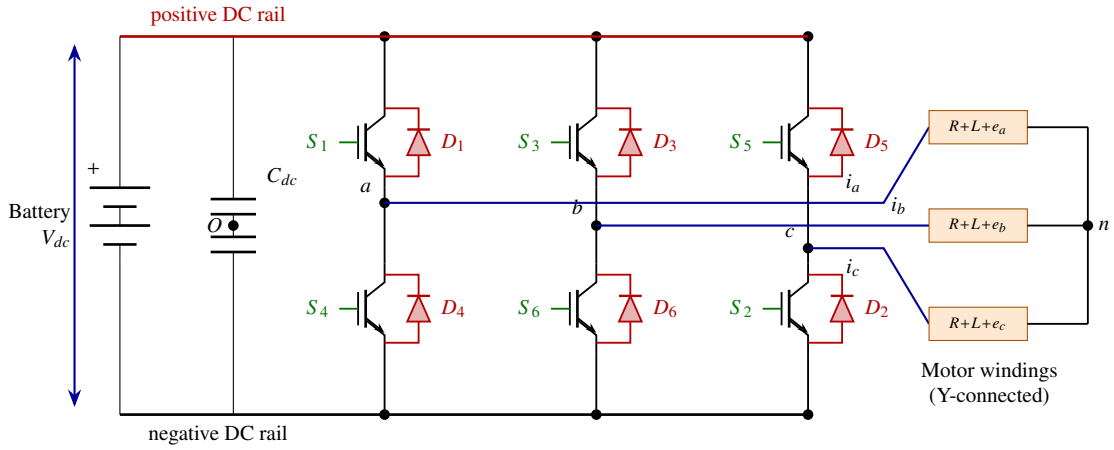


Figure 1.21: Two-level three-phase voltage source inverter feeding a Y-connected motor.

1.8.9 Six Power Switches and Antiparallel Diodes

Each leg of Fig. 1.21 comprises two switches, drawn as IGBTs, numbered so that their conduction sequence in six-step operation is $S_1, S_2, \ldots, S_6$. Switches S_1, S_3 and S_5 connect the phases to the positive rail, and S_4, S_6 and S_2 to the negative rail. Antiparallel diodes D_1 to D_6 carry current when the controlled switch cannot, in the reverse direction. They are essential for two reasons. The motor is inductive, so its current cannot be interrupted instantly: when a switch turns off, the current freewheels through the opposite diode, which prevents a destructive voltage spike. And in regenerative operation, the diodes conduct the rectified current back to the DC rails. MOSFET bridges, including SiC, use the body diode or synchronous conduction for the same function.

1.8.10 Switching States

Each leg has two valid states, so the bridge has $2^3 = 8$ valid states. Define a switching function for each leg:

$$S_x = \begin{cases} 1, & \text{upper switch on, lower off,} \\ 0, & \text{lower switch on, upper off,} \end{cases} \qquad x \in \{a, b, c\}. \tag{1.66}$$

Figure 1.22 shows the eight states, drawing each leg's connection to a DC rail. Six are *active* states, in which the motor is connected between the rails and receives voltage. The two remaining are *zero* states, in which all three terminals connect to one rail so that line voltages vanish and motor current freewheels. The conducting path of a device depends on the current sign: a closed S_1 with negative phase current is actually conducted by D_1.

1.8.11 Upper and Lower Switching Devices

The upper and lower switches in a leg are driven by complementary signals. Since real devices take finite time to turn off, a short *dead time*, commonly about one microsecond for IGBTs and less for SiC, is inserted during which both are off. It prevents shoot-through but causes a small voltage error dependent on current direction, which control software compensates. Upper-

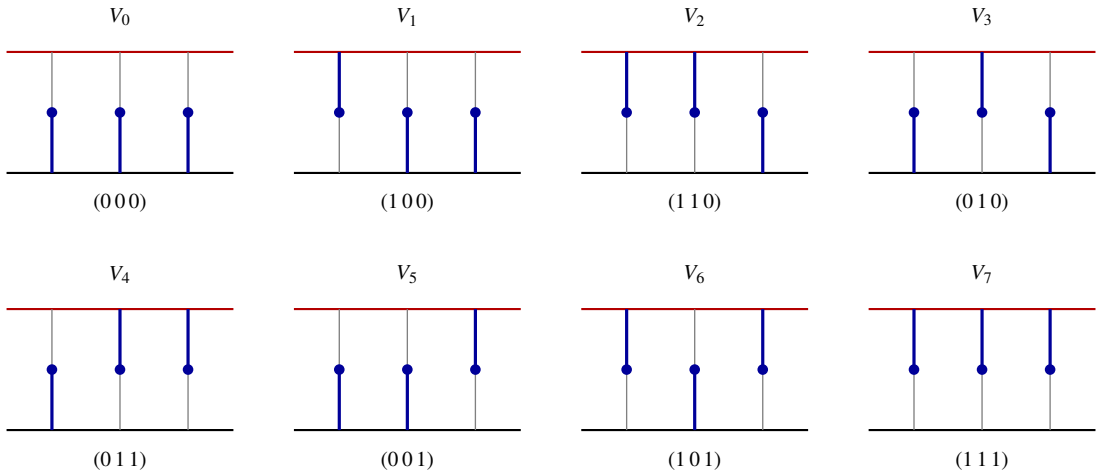


Figure 1.22: The eight switching states of a three-phase VSI. Each leg joins the upper or lower rail according to $(S_a\, S_b\, S_c)$.

switch gate drivers must also be isolated, since the upper emitter moves with the pole voltage. Desaturation detection in the driver protects against short circuits.

1.8.12 Pole Voltage, Phase Voltage and Line Voltage

Three voltage definitions are needed. The *pole voltage* v_{aO} is measured from a leg's midpoint to the DC midpoint O. It takes only two values:

$$v_{xO} = V_{dc}\left(S_x - \tfrac{1}{2}\right), \qquad x \in \{a, b, c\}. \tag{1.67}$$

The *line voltage* is the difference between pole voltages, so the midpoint vanishes:

$$v_{ab} = v_{aO} - v_{bO} = V_{dc}(S_a - S_b), \tag{1.68}$$

and similarly for v_{bc} and v_{ca}. The *phase voltage* is measured to motor neutral. To relate it to pole voltages, introduce the neutral-to-midpoint voltage v_{nO}:

$$v_{an} = v_{aO} - v_{nO}, \quad v_{bn} = v_{bO} - v_{nO}, \quad v_{cn} = v_{cO} - v_{nO}. \tag{1.69}$$

For a balanced star load with no neutral connection, $i_a + i_b + i_c = 0$, and with balanced back-EMF, $v_{an} + v_{bn} + v_{cn} = 0$. Adding the three parts of Eq. (1.69) then gives

$$v_{nO} = \tfrac{1}{3}\left(v_{aO} + v_{bO} + v_{cO}\right). \tag{1.70}$$

This v_{nO} is the *common-mode voltage*, a source of bearing currents and EMI in real drives. Substituting Eq. (1.70) into Eq. (1.69),

$$\begin{bmatrix} v_{an} \\ v_{bn} \\ v_{cn} \end{bmatrix} = \frac{V_{dc}}{3} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} S_a \\ S_b \\ S_c \end{bmatrix}. \tag{1.71}$$

Table 1.14 evaluates Eqs. (1.68) and (1.71) for all eight states. Phase voltage takes only the values 0, $\pm V_{dc}/3$ and $\pm 2V_{dc}/3$, whereas line voltage takes 0 and $\pm V_{dc}$. Both zero states give zero output, which is the redundancy later exploited in SVPWM.

Table 1.14: Three-phase VSI switching states (phase and line voltages in units of V_{dc}).

State	$(S_a S_b S_c)$	On switches	v_{an}	v_{bn}	v_{cn}	v_{ab}	v_{bc}	v_{ca}	Type
V_0	000	S_4, S_6, S_2	0	0	0	0	0	0	Zero
V_1	100	S_1, S_6, S_2	2/3	−1/3	−1/3	1	0	−1	Active
V_2	110	S_1, S_3, S_2	1/3	1/3	−2/3	0	1	−1	Active
V_3	010	S_4, S_3, S_2	−1/3	2/3	−1/3	−1	1	0	Active
V_4	011	S_4, S_3, S_5	−2/3	1/3	1/3	−1	0	1	Active
V_5	001	S_4, S_6, S_5	−1/3	−1/3	2/3	0	−1	1	Active
V_6	101	S_1, S_6, S_5	1/3	−2/3	1/3	1	−1	0	Active
V_7	111	S_1, S_3, S_5	0	0	0	0	0	0	Zero

1.8.13 DC-Link Voltage

The DC-link voltage V_{dc} limits all output voltages. Every output is a fraction of it, as Table 1.14 makes clear, so it sets the highest motor speed before field weakening is needed. In a BEV the link voltage is not constant: it varies with SoC and load, between 330 and 462 V for the pack in Example 6. The drive must therefore be designed to deliver its torque at minimum voltage while its devices are rated for maximum voltage. Capacitor sizing limits voltage ripple ΔV for a ripple current; a first estimate of required capacitance for a current step ΔI over a switching period is

$$C_{dc} \geq \frac{\Delta I \, T_s}{\Delta V}, \tag{1.72}$$

where T_s (s) is the switching period. Film capacitors are preferred for their ripple-current capability and long life.

1.8.14 Inverter Output Voltage Waveforms

The simplest way to obtain AC output is to sequence the six active states of Table 1.14 in order, staying one sixth of a period at each. Figure 1.23 shows the resulting waveforms in this *six-step* operation. The pole voltages are square waves of amplitude $V_{dc}/2$, displaced by 120°. Their differences give the quasi-square line voltage v_{ab} with 120° pulses. The phase voltage v_{an} shows the characteristic six-stepped shape with levels at $\pm V_{dc}/3$ and $\pm 2V_{dc}/3$.

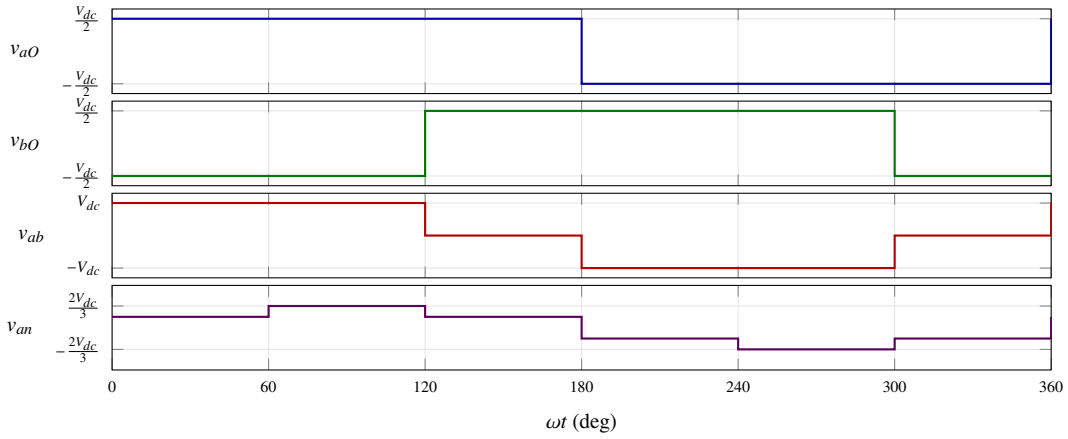


Figure 1.23: Six-step (180° conduction) waveforms of pole, line and phase voltages.

1.8.15 120-Degree and 180-Degree Conduction Modes

180-degree conduction. Each switch conducts for 180° of the period, and three switches are on at any time, one per leg. This is the mode of Fig. 1.23: complete complementary leg operation with all terminals tied to a rail. Output voltage does not depend on load, as long as both devices of each leg are driven. A switch turning off and its leg partner turning on are simultaneous, so a dead time is required.

120-degree conduction. Each switch conducts for 120°, and only two switches are on at any instant, one upper and one lower in different legs. The third phase floats for 60° intervals, which gives a natural commutation gap between switches in one leg and hence a safety margin. The phase voltage is a quasi-square wave of amplitude $V_{dc}/2$ for a resistive star load, and the floating terminal voltage depends on the load. This mode is the basis of trapezoidal brushless DC drives, where the third phase's back-EMF is used to sense the rotor position. For the same V_{dc}, however, the mode gives a smaller fundamental, as derived next.

1.8.16 Fundamental Output Voltage

A Fourier expansion of the pole voltage in Fig. 1.23 gives, for a square wave of amplitude $V_{dc}/2$,

$$v_{aO}(t) = \sum_{n=1,3,5,\ldots} \frac{2V_{dc}}{n\pi} \sin(n\omega t). \tag{1.73}$$

Line voltage is obtained by subtraction, so that triplen ($3n$) harmonics, being in phase across all legs, cancel:

$$v_{ab}(t) = \sum_{n=6k\pm1} \frac{2\sqrt{3}\,V_{dc}}{n\pi} \sin\left(n\omega t + \frac{n\pi}{6}\right). \tag{1.74}$$

Thus only harmonics of order $n = 1, 5, 7, 11, 13, \ldots$ appear. Phase voltage has the same harmonics. For 180° conduction its fundamental peak and the corresponding values are

$$\hat{V}_{an1} = \frac{2}{\pi} V_{dc} = 0.637\, V_{dc}, \qquad \hat{V}_{ab1} = \frac{2\sqrt{3}}{\pi} V_{dc} = 1.103\, V_{dc}, \tag{1.75}$$

with total rms values $V_{an} = \frac{\sqrt{2}}{3} V_{dc} = 0.471\, V_{dc}$ and $V_{ab} = \sqrt{\frac{2}{3}} V_{dc} = 0.816\, V_{dc}$.

For 120° conduction with a star-connected resistive load, the phase voltage is a quasi-square wave of amplitude $V_{dc}/2$ and width 120°, and its fundamental peak is

$$\hat{V}_{an1} = \frac{4}{\pi} \frac{V_{dc}}{2} \sin 60^\circ = \frac{\sqrt{3}}{\pi} V_{dc} = 0.551\, V_{dc}. \tag{1.76}$$

The 180° mode has about 15% more fundamental voltage for the same bus. It is the highest fundamental any two-level VSI can produce, and so represents the ceiling of the voltage in Fig. 1.26.

Example 7: Three-phase VSI output voltage

Problem. A VSI operates in six-step mode from a 400 V battery. Determine the phase voltage levels and, for a star-connected resistive load, compare the fundamental output in 180° and 120° modes.

Given. $V_{dc} = 400$ V.

Required. Levels; $\hat{V}_{an1}$, $\hat{V}_{ab1}$; rms values.

Governing equations. Eqs. (1.71), (1.75), (1.76).

Solution. Phase-voltage levels are $V_{dc}/3 = 133.3$ V and $2V_{dc}/3 = 266.7$ V.

180° mode:

$$\begin{aligned} \hat{V}_{an1} &= 0.637 \times 400 = 254.6 \text{ V peak}, \quad 180.1 \text{ V rms}, \\ \hat{V}_{ab1} &= 1.103 \times 400 = 441.1 \text{ V peak}, \quad 311.9 \text{ V rms}, \\ V_{ab} &= 0.8165 \times 400 = 326.6 \text{ V rms total}, \quad V_{an} = 188.6 \text{ V rms total}. \end{aligned}$$

120° mode:

$$\hat{V}_{an1} = 0.551 \times 400 = 220.5 \text{ V peak}, \ 155.9 \text{ V rms}; \quad \hat{V}_{ab1} = \sqrt{3}\,(220.5) = 382.0 \text{ V peak}.$$

Result. The 180° mode gives 254.6 V, against 220.5 V in the 120° mode.

Interpretation. The line-voltage fundamental is $311.9/326.6 = 95.5\%$ of total rms, corresponding to a THD of about 31%. High harmonic content at low order is why six-step alone is unsuitable for smooth low-speed traction, although field-weakened drives approach it at high

speed to use the full bus.

1.8.17 Harmonics and Switching Losses

Harmonics. Total harmonic distortion measures the waveform's departure from a sinusoid:

$$\mathrm{THD} = \frac{\sqrt{\sum_{n\neq 1} V_n^2}}{V_1} = \frac{\sqrt{V_{\mathrm{rms}}^2 - V_1^2}}{V_1}. \tag{1.77}$$

Harmonics do not produce useful average torque. They create extra copper and iron loss and torque ripple. Because the winding impedance $n\omega L$ rises with harmonic order, voltage harmonics at high order create little current. This is the central reason for pushing harmonic energy to high frequency through PWM.

Conduction loss. A conducting device has voltage drop $V_{ce} = V_0 + r_0 i$, so

$$P_{\mathrm{cond}} = V_0 I_{\mathrm{avg}} + r_0 I_{\mathrm{rms}}^2, \tag{1.78}$$

where V_0 (V) is the threshold voltage and r_0 (Ω) the on-state resistance, and I_{avg}, I_{rms} (A) are the device currents.

Switching loss. During a transition both voltage across and current through a device are significant. Assuming a linear transition,

$$P_{\mathrm{sw}} = \tfrac{1}{2}\, V_{dc}\, I\, (t_{\mathrm{on}} + t_{\mathrm{off}})\, f_s = (E_{\mathrm{on}} + E_{\mathrm{off}})\, f_s, \tag{1.79}$$

where t_{on} and t_{off} (s) are transition times, f_s (Hz) the switching frequency, and E_{on}, E_{off} (J) the energies per transition from data sheets. Equation (1.79) presents the central trade-off in inverter design: raising f_s lowers current ripple and audible noise, but raises switching loss linearly. Traction IGBT inverters typically run at about 5–10 kHz, and SiC allows higher values at the same or lower loss.

1.8.18 Applications of Three-Phase Inverters in EV Motor Drives

Three-phase VSIs are used in each of the following functions in an electrified vehicle.

- Traction drives for PMSM, induction and wound-rotor machines, from scooters to heavy trucks.
- Generator and traction machines in series-parallel hybrids (Fig. 1.5).
- Compressors for air conditioning, and coolant pumps, at lower power.

- Bidirectional active rectifiers in on-board chargers, and integrated charging, in which the traction inverter and motor windings double as charger hardware.

The next three sections consider the most important question left open here, namely how the switches are to be driven.

1.9 PWM Techniques

1.9.1 Introduction to Pulse Width Modulation

Pulse width modulation (PWM) controls the *average* output over a short period by varying the fraction of time for which a switch is on. If a leg's upper switch is on for time t_{on} in a period T_s, the mean pole voltage is

$$\bar{v}_{aO} = \frac{1}{T_s}\int_0^{T_s} v_{aO}\,\mathrm{d}t = V_{dc}\left(d - \tfrac{1}{2}\right), \qquad d = \frac{t_{\text{on}}}{T_s}, \tag{1.80}$$

where d (dimensionless) is the duty ratio. Letting d vary slowly and sinusoidally yields a sinusoidal average voltage. The high-frequency pulses are then filtered by the motor's inductance, so the current is nearly sinusoidal. The principle is known as the *volt-second balance*: over each switching period, the pulses must deliver the same volt-seconds as the desired waveform.

1.9.2 Importance of PWM in EV Motor Drives

PWM is what allows the fixed battery voltage to become a smooth, adjustable AC supply. It matters in EV drives for these reasons.

- Amplitude and frequency of the output are controlled independently, which allows fast torque and flux control.
- Low-order harmonics are eliminated and distortion is moved near the switching frequency, reducing torque ripple and loss.
- Fast response is obtained, since the voltage is updated every carrier period.
- It is available from any automotive microcontroller via dedicated timer peripherals.

1.9.3 Switching Frequency

The switching frequency $f_s = 1/T_s$ (Hz) is the rate at which a leg's devices switch, set in carrier PWM by the carrier frequency, so $f_s = f_c$. It trades ripple against loss, Eq. (1.79). It also influences audible noise: values within the human hearing band, up to 10 kHz or so, can be objectionable, especially in quiet EVs. Control bandwidth is limited to a small fraction of f_s, usually by a factor of about 10.

1.9.4 Modulation Index

The modulation index expresses how much of the available DC bus is being used. For carrier-based PWM it is the amplitude ratio defined below. Alternatively, it may be normalised to the maximum six-step fundamental, as

$$M = \frac{\hat{V}_1}{\frac{2}{\pi} V_{dc}}, \tag{1.81}$$

so that $M = 1$ means six-step operation. Care is needed, since different textbooks use different definitions: this unit uses m_a for carrier PWM and states the definition for SVPWM in Section 1.10.

1.9.5 Frequency Ratio

The *frequency modulation ratio* is

$$m_f = \frac{f_c}{f_1}, \tag{1.82}$$

where f_c (Hz) is the carrier frequency and f_1 (Hz) the fundamental output frequency. A large m_f pushes harmonics to high frequency. In the *synchronous* mode, m_f is an integer and in three-phase drives an odd multiple of 3, to ensure quarter-wave symmetry and cancel harmonics at m_f in line voltage. In the *asynchronous* mode, f_c stays fixed while f_1 varies, which is acceptable with large m_f, typically above about 21 [4]. EVs at high speed have only limited m_f: at 12 000 rpm, a four-pole-pair machine has $f_1 = 800$ Hz, so $f_c = 10$ kHz gives only $m_f = 12.5$. Therefore high-speed operation needs careful, usually synchronised, modulation.

1.9.6 Amplitude Ratio

The *amplitude modulation ratio* is

$$m_a = \frac{\hat{V}_r}{\hat{V}_c}, \tag{1.83}$$

where $\hat{V}_r$ (V) is the peak reference (modulating) signal and $\hat{V}_c$ (V) the peak carrier. For $m_a \leq 1$ the modulation is linear; beyond that the reference exceeds the carrier for some pulses, and these are dropped, an operating regime called *overmodulation*.

1.9.7 Carrier-Based PWM

In carrier-based PWM, a reference signal is compared with a high-frequency triangular carrier, and the comparator output directly drives a leg. A triangular carrier ensures that the pulse width is linearly related to the reference. Consider one carrier period in which the reference is a constant v_r, with carrier swinging between $\pm\hat{V}_c$. The upper switch is on when $v_r > v_{\text{tri}}$. Since the carrier is linear, the fraction of time that v_{tri} lies below v_r is

$$d = \frac{1}{2}\left(1 + \frac{v_r}{\hat{V}_c}\right). \tag{1.84}$$

Substituting Eq. (1.84) in Eq. (1.80) gives

$$\bar{v}_{aO} = \frac{v_r}{\hat{V}_c}\,\frac{V_{dc}}{2}, \tag{1.85}$$

which is exactly linear. The modulator behaves as a linear amplifier of gain $V_{dc}/(2\hat{V}_c)$, a powerful result since the control system can treat the inverter as a simple gain.

1.9.8 Sinusoidal PWM (SPWM)

SPWM uses a sinusoidal reference, $v_r(t) = \hat{V}_r \sin \omega t$. Applying Eq. (1.85) at each instant yields the fundamental of the pole voltage:

$$\hat{V}_{aO1} = m_a \frac{V_{dc}}{2}, \qquad m_a \le 1. \tag{1.86}$$

The fundamental line voltage is then $\sqrt{3}$ times the phase value, so in rms terms

$$V_{LL1,\mathrm{rms}} = \frac{\sqrt{3}}{\sqrt{2}}\, m_a \frac{V_{dc}}{2} = \frac{\sqrt{3}}{2\sqrt{2}}\, m_a V_{dc} = 0.612\, m_a V_{dc}. \tag{1.87}$$

The maximum linear line voltage in SPWM is thus $0.612\,V_{dc}$ rms, or $0.866\,V_{dc}$ peak. This is about 78.5% of the six-step fundamental: a significant shortfall, since the DC bus is costly.

1.9.9 Unipolar and Bipolar PWM Concepts

The terms bipolar and unipolar arise chiefly in the single-phase full bridge.

- **Bipolar PWM.** Diagonal switch pairs are driven together, so the output swings between $+V_{dc}$ and $-V_{dc}$. Harmonics appear around f_c.
- **Unipolar PWM.** Each leg has its own reference, of opposite sign, compared with the same carrier. Output steps between 0 and $\pm V_{dc}$, and the effective ripple frequency at the output is doubled to $2f_c$, while the same switching loss is maintained.

In a three-phase VSI, each leg is inherently two-level, or bipolar with respect to O. But the line voltage, as the difference of two poles, has unipolar character: it takes the values $0, \pm V_{dc}$ within any half cycle, which is visible in the v_{ab} trace of Fig. 1.25.

1.9.10 Three-Phase Sinusoidal PWM

Three-phase SPWM employs three references, displaced by 120°, and a shared carrier:

$$v_{ra} = m_a \sin \omega t, \quad v_{rb} = m_a \sin\left(\omega t - \tfrac{2\pi}{3}\right), \quad v_{rc} = m_a \sin\left(\omega t + \tfrac{2\pi}{3}\right), \tag{1.88}$$

all normalised to a unit carrier. Figure 1.24 shows these for $m_a = 0.8$ and $m_f = 15$. Each leg is modulated independently. At each intersection of the carrier with a reference, the corresponding leg changes state.

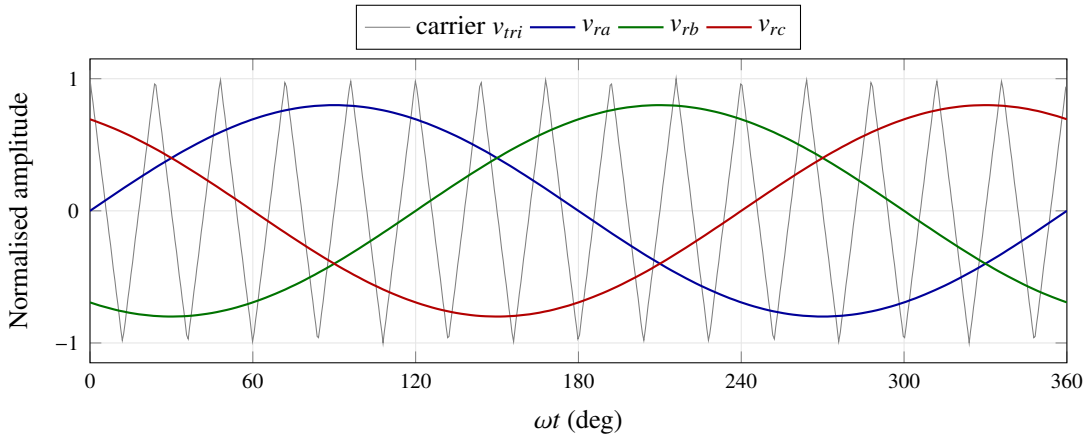


Figure 1.24: Three-phase SPWM reference waveforms and triangular carrier ($m_a = 0.8$, $m_f = 15$).

1.9.11 Switching Signal Generation

The comparator logic is simply

$$S_x = \begin{cases} 1, & v_{rx} \geq v_{\text{tri}}, \\ 0, & v_{rx} < v_{\text{tri}}, \end{cases} \qquad x \in \{a, b, c\}, \tag{1.89}$$

and the complementary lower gate signal is $\overline{S_x}$, with dead time. Figure 1.25 shows the resulting pole voltages for all three legs and the line voltage v_{ab}. It is evident that the pulse width in v_{aO} is greatest when v_{ra} peaks, and the line voltage pulses are concentrated around the peaks of the sinusoidal fundamental, as expected. Digital controllers replace the analogue comparison by a timer counting up and down, with a compare register holding the reference. Algorithm 4 gives the steps.

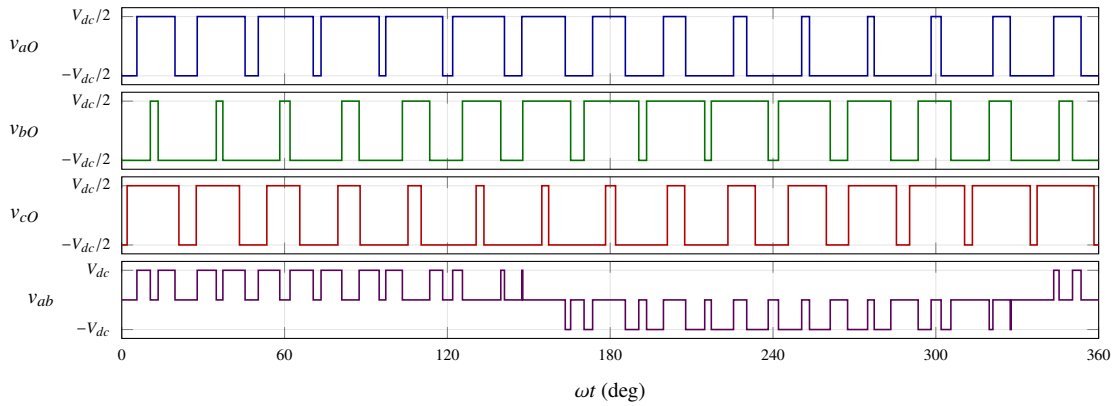


Figure 1.25: SPWM switching pulses: pole voltages of the three legs, and line voltage v_{ab} ($m_a = 0.8$, $m_f = 15$).

1.9.12 Output Voltage Characteristics

Figure 1.26 shows how the fundamental output relates to m_a. In the linear range it is a straight line reaching unity at $m_a = 1$. Into overmodulation, the curve flattens, and the output rises

Algorithm 4 SPWM generation

Require: V_{dc}, reference peak $\hat{V}^*$, frequency f_1, carrier frequency f_c, timer period N_{PRD}
Ensure: Gate signals for $S_1, \ldots, S_6$
Initialise angle $\vartheta \leftarrow 0$; set $T_s \leftarrow 1/f_c$
loop (at each carrier period)
  $m_a \leftarrow \min\left(2\hat{V}^*/V_{dc},\ 1\right)$
  $\vartheta \leftarrow (\vartheta + 2\pi f_1 T_s) \bmod 2\pi$
  for x in $\{a, b, c\}$, phase offset $\phi_x \in \{0,\ 2\pi/3,\ -2\pi/3\}$ **do**
    $v_{rx} \leftarrow m_a \sin(\vartheta - \phi_x)$
    $\mathrm{CMP}_x \leftarrow \frac{1}{2}(1 + v_{rx})\, N_{\mathrm{PRD}}$
    if timer count $\le \mathrm{CMP}_x$ **then**
      $S_x \leftarrow 1$
    else
      $S_x \leftarrow 0$
    end if
    Apply S_x to upper gate and $\overline{S_x}$ to lower gate, with dead time
  end for
end loop

slowly towards the six-step limit $4/\pi = 1.273$, at the expense of larger low-order harmonics and a loss of linearity, which makes closed-loop control harder. The figure also shows the higher linear limit of 1.155 available with SVPWM, discussed in Section 1.10.

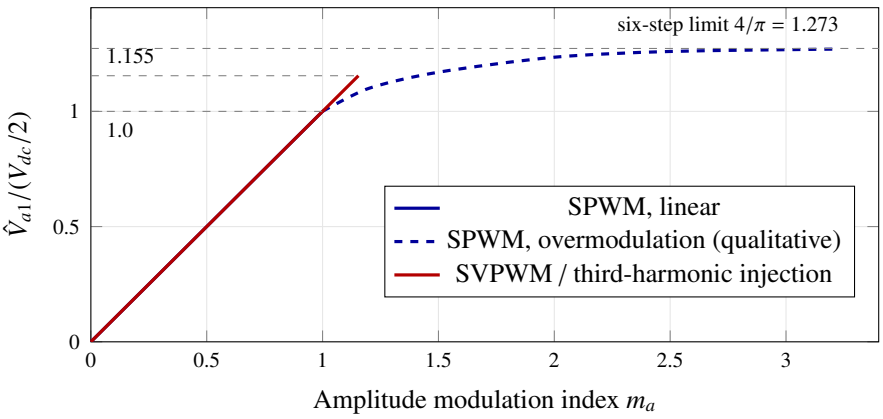


Figure 1.26: Fundamental output voltage against modulation index. The overmodulation curve is qualitative.

Example 8: SPWM modulation index

Problem. A 400 V VSI using SPWM must deliver 200 V rms line-to-line at 200 Hz. The triangular carrier has 5 V peak and a frequency near 10 kHz. Find m_a, the reference peak, the fundamental phase voltage, and a suitable m_f. What is the maximum linear line voltage?

Given. $V_{dc} = 400$ V; $V_{LL1} = 200$ V; $\hat{V}_c = 5$ V; $f_1 = 200$ Hz.

Required. m_a, $\hat{V}_r$, $\hat{V}_{an1}$, m_f, $V_{LL,\max}$.

Governing equations. Eqs. (1.82), (1.83), (1.86), (1.87).

Solution.

$$
\begin{aligned}
m_a &= \frac{V_{LL1}}{0.6124\,V_{dc}} = \frac{200}{0.6124 \times 400} = 0.8165, \\
\hat{V}_r &= m_a \hat{V}_c = 0.8165 \times 5 = 4.08 \text{ V}, \\
\hat{V}_{an1} &= 0.8165 \times \frac{400}{2} = 163.3 \text{ V peak}, \\
m_f &= \frac{10\,000}{200} = 50 \;\Rightarrow\; \text{choose 51 (odd multiple of 3), so } f_c = 10.2 \text{ kHz}, \\
V_{LL,\max} &= 0.6124 \times 400 \times 1 = 244.9 \text{ V rms}.
\end{aligned}
$$

Result. $m_a = 0.82$, $\hat{V}_r = 4.08$ V, $\hat{V}_{an1} = 163$ V, $m_f = 51$, $V_{LL,\max} = 245$ V rms.

Interpretation. The operating point lies safely in the linear range. The drive has 18% headroom in SPWM, for transient current control. SVPWM would raise the linear limit to 283 V, as shown in Section 1.10.

1.9.13 Harmonic Performance

For large m_f, the pole voltage harmonics are found at sidebands of the carrier and its multiples, at orders

$$h = j\,m_f \pm k, \tag{1.90}$$

with $j + k$ odd for the pole voltage [4, 6]. Figure 1.27 shows the computed spectrum of a pole voltage, for $m_a = 0.8$ and $m_f = 21$. The fundamental is exactly m_a, as Eq. (1.86) predicts. Next come a strong carrier component at $h = 21$ and sidebands at $m_f \pm 2$, then the $2m_f \pm 1$, $2m_f \pm 3$ family, and higher groups. Crucially, no significant low-order harmonics exist below about $m_f - 2$. In a three-phase system, the component at m_f and any triplen sidebands are common to the three legs and so vanish from the line voltage. The current ripple is then limited by motor inductance, as it is at high frequency.

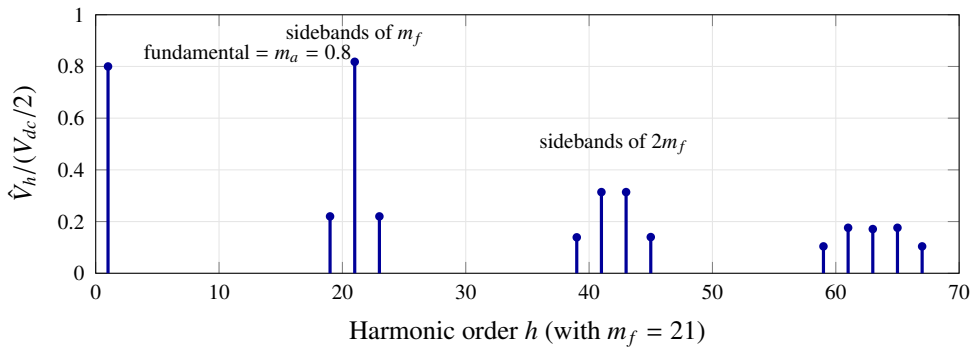


Figure 1.27: Computed harmonic spectrum of the SPWM pole voltage ($m_a = 0.8$, $m_f = 21$, normalised to $V_{dc}/2$).

1.9.14 Advantages and Limitations of SPWM

Advantages.

- Simple to implement, with one comparison per leg and low processor load.
- Linear voltage gain up to $m_a = 1$.
- Fixed switching frequency and a predictable harmonic spectrum.

Limitations.

- Only 78.5% of the six-step fundamental is available in the linear range.
- Each phase is treated independently, so the three-phase structure of the load is not exploited.
- More switching and more current ripple than optimised sequences at the same frequency for high modulation indices.
- Non-linear overmodulation behaviour.

The bus limitation can be overcome by adding a common-mode (zero-sequence) offset to all references. Because the motor neutral is isolated, such an offset leaves line voltages unchanged. Third-harmonic injection and the min-max offset are examples [10], and the latter is the carrier-based equivalent of SVPWM.

1.10 Space Vector Pulse Width Modulation (SVPWM)

1.10.1 Introduction to Space Vector Representation

Rather than modulating each phase independently, SVPWM treats the inverter and motor as a single three-phase system [9]. It represents the three phase voltages by one rotating vector and synthesises it from the eight states of Table 1.14. This viewpoint is natural for AC machines, because the three stator currents produce a single rotating magnetic field, whose direction and magnitude are what really matter. It also fits field-oriented control, which outputs a voltage vector, and gains a higher DC-bus utilisation than SPWM.

1.10.2 Three-Phase Voltage Space Vector

The three windings of a machine are spaced 120° apart in space. The *space vector* of three quantities is defined by the complex sum

$$\vec{v} = \frac{2}{3}\left(v_a + \alpha\, v_b + \alpha^2\, v_c\right), \qquad \alpha = e^{\,j2\pi/3}, \tag{1.91}$$

where v_a, v_b, v_c (V) are the phase voltages and α is the 120° rotation operator. The factor 2/3 makes the transformation *amplitude invariant*. For a balanced sinusoidal set, with $v_a = \hat{V}\cos\omega t$ and the others displaced by $\mp 120°$, direct substitution with Euler's formula gives

$$\vec{v} = \hat{V}\, e^{\,j\omega t}. \tag{1.92}$$

Hence a balanced three-phase sinusoidal set is just a vector of constant length $\hat{V}$ rotating at ω. Equation (1.92) is the key to SVPWM: to create sinusoidal phase voltages, one only needs a vector that traces a circle.

1.10.3 Clarke Transformation

The Clarke transformation expresses Eq. (1.91) in real and imaginary parts. Using $\alpha = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$ and $\alpha^2 = -\frac{1}{2} - j\frac{\sqrt{3}}{2}$, write $\vec{v} = v_\alpha + jv_\beta$:

$$\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = \frac{2}{3}\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix}. \tag{1.93}$$

For $v_a + v_b + v_c = 0$, this reduces to $v_\alpha = v_a$ and $v_\beta = (v_b - v_c)/\sqrt{3}$. The inverse is

$$v_a = v_\alpha, \quad v_b = -\tfrac{1}{2}v_\alpha + \tfrac{\sqrt{3}}{2}v_\beta, \quad v_c = -\tfrac{1}{2}v_\alpha - \tfrac{\sqrt{3}}{2}v_\beta. \tag{1.94}$$

1.10.4 Alpha-Beta Reference Frame

The $\alpha\beta$ frame is a stationary, orthogonal, two-axis frame in the plane of the stator, with α aligned to phase a. The transformation removes redundancy: a balanced three-wire system has only two independent variables. In field-oriented control, the measured currents pass through the Clarke and Park transformations to a rotor-fixed dq frame, in which torque and flux are nearly DC quantities and are regulated by PI controllers. The resulting voltage command returns to $\alpha\beta$, which is the input to SVPWM.

1.10.5 Eight Switching States

Substituting phase voltages from Eq. (1.71) into Eq. (1.91) gives a compact result in terms of switch states:

$$\vec{v} = \frac{2}{3}V_{dc}\left(S_a + \alpha S_b + \alpha^2 S_c\right). \tag{1.95}$$

This holds because the common $(S_a + S_b + S_c)$ term in the phase voltages multiplies $1 + \alpha + \alpha^2 = 0$. Evaluating for each of the eight states yields Table 1.15. For example, state (1 0 0) gives $\frac{2}{3}V_{dc}$ along α, and state (1 1 0) gives $\frac{2}{3}V_{dc}(1 + \alpha) = \frac{2}{3}V_{dc}e^{j\pi/3}$.

Table 1.15: Space vectors of the eight switching states.

Vector	$(S_a S_b S_c)$	v_α / V_{dc}	v_β / V_{dc}	Magnitude	Angle
$\vec{V}_0$	000	0	0	0	–
$\vec{V}_1$	100	2/3	0	$2V_{dc}/3$	0°
$\vec{V}_2$	110	1/3	$1/\sqrt{3}$	$2V_{dc}/3$	60°
$\vec{V}_3$	010	−1/3	$1/\sqrt{3}$	$2V_{dc}/3$	120°
$\vec{V}_4$	011	−2/3	0	$2V_{dc}/3$	180°
$\vec{V}_5$	001	−1/3	$-1/\sqrt{3}$	$2V_{dc}/3$	240°
$\vec{V}_6$	101	1/3	$-1/\sqrt{3}$	$2V_{dc}/3$	300°
$\vec{V}_7$	111	0	0	0	–

1.10.6 Six Active Vectors and Two Zero Vectors

The general expression for the six active vectors is

$$\vec{V}_k = \frac{2}{3} V_{dc}\, e^{\,j(k-1)\pi/3}, \qquad k = 1, \ldots, 6. \tag{1.96}$$

They have equal magnitude and are spaced at 60°. The two zero vectors $\vec{V}_0$ and $\vec{V}_7$ lie at the origin. They have the same effect on line voltage but differ in the rail to which the machine is clamped, which gives one degree of freedom in distributing switching and in common-mode behaviour.

1.10.7 Space Vector Hexagon

Plotting the vectors in the $\alpha\beta$ plane gives Fig. 1.28. The tips of the six active vectors form a regular hexagon, with both zero vectors at the centre. Any point within the hexagon can be reached on average by time-weighting two adjacent active vectors and a zero vector. Six-step operation corresponds to jumping between hexagon corners, and the average output voltage in the shaded sector is the reference vector drawn in red. The dashed circle inscribed in the hexagon is the largest circular trajectory, and it defines the linear modulation limit.

1.10.8 Six Sectors

The active vectors divide the plane into six 60° sectors, marked I to VI in Fig. 1.28. Sector I lies between $\vec{V}_1$ and $\vec{V}_2$, sector II between $\vec{V}_2$ and $\vec{V}_3$, and so on. Within a sector, the two bounding vectors are the most effective choice, as they lead to the smallest ripple.

1.10.9 Reference Voltage Vector

The reference vector comes from the current controller, or for open-loop drives from the commanded magnitude and angle:

$$\vec{V}_{ref} = V_\alpha + jV_\beta = V_{ref}\, e^{\,j\theta}, \quad V_{ref} = \sqrt{V_\alpha^2 + V_\beta^2}, \quad \theta = \operatorname{atan2}(V_\beta, V_\alpha). \tag{1.97}$$

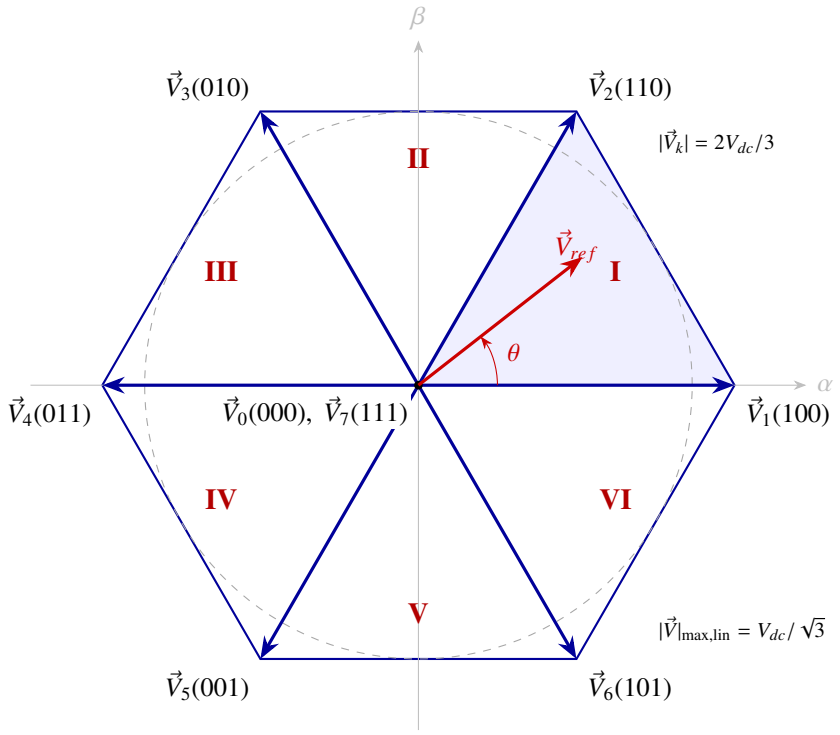


Figure 1.28: Space-vector hexagon with six active vectors, two zero vectors, the six sectors and a reference vector $\vec{V}_{ref}$.

It is sampled once per switching period T_s, on the assumption that T_s is much shorter than the fundamental period, so that $\vec{V}_{ref}$ is effectively constant over T_s.

1.10.10 Sector Identification

With θ wrapped to $[0, 2\pi)$, the sector number is

$$k = \left\lfloor \frac{\theta}{\pi/3} \right\rfloor + 1. \tag{1.98}$$

A trigonometric function is not needed, as sign tests alone suffice. Define

$$A = V_\beta, \quad B = \tfrac{1}{2}\left(\sqrt{3}V_\alpha - V_\beta\right), \quad C = \tfrac{1}{2}\left(-\sqrt{3}V_\alpha - V_\beta\right), \tag{1.99}$$

and $N = \operatorname{sgn}^+(A) + 2\operatorname{sgn}^+(B) + 4\operatorname{sgn}^+(C)$, where $\operatorname{sgn}^+(x) = 1$ for $x > 0$ and 0 otherwise. Then $N = 3, 1, 5, 4, 6, 2$ correspond to sectors I, II, III, IV, V, VI respectively.

1.10.11 Calculation of Switching Times

SVPWM rests on volt-second balance. Over the period T_s, the average voltage vector must equal the reference. For a reference in sector I, apply $\vec{V}_1$ for T_1, $\vec{V}_2$ for T_2, and a zero vector for T_0:

$$\vec{V}_{ref}\,T_s = \vec{V}_1 T_1 + \vec{V}_2 T_2 + \vec{V}_0 T_0, \qquad T_1 + T_2 + T_0 = T_s. \tag{1.100}$$

Figure 1.29 is the geometric meaning of Eq. (1.100): $\vec{V}_{ref}$ is the sum of the two weighted vectors along the sector boundaries.

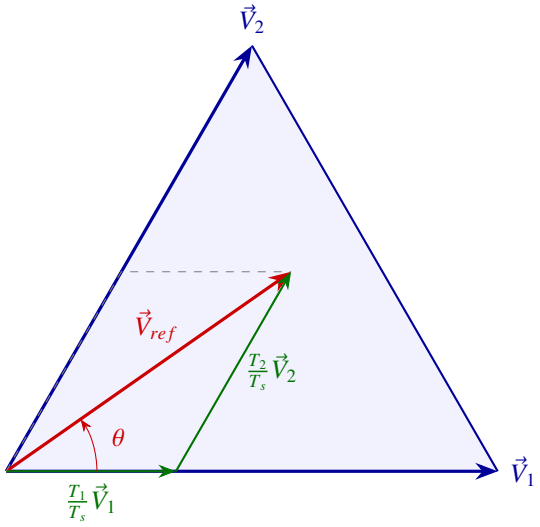


Figure 1.29: Decomposition of the reference vector into its adjacent active vectors in sector I.

1.10.12 Active Vector Durations

Resolving Eq. (1.100) into its α and β components, with values from Table 1.15, gives

$$\alpha: \quad V_{ref}\cos\theta\, T_s = \tfrac{2}{3}V_{dc}\, T_1 + \tfrac{1}{3}V_{dc}\, T_2, \tag{1.101}$$

$$\beta: \quad V_{ref}\sin\theta\, T_s = \tfrac{1}{\sqrt{3}}V_{dc}\, T_2. \tag{1.102}$$

From Eq. (1.102),

$$T_2 = \frac{\sqrt{3}\, V_{ref} T_s}{V_{dc}}\, \sin\theta. \tag{1.103}$$

Substituting Eq. (1.103) into Eq. (1.101) and using $\sin(60^\circ - \theta) = \frac{\sqrt{3}}{2}\cos\theta - \frac{1}{2}\sin\theta$,

$$T_1 = \frac{3V_{ref}T_s}{2V_{dc}}\left(\cos\theta - \frac{\sin\theta}{\sqrt{3}}\right) = \frac{\sqrt{3}\, V_{ref}T_s}{V_{dc}}\, \sin\left(60^\circ - \theta\right). \tag{1.104}$$

Defining the SVPWM *modulation index* as

$$m = \frac{\sqrt{3}\, V_{ref}}{V_{dc}}, \tag{1.105}$$

the times in any sector k with active vectors $\vec{V}_k$, $\vec{V}_{k+1}$ are

$$T_a = m\, T_s\, \sin\left(\frac{k\pi}{3} - \theta\right), \qquad T_b = m\, T_s\, \sin\left(\theta - \frac{(k-1)\pi}{3}\right), \tag{1.106}$$

where θ is measured from the α axis. For $k = 1$, $T_a = T_1$ and $T_b = T_2$.

1.10.13 Zero Vector Duration

The remainder of the period is taken by the zero vectors:

$$T_0 = T_s - T_a - T_b. \tag{1.107}$$

A short zero period means the vector lies near the hexagon boundary. For the common symmetric scheme T_0 is split equally between $\vec{V}_0$ and $\vec{V}_7$, which gives the best harmonic performance. The split can instead be biased towards one zero vector: *discontinuous* PWM omits one of them in parts of the fundamental period, to clamp a leg and reduce switching loss by up to about one third, with increased ripple at low modulation [10].

Example 9: SVPWM switching times for a reference vector

Problem. A 400 V inverter switches at 10 kHz. The controller requests $V_{ref} = 150$ V at $\theta = 40^\circ$. Find the sector, modulation index, vector durations and the on-time of each upper switch, and verify the volt-seconds.

Given. $V_{dc} = 400$ V; $T_s = 100\,\mu$s; $V_{ref} = 150$ V; $\theta = 40^\circ$.

Required. k, m, T_1, T_2, T_0; on-times.

Governing equations. Eqs. (1.98), (1.105), (1.106), (1.107).

Solution. As $0^\circ \le 40^\circ < 60^\circ$, the vector is in sector I.

$$\begin{aligned} m &= \frac{\sqrt{3} \times 150}{400} = 0.6495, \\ T_1 &= 0.6495 \times 100 \times \sin 20^\circ = 22.2\,\mu\text{s}, \\ T_2 &= 0.6495 \times 100 \times \sin 40^\circ = 41.8\,\mu\text{s}, \\ T_0 &= 100 - 22.2 - 41.8 = 36.0\,\mu\text{s}. \end{aligned}$$

With sequence $V_0 V_1 V_2 V_7 V_2 V_1 V_0$ (Section 1.10.14):

$$\begin{aligned} t_{S_1} &= T_1 + T_2 + T_0/2 = 82.0\,\mu\text{s}, \\ t_{S_3} &= T_2 + T_0/2 = 59.8\,\mu\text{s}, \\ t_{S_5} &= T_0/2 = 18.0\,\mu\text{s}. \end{aligned}$$

Check: using Table 1.15, $V_\alpha = (266.7 \times 22.2 + 133.3 \times 41.8)/100 = 114.9$ V, and $150 \cos 40^\circ = 114.9$ V. Also $V_\beta = (230.9 \times 41.8)/100 = 96.5$ V, against $150 \sin 40^\circ = 96.4$ V.

Result. Sector I; $m = 0.65$; $T_1 = 22.2\,\mu$s; $T_2 = 41.8\,\mu$s; $T_0 = 36.0\,\mu$s.

Interpretation. $T_2 > T_1$, since 40° lies nearer $\vec{V}_2$. A linear range remains, since $T_0 > 0$: the same drive could produce up to $400/\sqrt{3} = 231$ V.

1.10.14 Switching Sequence

Time durations alone do not specify the sequence. A good sequence changes only one leg at each transition, so as to minimise the switching loss, and is symmetric in T_s, to reduce harmonics. The seven-segment pattern achieves this. Figure 1.30 shows sector I: $V_0 \rightarrow V_1 \rightarrow V_2 \rightarrow V_7 \rightarrow V_2 \rightarrow V_1 \rightarrow V_0$, in other words 000 → 100 → 110 → 111 → 110 → 100 → 000. Each of the three legs switches once on and once off per period, so the waveform is equivalent to centre-aligned PWM with a single carrier period. The switching order in the alternate sectors is reversed, as in Table 1.16, to retain single-leg transitions.

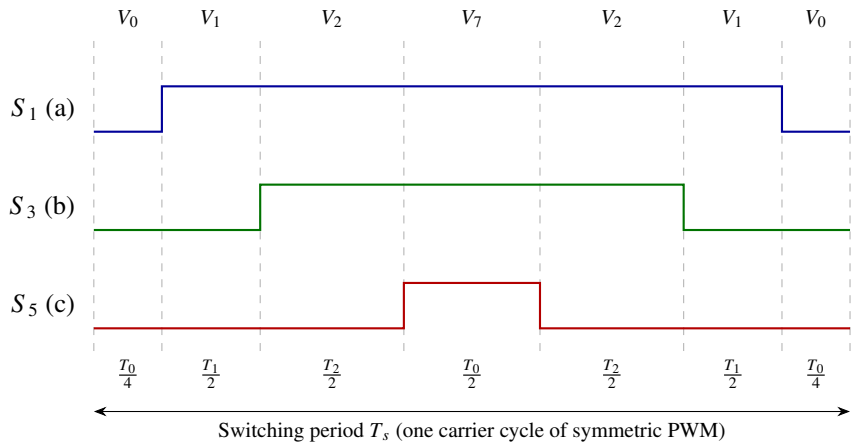


Figure 1.30: Symmetric seven-segment SVPWM switching sequence in sector I.

Table 1.16: Seven-segment switching sequence for each sector (first half of T_s; the second half mirrors it).

Sector	Segment 1	Segment 2	Segment 3	Segment 4 (centre)
I	000	100	110	111
II	000	010	110	111
III	000	010	011	111
IV	000	001	011	111
V	000	001	101	111
VI	000	100	101	111

1.10.15 SVPWM Implementation Algorithm

Algorithm 5 gives the complete procedure for one switching period. On a microcontroller, the three compare registers of a centre-aligned PWM timer are loaded with the leg on-times, and the hardware then produces the sequence automatically, including dead time.

Algorithm 5 SVPWM implementation

Require: V_α, V_β, measured V_{dc}, switching period T_s
Ensure: Upper-switch on-times t_a, t_b, t_c and gate signals

$V_{ref} \leftarrow \sqrt{V_\alpha^2 + V_\beta^2}$; $\theta \leftarrow \operatorname{atan2}(V_\beta, V_\alpha)$; **if** $\theta < 0$ **then** $\theta \leftarrow \theta + 2\pi$
$m \leftarrow \sqrt{3} V_{ref} / V_{dc}$
if $m > 1$ **then** ▷ limit to the inscribed circle
    $m \leftarrow 1$
end if
$k \leftarrow \lfloor 3\theta/\pi \rfloor + 1$
$T_a \leftarrow m T_s \sin(k\pi/3 - \theta)$; $T_b \leftarrow m T_s \sin(\theta - (k-1)\pi/3)$
$T_0 \leftarrow T_s - T_a - T_b$
From Table 1.16, identify which leg turns on first, second and third
if k is odd **then**
    $t_{\text{first}} \leftarrow T_a + T_b + T_0/2$; $t_{\text{second}} \leftarrow T_b + T_0/2$
else
    $t_{\text{first}} \leftarrow T_a + T_b + T_0/2$; $t_{\text{second}} \leftarrow T_a + T_0/2$
end if
$t_{\text{third}} \leftarrow T_0/2$
Assign these to t_a, t_b, t_c according to sector and load compare registers
Timer generates the centre-aligned pulses with dead time

1.10.16 Modulation Index

Equation (1.105) defines m so that $m = 1$ at the linear limit. With m as defined, the fundamental phase voltage peak is

$$\hat{V}_{an1} = V_{ref} = \frac{m\, V_{dc}}{\sqrt{3}}. \tag{1.108}$$

Relative to Eq. (1.81), $M = \pi m/(2\sqrt{3})$, and so $M = 0.907$ at the limit.

1.10.17 Linear Modulation Region

Linear operation requires a circular trajectory of $\vec{V}_{ref}$ inside the hexagon, so that $T_0 \geq 0$ at every angle. The worst case is at the middle of a sector, $\theta = 30^\circ$, where

$$T_a + T_b = m\, T_s\, [\sin(60^\circ - \theta) + \sin\theta] = m\, T_s \cos(\theta - 30^\circ) \leq T_s, \tag{1.109}$$

so that $m \leq 1$. Geometrically, the inscribed circle radius in Fig. 1.28 is the hexagon apothem, $\frac{2}{3} V_{dc} \cos 30^\circ = V_{dc}/\sqrt{3}$. Therefore

$$V_{ref,\max} = \frac{V_{dc}}{\sqrt{3}} = 0.577\, V_{dc}, \qquad V_{LL,\text{rms},\max} = \frac{V_{dc}}{\sqrt{2}} = 0.707\, V_{dc}. \tag{1.110}$$

Beyond this, in overmodulation, the reference runs into the hexagon boundary. The trajectory can be deformed gradually until the output reaches six-step at the hexagon corners.

1.10.18 Comparison with SPWM

Compare Eq. (1.110) with the SPWM limit from Eq. (1.86):

$$\frac{\hat{V}_{\text{SVPWM}}}{\hat{V}_{\text{SPWM}}} = \frac{V_{dc}/\sqrt{3}}{V_{dc}/2} = \frac{2}{\sqrt{3}} = 1.1547. \tag{1.111}$$

SVPWM provides 15.47% more linear output for the same DC bus, as Fig. 1.26 displays. The physical explanation is common-mode injection. The phase voltage is what matters, not the pole voltage, so the two zero vectors move the neutral potential in a way that leaves the line voltages intact, but flattens the pole-voltage references. As shown in Ref. [10], SVPWM is equivalent to SPWM with the offset

$$v_{\text{off}} = -\tfrac{1}{2}\left[\max(v_a^*, v_b^*, v_c^*) + \min(v_a^*, v_b^*, v_c^*)\right] \tag{1.112}$$

added to each of the references. Table 1.17 compares the two. The important point is that SVPWM gains a significant DC-bus benefit for little additional cost, on any modern automotive processor.

Table 1.17: Comparison of SPWM and SVPWM.

Parameter	SPWM	SVPWM
Basis	Independent per-phase comparison	Single rotating vector
Max linear phase peak	$0.5\,V_{dc}$	$0.577\,V_{dc}$
Max linear line rms	$0.612\,V_{dc}$	$0.707\,V_{dc}$
Fraction of six-step	78.5%	90.7%
Current ripple	Higher at high m	Lower, near-optimal
Implementation	Very simple	Moderate
Zero-vector control	None	Full (freedom of V_0/V_7)
Fit with FOC	Via inverse Clarke	Direct ($\alpha\beta$ input)

1.10.19 Advantages and Limitations of SVPWM

Advantages.

- 15.47% higher linear output voltage than SPWM.
- Lower current ripple and harmonic distortion, particularly at high modulation.
- Single-leg transitions, and the freedom of zero-vector placement.
- Natural pairing with field-oriented control, and fully digital implementation.

Limitations.

- More computation than SPWM, although this is modest on modern processors.

- Fixed-frequency sampling adds a delay of about one and a half switching periods.
- Common-mode voltage and sensitivity to dead time remain, the latter especially at low modulation.

1.10.20 Applications in EV Traction Drives

SVPWM or its carrier-based equivalent is the dominant modulator in PMSM traction drives under FOC. In the example drive of Example 6, the extra 15% voltage increases the speed at which field weakening starts, lowers the current needed at top speed, and maintains torque at low SoC. Discontinuous variants, used at high modulation, reduce inverter loss, and overmodulation up to six-step is used in the field-weakening range, where bus utilisation is at a premium. Reference [5] gives an extensive treatment of these drive applications.

1.11 Hysteresis Current Control

1.11.1 Introduction to Hysteresis Current Control

PWM methods generate a *voltage* and rely on an outer loop to regulate current. Motor torque depends on current, so an alternative is to control the current directly. Hysteresis current control (HCC), or bang-bang control, does this with no modulator at all: it compares the measured phase current with its reference and switches the leg whenever the error leaves a tolerance band [11]. It has the fastest possible response for a given inverter, since it acts on each error as it occurs. Its drawback, examined below, is a variable switching frequency.

1.11.2 Operating Principle

Consider phase a of Fig. 1.21. With S_1 on, a positive voltage across the winding drives current up; with S_4 on, a negative voltage drives it down. The controller holds the current within a band around the reference by choosing between these two states. It behaves in the same way as a thermostat, which allows the room temperature to wander within limits. Figure 1.31 shows the mechanism over a few switching cycles.

1.11.3 Reference Current and Actual Current

The reference current $i_x^*(t)$ for each phase comes from the torque controller, usually as a sinusoid that is synchronised with rotor position:

$$i_a^* = \hat{I}\sin\omega t, \quad i_b^* = \hat{I}\sin\left(\omega t - \tfrac{2\pi}{3}\right), \quad i_c^* = \hat{I}\sin\left(\omega t + \tfrac{2\pi}{3}\right), \tag{1.113}$$

where $\hat{I}$ (A) is set by the torque demand and ω (rad/s) by rotor speed. The actual current i_x (A) is measured with Hall-effect sensors or shunts, usually at high bandwidth. The tracking error is

$$e_x(t) = i_x^*(t) - i_x(t). \tag{1.114}$$

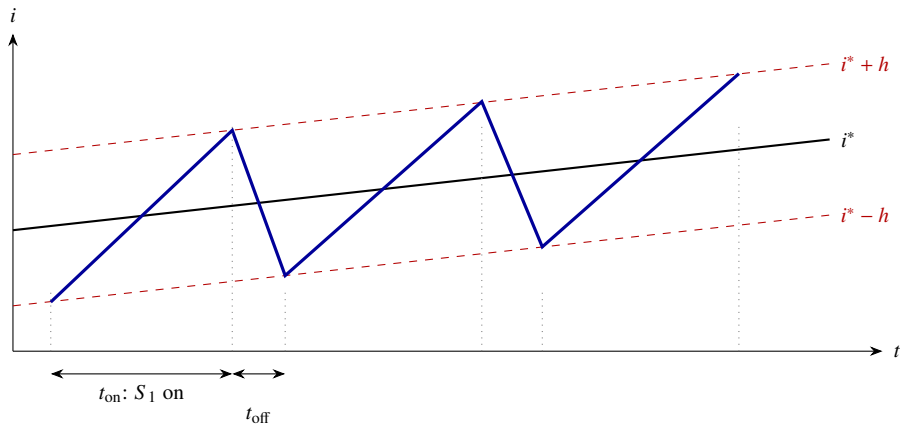


Figure 1.31: Hysteresis band operation over a few switching cycles.

1.11.4 Hysteresis Band

The hysteresis band is the permitted error range $\pm h$, with total width $\Delta I = 2h$ (A). As h decreases, tracking improves, but the switching frequency increases, so h is chosen by trading off current ripple against inverter switching loss. It must also be large compared to current sensor noise, or spurious switching will result.

1.11.5 Upper and Lower Switching Limits

The band defines two moving limits,

$$i_{\text{upper}}(t) = i^*(t) + h, \qquad i_{\text{lower}}(t) = i^*(t) - h. \tag{1.115}$$

In Fig. 1.31, the current ramps up with S_1 on until it reaches i_{upper}. Then it ramps down with S_4 on (in practice, D_4 conducts for positive current) until it meets i_{lower}. The slopes differ, as derived below, so the times t_{on} and t_{off} are generally unequal.

1.11.6 Switching Logic

The rule in phase x is a relay with memory:

$$S_x(t) = \begin{cases} 1, & e_x \geq +h, \\ 0, & e_x \leq -h, \\ S_x(t^-), & -h < e_x < +h, \end{cases} \tag{1.116}$$

where $S_x(t^-)$ is the previous state. The third line is the hysteresis itself: inside the band, no change occurs, which prevents chattering. Table 1.18 lists the logic, and Algorithm 6 gives its implementation.

Table 1.18: Hysteresis current controller switching logic for phase a.

Condition	Meaning	S_a	Upper S_1	Lower S_4	Effect on i_a
$e_a \geq +h$	$i_a \leq i_a^* - h$	1	ON	OFF	Rises
$e_a \leq -h$	$i_a \geq i_a^* + h$	0	OFF	ON	Falls
$\lvert e_a \rvert < h$	Within band	Hold	Hold	Hold	Continues

Algorithm 6 Hysteresis current control switching logic

Require: Reference i_x^*, measured i_x, band h, dead time t_d
Ensure: Gate signals of each leg

Initialise $S_x \leftarrow 0$ for $x \in \{a, b, c\}$
loop (at the sampling rate, or continuously in analogue hardware)
    for x in $\{a, b, c\}$ **do**
        $e_x \leftarrow i_x^* - i_x$
        if $e_x \geq +h$ **then**
            $S_x \leftarrow 1$
        else if $e_x \leq -h$ **then**
            $S_x \leftarrow 0$
        end if ▷ otherwise keep previous state
        Drive the upper gate with S_x, and the lower gate with $\overline{S_x}$, after dead time t_d
    end for
    if $\lvert i_x \rvert > I_{\max}$ for any phase **then**
        Disable all gates; report over-current fault
    end if
end loop

1.11.7 Three-Phase Hysteresis Current Controller

Figure 1.32 shows the complete three-phase scheme. The torque controller generates i_a^*, i_b^*, i_c^*. Each phase has an error summer and a hysteresis comparator, whose output drives its inverter leg. The measured currents close a very direct feedback loop, without any PI regulator or carrier. Since no voltage reference exists, HCC does not need motor parameters, and it inherently compensates for variations in DC-link voltage and back-EMF.

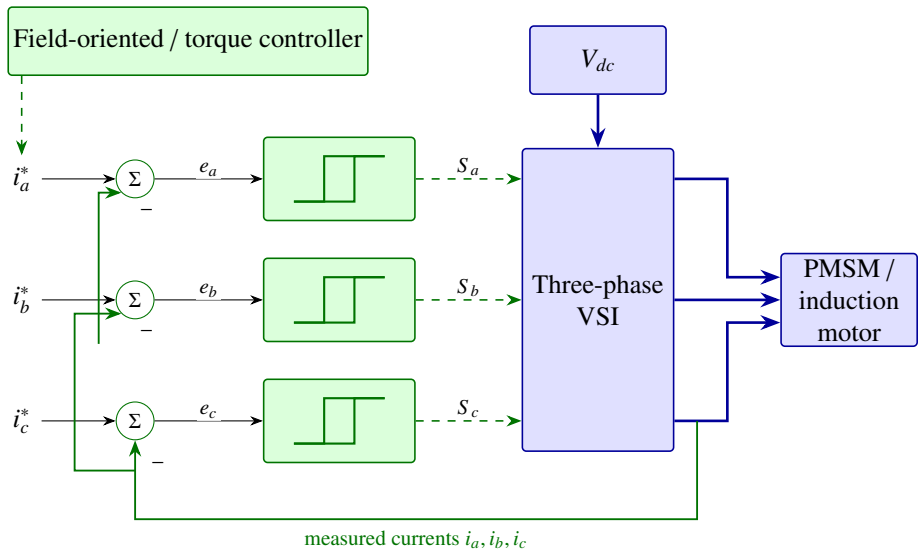


Figure 1.32: Three-phase hysteresis current controller.

With an isolated neutral, the three currents sum to zero, so only two are independent. Therefore each leg's switching changes the neutral potential and disturbs the other two phases. As a result, the three controllers interact, and the instantaneous error in a phase can reach up to twice the band [12]. Practical schemes can use a shared logic or suitable tuning to reduce the problem.

1.11.8 Current Tracking Performance

Figure 1.33 shows a simulated phase current, using a 10 A, 50 Hz reference with $h = 1$ A. The current stays inside the band over the whole cycle, in contrast to SPWM, where error depends on the load. The current therefore needs no phase compensation, since the error is bounded by the band and does not build up with the speed. The lower panel of the figure also indicates that the pulses vary in width and density through the cycle, showing the variable-frequency switching treated next.

1.11.9 Switching Frequency Characteristics

Treat the phase with the midpoint-referenced single-phase model. Neglecting resistance, the winding sees the pole voltage $\pm V_{dc}/2$ minus the back-EMF e_b (V), so with phase inductance L (H),

$$L\frac{\mathrm{d}i}{\mathrm{d}t} = \pm\frac{V_{dc}}{2} - e_b. \tag{1.117}$$

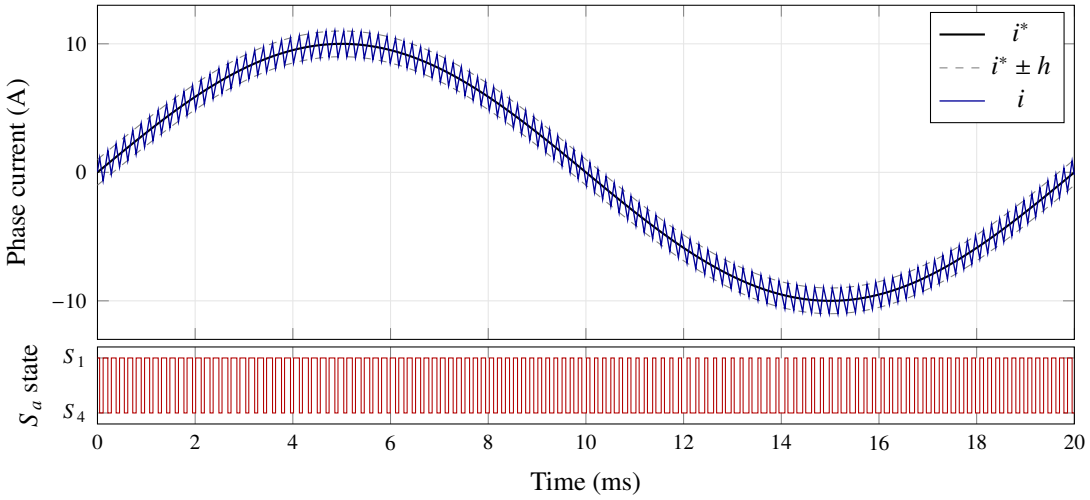


Figure 1.33: Simulated hysteresis band current tracking waveform ($\hat{I}$ = 10 A, 50 Hz, h = 1 A, single phase model with sinusoidal back-EMF).

Assuming the reference slope is small during one switching cycle, the current changes by $\Delta I = 2h$ in each direction, with times

$$t_{\text{on}} = \frac{L\Delta I}{\frac{V_{dc}}{2} - e_b}, \qquad t_{\text{off}} = \frac{L\Delta I}{\frac{V_{dc}}{2} + e_b}. \tag{1.118}$$

Summing and inverting gives the switching frequency:

$$f_s = \frac{1}{t_{\text{on}} + t_{\text{off}}} = \frac{\frac{V_{dc}^2}{4} - e_b^2}{2\,V_{dc}\,L\,h}. \tag{1.119}$$

Equation (1.119) is the key to the characteristics of HCC. Frequency is not constant: it varies with the back-EMF, hence with rotor position and speed, and it is highest at low back-EMF,

$$f_{s,\max} = \frac{V_{dc}}{8\,L\,h} \qquad (e_b = 0). \tag{1.120}$$

It falls as e_b rises, so the switching frequency is reduced at high speed. Adaptive-band methods vary h in step with the operating conditions to hold f_s nearly constant [13]. Variable frequency produces a wide harmonic spectrum, which complicates filter design and EMC compliance, and can increase acoustic noise.

Example 10: Hysteresis current controller switching operation

Problem. A leg of a 400 V inverter drives a phase with L = 2 mH. The instantaneous reference is 100 A (treat as constant for a few cycles), the band h = 2.5 A, and the back-EMF 100 V. Starting at the lower limit, list the switching sequence and find the on-time, off-time, frequency and average pole voltage. Neglect resistance.

Given. V_{dc} = 400 V; L = 2 mH; i^* = 100 A; h = 2.5 A; e_b = 100 V.

Required. Sequence; t_{on}, t_{off}, f_s, $\bar{v}_{aO}$.

Governing equations. Eqs. (1.116), (1.118), (1.119).

Solution. Limits are 102.5 A and 97.5 A. The slopes are

$$\begin{aligned} \text{rise:} \quad & \frac{200 - 100}{0.002} = 50\,000 \text{ A/s}, \\ \text{fall:} \quad & \frac{-200 - 100}{0.002} = -150\,000 \text{ A/s}. \end{aligned}$$

$$t_{\text{on}} = \frac{5}{50\,000} = 100\ \mu\text{s}, \qquad t_{\text{off}} = \frac{5}{150\,000} = 33.3\ \mu\text{s},$$
$$f_s = \frac{1}{133.3\ \mu\text{s}} = 7.5 \text{ kHz}.$$

Equation (1.119) confirms this: $(40\,000 - 10\,000)/(2 \times 400 \times 0.002 \times 2.5) = 7500$ Hz.

Time (μs)	i_a (A)	e_a (A)	S_a	Event
0	97.5	+2.5	1	Lower limit reached, S_1 ON
100	102.5	−2.5	0	Upper limit reached, S_4 ON
133.3	97.5	+2.5	1	Lower limit reached, S_1 ON
233.3	102.5	−2.5	0	Upper limit reached, S_4 ON

Duty is $d = 100/133.3 = 0.75$, so $\bar{v}_{aO} = (0.75 - 0.25) \times 200 = 100$ V.

Result. $t_{\text{on}} = 100\ \mu$s; $t_{\text{off}} = 33.3\ \mu$s; $f_s = 7.5$ kHz; $\bar{v}_{aO} = 100$ V.

Interpretation. The controller automatically generates the average voltage equal to the back-EMF, with no model or modulator. At $e_b = 0$, Eq. (1.120) gives 10 kHz; thus f_s varies with the operating point. A narrower band of $h = 1.25$ A would double every frequency.

1.11.10 Advantages and Limitations

Advantages.

- Very fast dynamic response, limited only by inductance and available voltage.
- Simple, robust structure that does not require a motor model.
- Inherent peak-current limitation, which protects devices.
- Automatic compensation for DC-bus variation, back-EMF and dead time.

Limitations.

- Variable switching frequency, with a spread harmonic spectrum and acoustic noise.
- Switching loss cannot be fixed, making thermal design more conservative.
- Interaction among phases for an isolated neutral.
- Digital implementation needs very high sampling rates: a loop at f_{sample} overshoots the band, by up to $T_{\text{sample}}\,\mathrm{d}i/\mathrm{d}t$.
- Higher ripple at low speed for a given band, and poor low-current behaviour near zero crossings.

1.11.11 Applications in Electric Vehicle Motor Drives

HCC is well-suited to drives in which torque response and simplicity matter more than harmonic quality. It has been used for switched-reluctance traction motors, in which phase current profiling is natural, and for low-cost brushless DC drives in light electric vehicles. In mainstream PMSM cars, fixed-frequency FOC with SVPWM is preferred, for reasons of NVH and EMC, but HCC concepts remain as a fast peak-current limit. Direct torque control is a related method, which applies hysteresis to torque and flux rather than to currents [5].

1.11.12 Comparison with Carrier-Based PWM and SVPWM

HCC differs from the previous two methods in one basic respect: SPWM and SVPWM are *voltage* modulators with a fixed period, inside a current loop; HCC is itself the *current* loop, with no fixed period. So carrier methods offer predictable spectra and losses, but response is limited by sampling and PI bandwidth. HCC reverses that trade-off. The complete comparison follows in Section 1.12.

1.12 Comparison of PWM Techniques

The three techniques of Sections 1.9–1.11 can now be compared directly. Figure 1.34 sets their control architectures side by side. In (a) SPWM, a current loop generates three phase voltage references, each compared with a triangular carrier. In (b) SVPWM, the same loop outputs a single $\alpha\beta$ vector, and times are computed. In (c) HCC, the error drives comparators directly, and no modulator or PI regulator intervenes. The structure explains most of the differences in Table 1.19: (a) and (b) have a fixed clock, so that they trade response for predictability, whereas (c) has none.

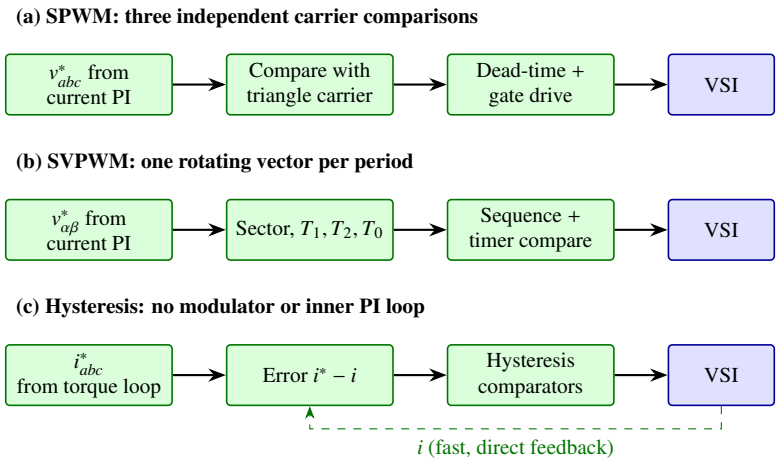


Figure 1.34: Comparative control architectures: (a) SPWM, (b) SVPWM, (c) hysteresis current control.

Table 1.19: Detailed comparison of SPWM, SVPWM and hysteresis current control.

Criterion	SPWM	SVPWM	Hysteresis control
Operating principle	Sine reference compared with triangular carrier in each phase	Rotating reference vector synthesised from adjacent active and zero vectors	Current error kept within a tolerance band
Modulation approach	Per-phase, voltage mode, carrier based	Whole-system, voltage mode, vector based	Per-phase, current mode, no modulator
DC bus utilisation	Line peak 0.866 V_{dc} (linear)	Line peak 1.0 V_{dc} (linear), 15.47% more	Follows the current; near bus limit tracking is lost
Harmonic performance	Clustered near m_f; good	Clustered; lowest THD at high modulation	Broad, spread spectrum
Switching frequency	Fixed, f_c	Fixed, $1/T_s$	Variable, Eq. (1.119)
Switching losses	Moderate; predictable	Moderate; reduced further by discontinuous forms	Varies with operating point; hard to bound
Current ripple	Moderate; rises with m_a	Lowest of the fixed-frequency methods	Bounded by the band, but with phase interaction
Implementation complexity	Low	Moderate	Low in analogue; high-rate in digital
Computational requirements	Very low	Moderate: sector, times, sequence	Low per sample but very fast sampling
Dynamic response	Limited by current loop and sampling	Limited by current loop and sampling	Fastest; limited by L and V_{dc}
Control flexibility	Limited	High: zero-vector freedom, overmodulation, FOC	Limited; only band adjustable

Table 1.19 continued

Criterion	SPWM	SVPWM	Hysteresis control
Suitability for EV traction	Low-cost and simple drives; auxiliaries	Standard for PMSM and induction traction	Niche: SRM, BLDC, fast current limiting
Advantages	Simple; low computation; predictable	Highest linear voltage; low ripple; digital fit	Very fast; parameter independent; current limiting
Disadvantages	Lower voltage; no joint phase treatment	More computation; sampling delay	Variable frequency; noise; EMC difficulty

1.12.1 Discussion of the Comparison

Three conclusions can be drawn from Table 1.19. First, SVPWM dominates SPWM in nearly every respect, at the small cost of a few extra calculations; and since a carrier-based form with a zero-sequence offset (Eq. (1.112)) performs identically, the two are best seen as members of one family. Second, HCC is a fundamentally different choice, rather than an improved version: its strength in current response is paid for in harmonic and thermal unpredictability. Third, DC bus utilisation carries particular value in a BEV, because battery voltage falls with SoC and the highest-speed capability depends on the voltage margin, as shown in Eq. (1.110).

1.12.2 Preferred Technique for Different EV Motor-Drive Operating Conditions

No single technique is best for every operating condition, and modern drives may change strategy with speed and load. Table 1.20 and the following notes relate the choice to the operating region.

Table 1.20: Preferred modulation technique by EV operating condition.

Operating condition	Preferred technique	Reason
Low speed, high torque (launch, crawl)	SVPWM, continuous	Low ripple and quiet operation when audible noise is exposed
Medium speed cruise	SVPWM, continuous or discontinuous	High efficiency at light load, low current ripple
High speed, field weakening	SVPWM with overmodulation, up to six-step	Maximum voltage extends constant-power range
Very low m_f at top speed	Synchronised SVPWM	Prevents sub-harmonics and beat effects
Peak current protection	Hysteresis limit in hardware	Fastest response to a fault or a transient
Low-cost auxiliary drive	SPWM	Simple, sufficient performance
SRM or trapezoidal BLDC drive	Hysteresis control	Direct shaping of current pulses

At *low speed and high torque*, back-EMF is small and modulation index is small, so voltage headroom is irrelevant, whereas current ripple and audible noise matter because little road or wind noise masks them. Continuous SVPWM at a fixed frequency is therefore preferred. SVPWM and SPWM behave similarly at this point.

In *urban and highway cruise*, the drive runs at part load for most of its life. Loss reduction determines the range, so discontinuous SVPWM is attractive at moderate to high modulation, where the clamped leg reduces switching loss.

At *high speed in the field-weakening region*, the inverter runs out of voltage. SVPWM's extra 15% is valuable, and overmodulation towards six-step gives the highest torque per ampere at the top of the speed range, at the expense of low-order harmonics. As m_f falls, synchronisation of the switching pattern with the fundamental is needed.

In *fast transients and faults*, the hysteresis principle retains its value. Even in drives using SVPWM, a hardware comparator commonly limits peak current on a cycle-by-cycle basis, because it responds in microseconds, without waiting for the next sampling instant.

For *cost-sensitive auxiliaries* such as coolant pumps and small compressors, SPWM's simplicity is an advantage, and bus utilisation is less important. For *switched-reluctance* or *trapezoidal BLDC* drives, the required current pulses suit hysteresis control, which is preferred.

For the 100 kW traction drive of Example 6, the appropriate choice is therefore field-oriented control with SVPWM at a fixed 10 kHz, using discontinuous modulation at higher modulation indices, overmodulation at top speed, and a hardware hysteresis comparator for peak-current protection.

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